UCS3461 Foundations of Artificial Intelligence

ADVERSARIAL SEARCH

Semester 4, B.E. Computer Science and Engineering
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Sakaya Milton R
SSN College of Engineering

Outline

- Games
- Minimax: Optimal Decisions in Games
- Minimax with α-β Pruning

Games

Games

- Multiagent environment
  - Actions of other agents affect the state of the world
- Agents have independent utilities (values on outcomes)
- Games: agents compete – their goals conflict
  - Adversarial search
- Agents may also cooperate – different problem

Types of Games

- Many different kinds of games!
- Axes
  - One, two, or more players?
  - Zero sum?
  - Perfect information (can you see the state)?
  - Deterministic or stochastic?
- Need algorithms for calculating a strategy (policy) which recommends a action/move in each state.

$$\text{policy} : \text{state} \rightarrow \text{action}$$

Zero-Sum Games

- Game theory – Economics
- Two agents act alternately (two-player turn-taking)
- Fully observable (perfect information)
- Deterministic
- Utilities at the end of the game are equal and opposite
  - Agents have opposite utilities (values on outcomes)
  - Adversarial, pure competition
  - Lets us think of a single value that one maximizes and the other minimizes
- Constant-sum game
  - Chess: $(0, 1)$, $(1, 0)$, $(\frac{1}{2}, \frac{1}{2})$

Games are Too Hard to Solve

Games are like real-world problems — too hard to solve

- Chess
- Branching factor $= 35$
- 50 moves by each player
- Search tree $= 35^{100} \approx 10^{154}$ nodes!
- Search graph has "only" 10^{40} nodes!!
- Optimal decision is infeasible
- Some decision is imperative, like real-world!

Two-player Game

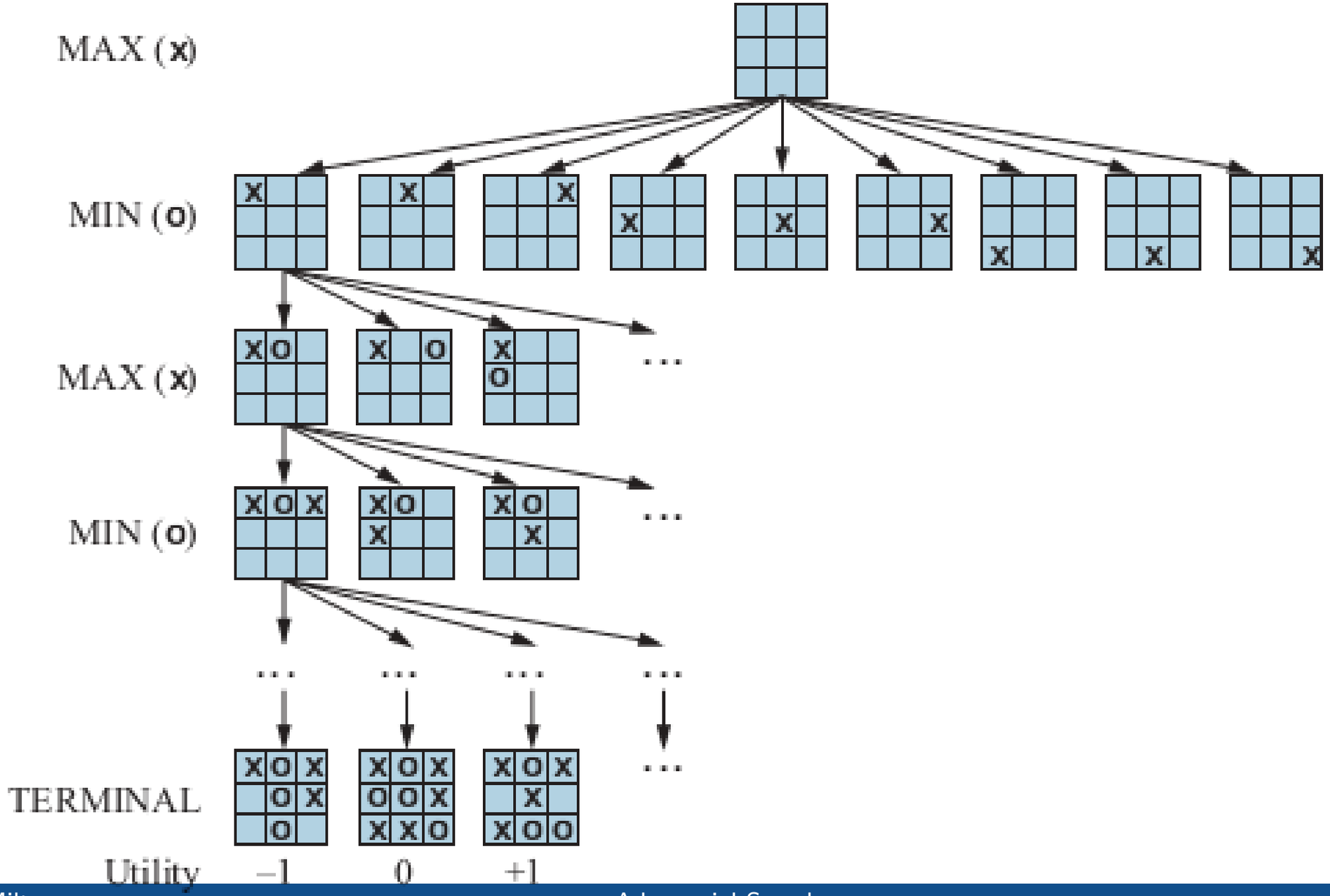
Search problem

- S_0: Initial state
- Player(s): which player has the move in state s
- Action(s): set of legal actions in state s
- Result(s, a): transition model, result of move (action) a in state s
- Terminal?(s): terminal states – true when the game is over
- Utility(s, p): utility function (objective function, payoff) – the numeric value of the game in terminal state s for player p when the game is over.
  - Zer-sum game
  - Chess: $(0, 1)$, $(1, 0)$, $(\frac{1}{2}, \frac{1}{2})$

Game Tree

- Initial state, Actions(), Result()
- Nodes $\equiv$ game states, edges $\equiv$ actions
- Search tree examines enough nodes of game tree to choose a move
- Tic-tac-toe
  - MAX moves first, MAX and MIN alternate
  - MAX places an X in an empty square
  - MIN places an O in an empty square
  - Leaves – terminal states
    - one player has 3 in a row
    - all the squares are filled
  - Utility values from the point of view of MAX
  - $9! = 362,880$ terminal nodes
- Chess: 10^{40} nodes

Tic-Tac-Toe Game Tree



Choose Good Move

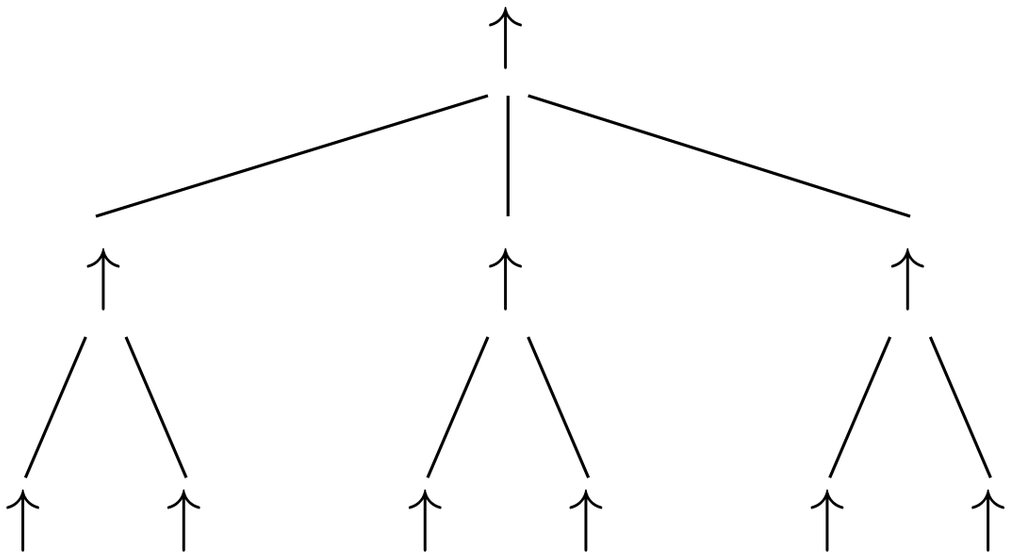
- Choose a good move when time is limited
- Pruning – ignore subtrees of search tree, which do not affect the final choice
- Evaluation function – heuristic function, to approximate the true utility without doing complete search

Optimal Decisions in Games

Optimal Decisions in Games

- In search, strategy = sequence of actions to a goal state
- In adversarial search, contingent strategy
  - MAX's move in the initial state
  - In states resulting from every possible response by MIN, MAX's moves
  - In states resulting from every possible response by MIN to those moves, MAX's moves

Value of States



$$\text{Value}(n) = \max_{c \in \text{Successors}(n)} \text{Value(c)}$$

Terminal state
$\text{Value}(n) = \text{Utility}(n)$

Maxvalue Algorithm

Algorithm: Maxvalue(n)

Input: n a node in a game tree
Output: A pair (value in node n, path that gives this value)

```
if n is a leaf node then  return evaluate(n)
max_val ← −∞
foreach child c of n do
    val ← Maxvalue(c)
    if max_val < val  then
        max_val ← val
return max_val
```

Maxvalue Algorithm

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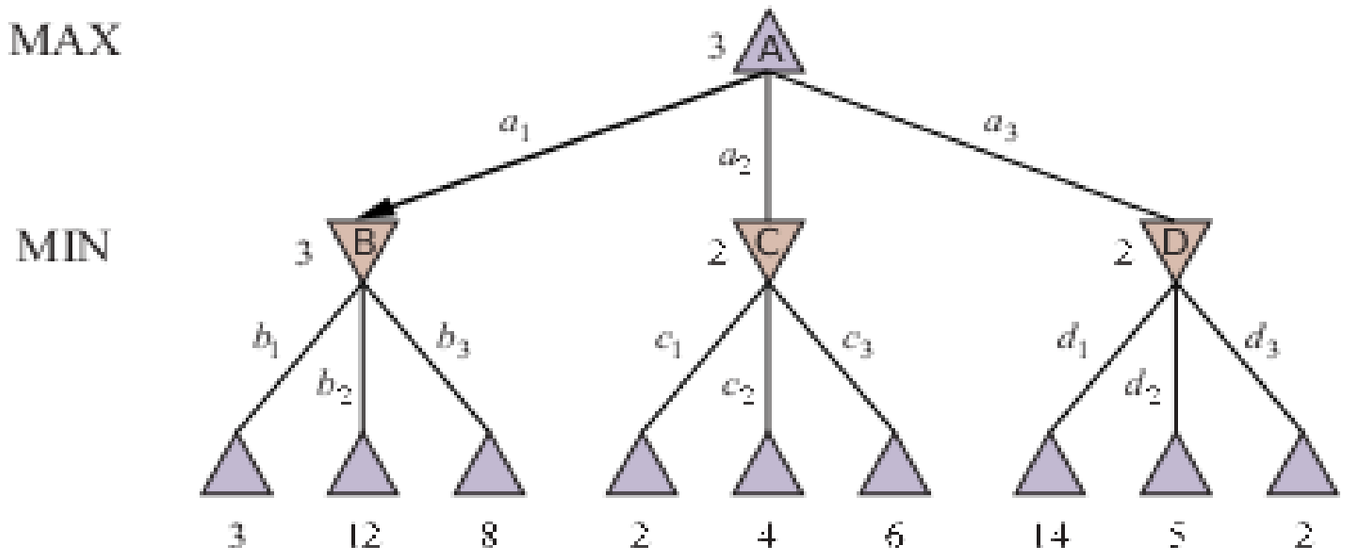
if n is a leaf node then  return evaluate(n), [ ]
max_val ← −∞
foreach child c of n do
    val, path ← Maxvalue(c)
    if max_val < val  then
        max_val ← val
        best ← [c] + path
return max_val, best
```

Maxvalue Example Trace



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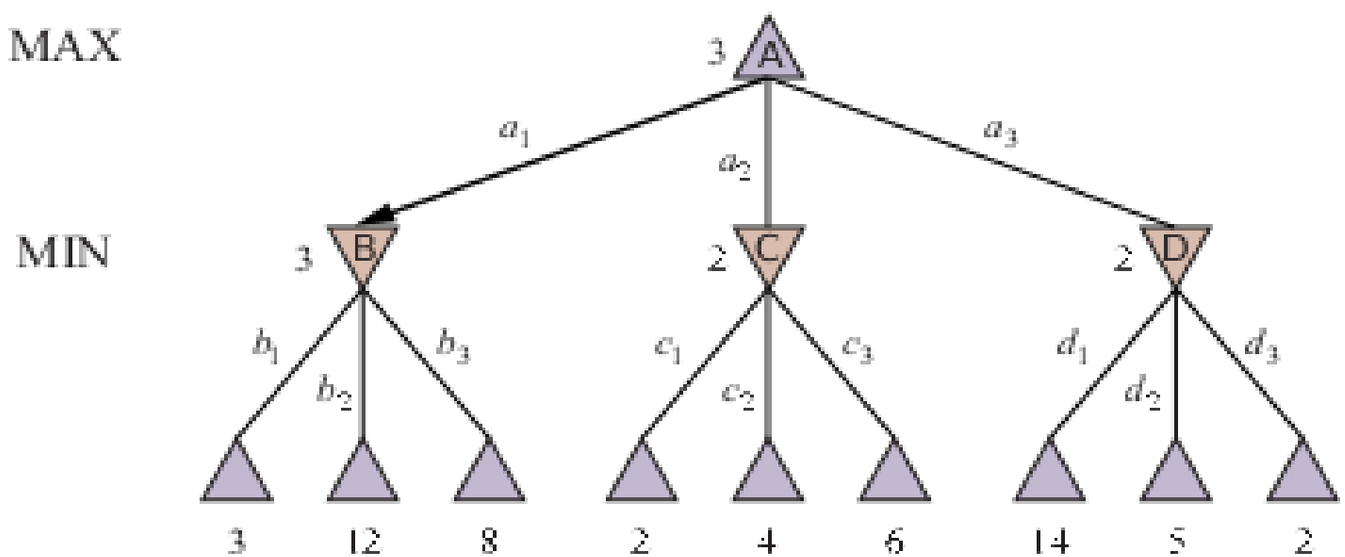

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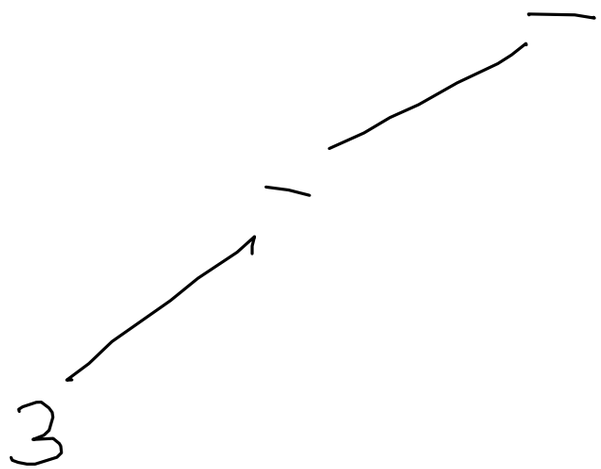

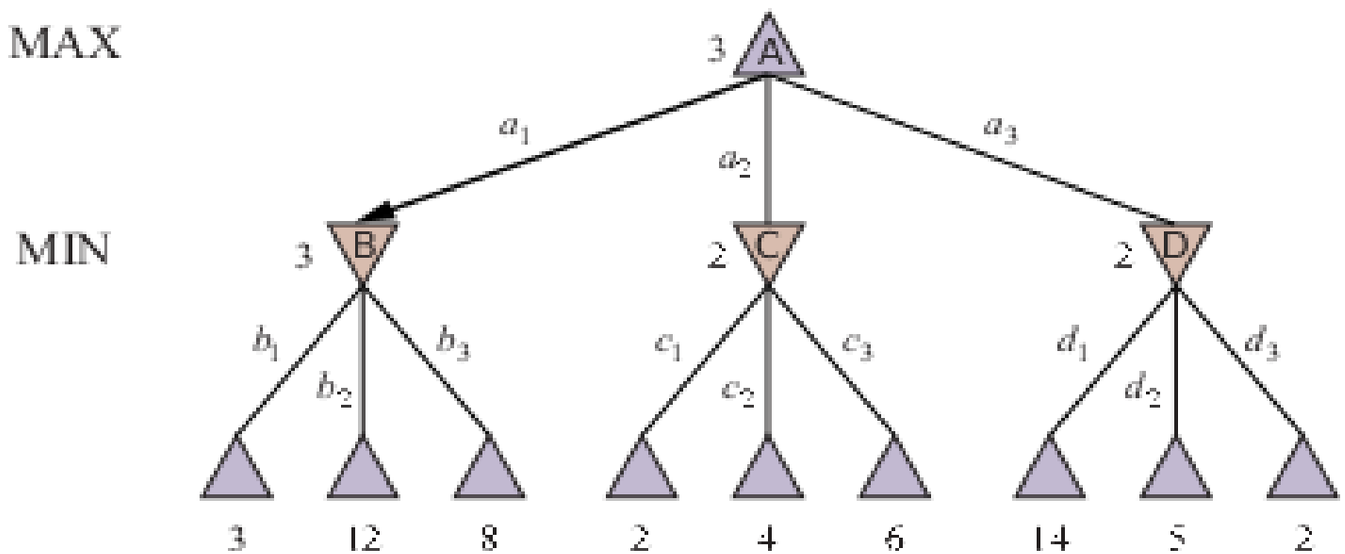

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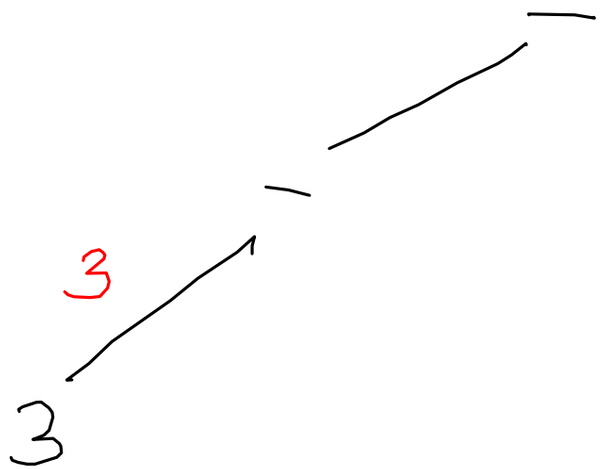

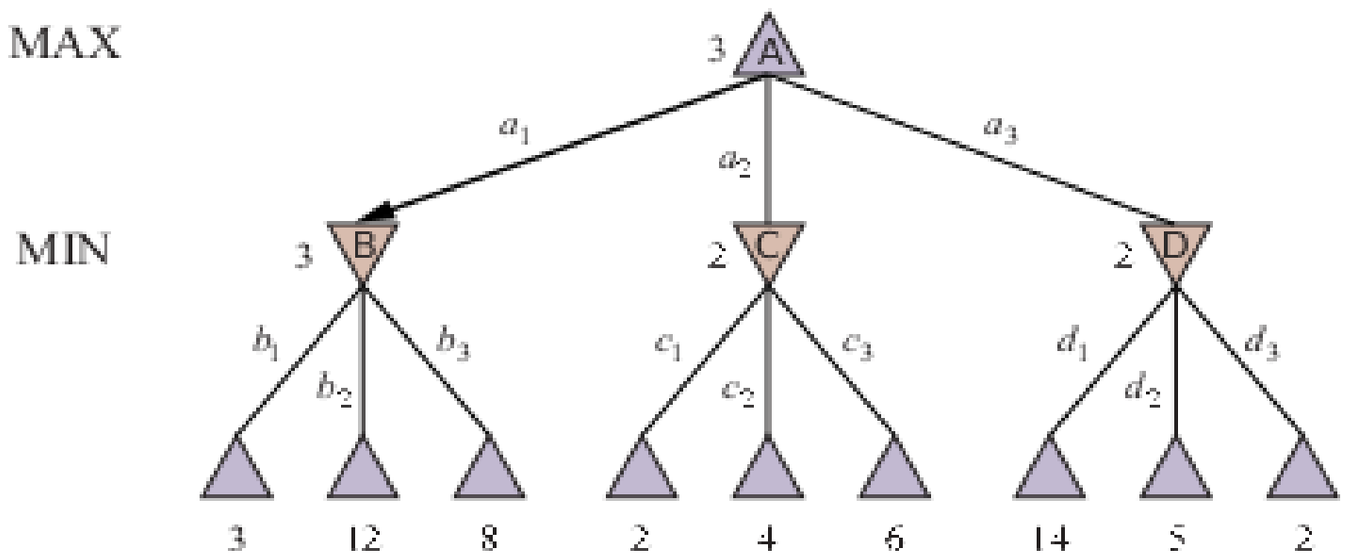

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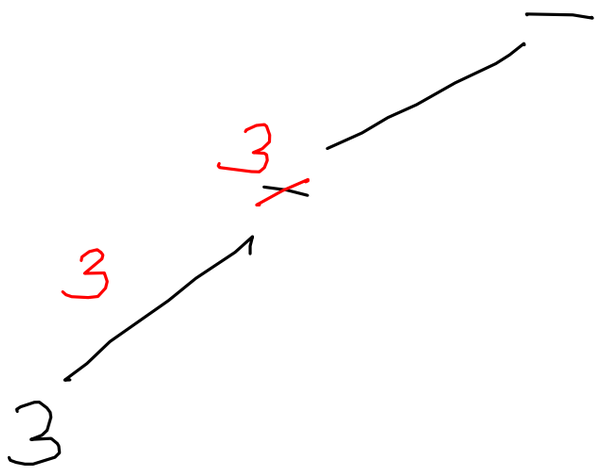

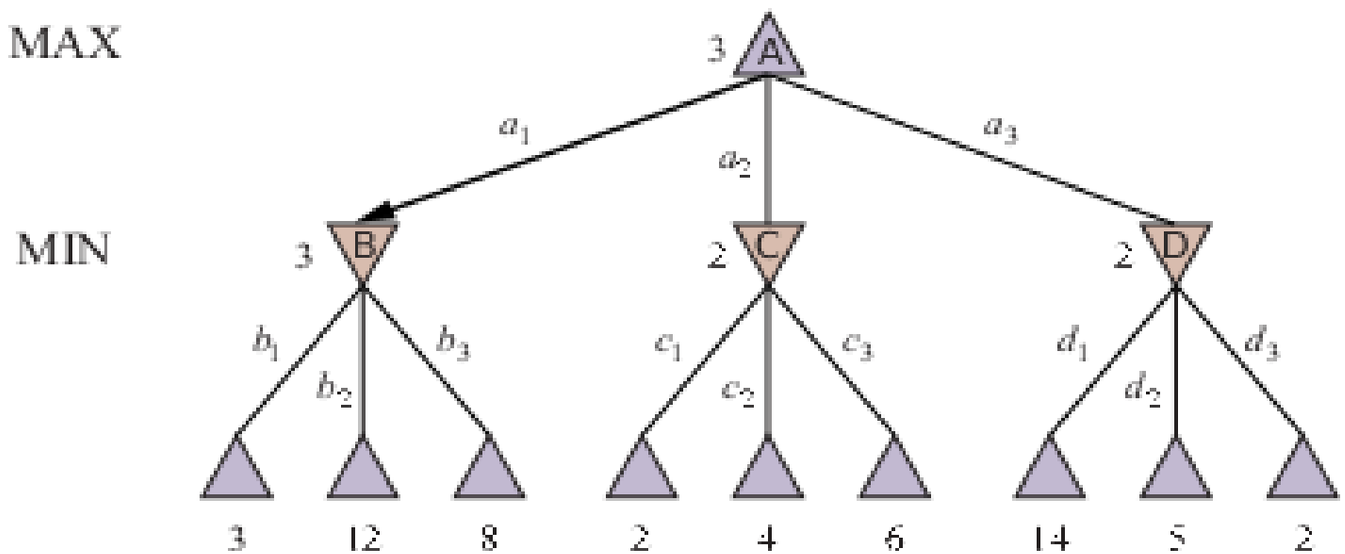

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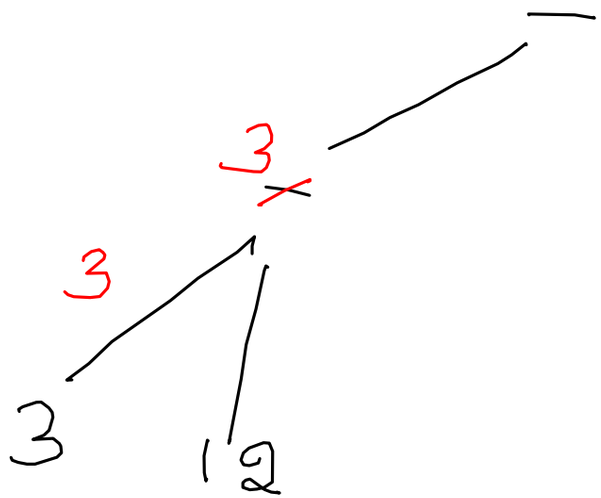

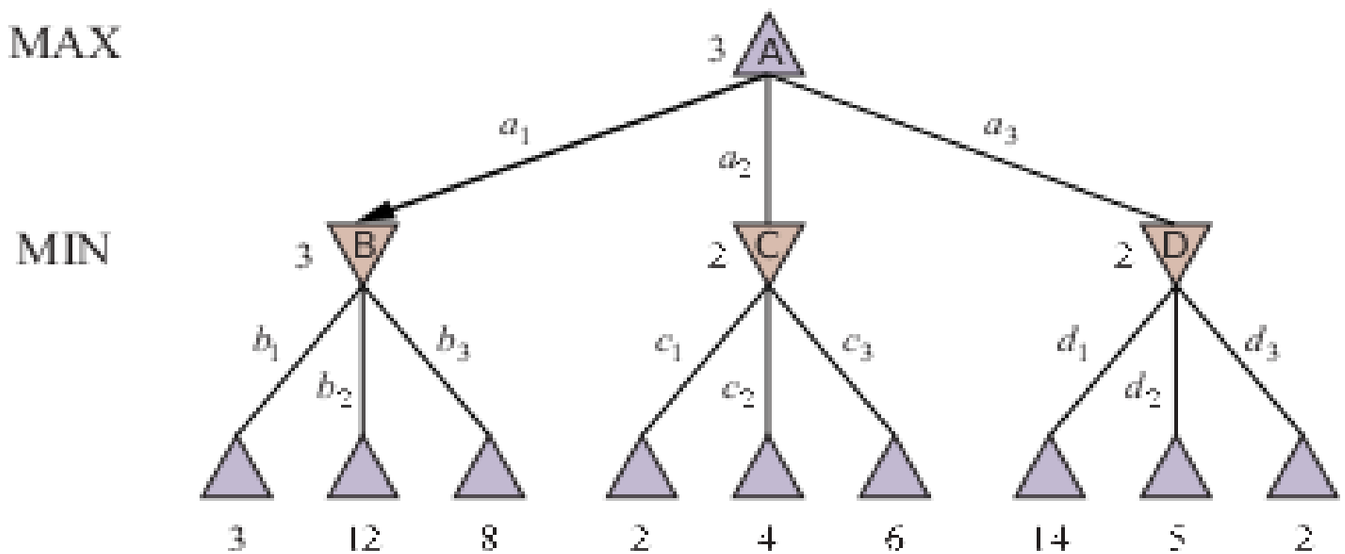

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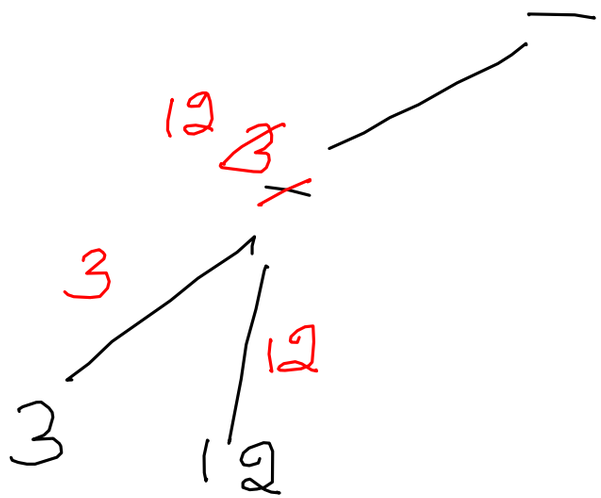

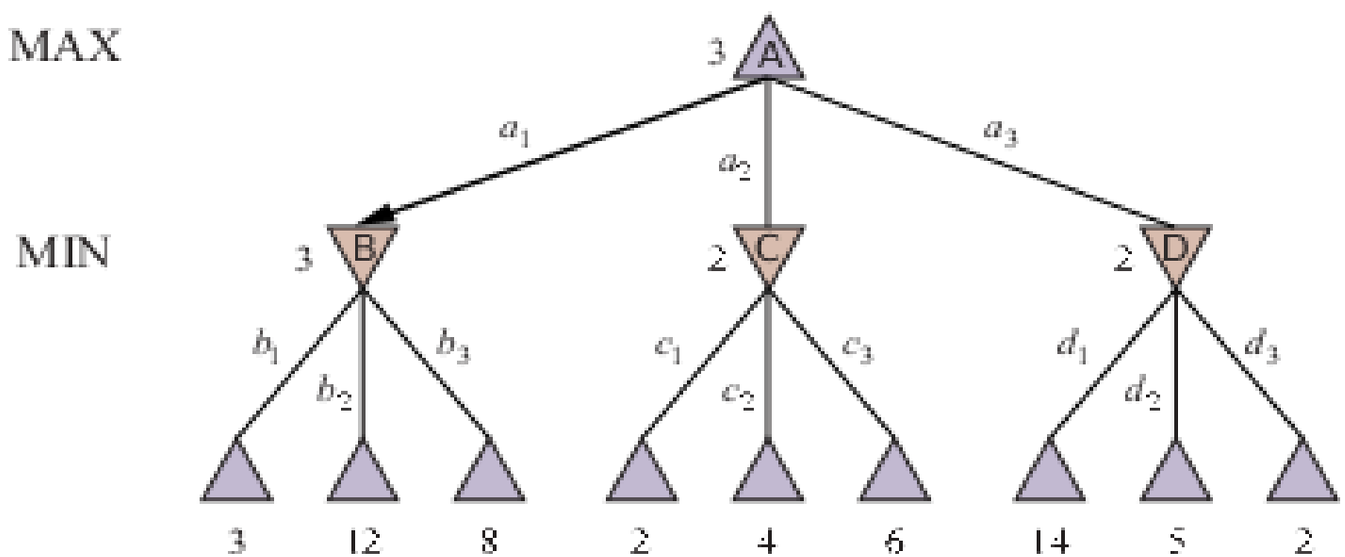

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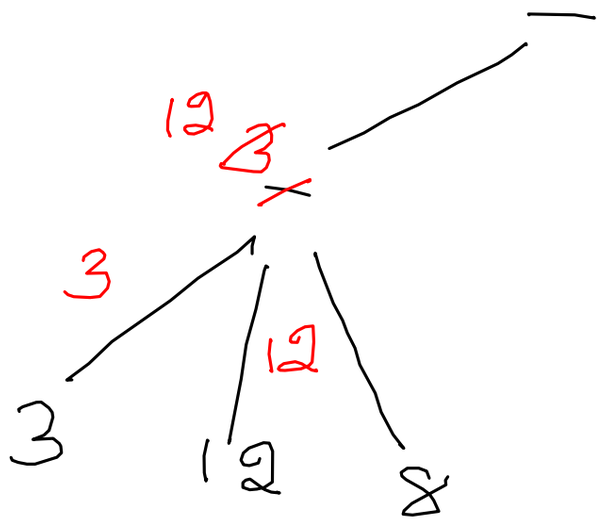

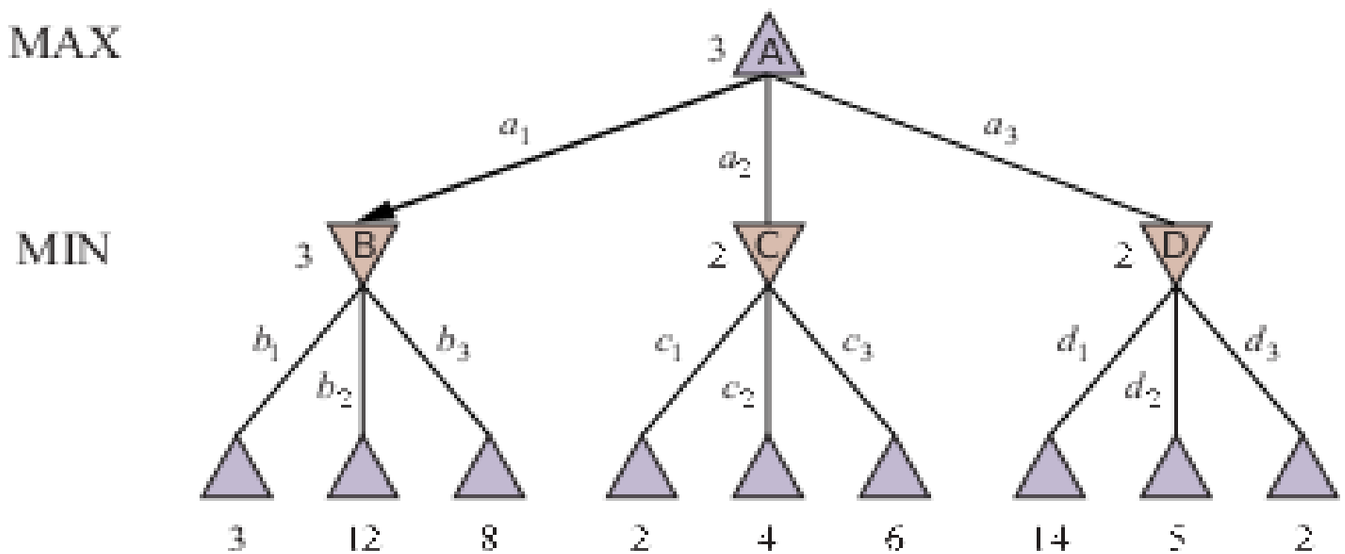

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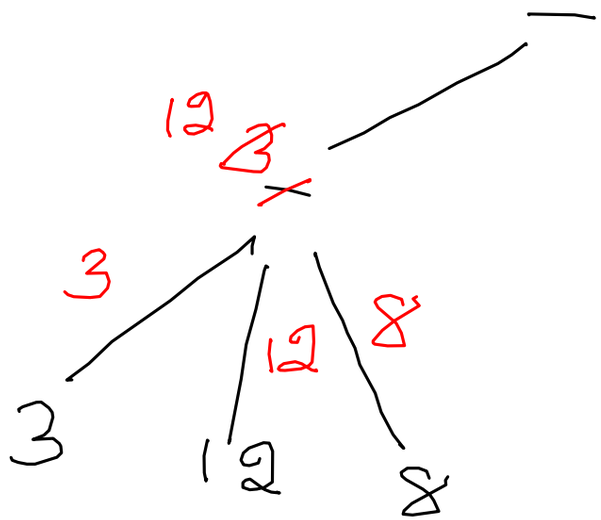

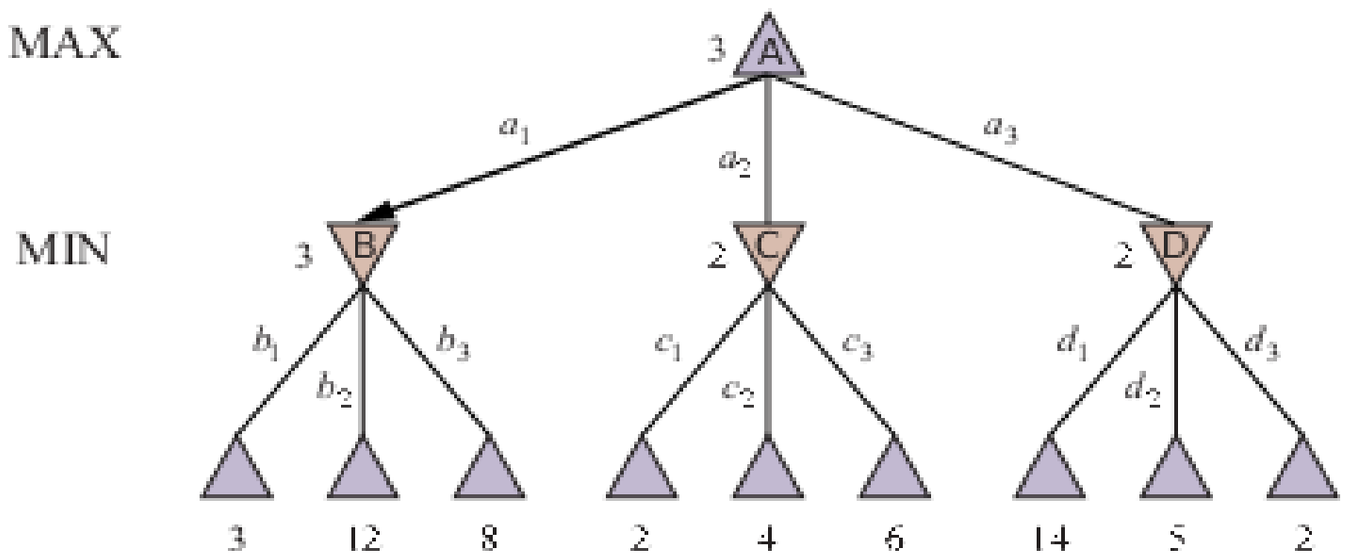

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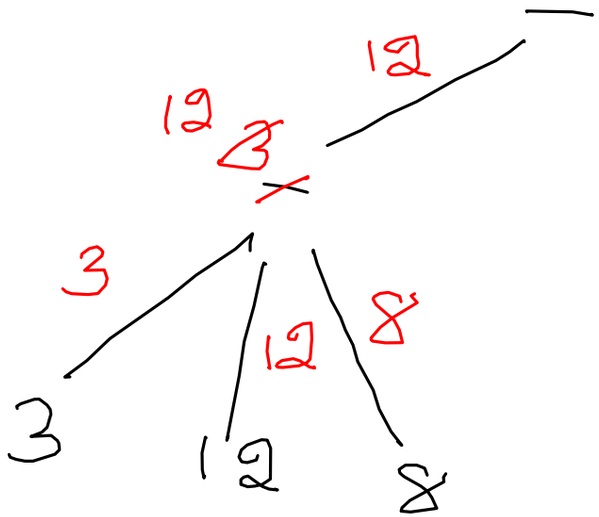

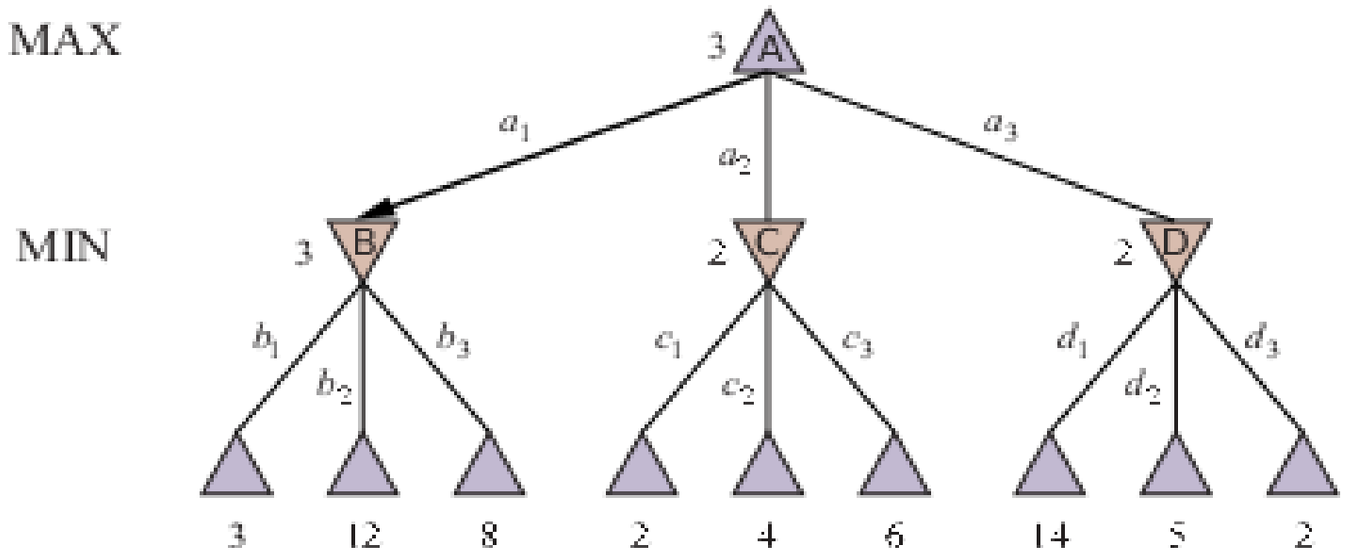

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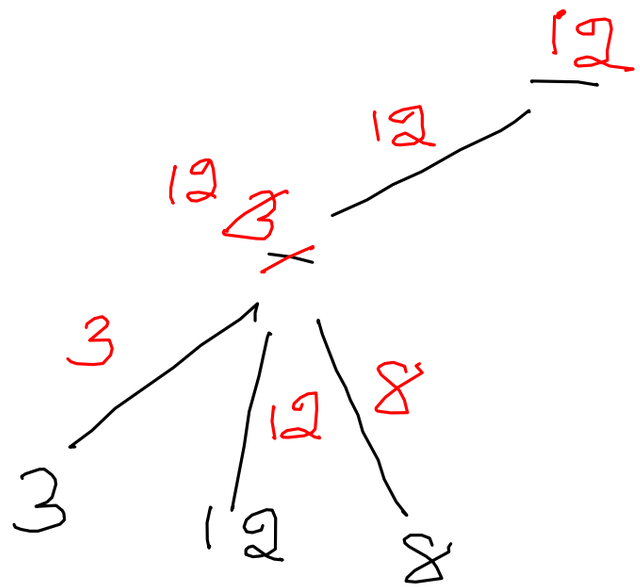

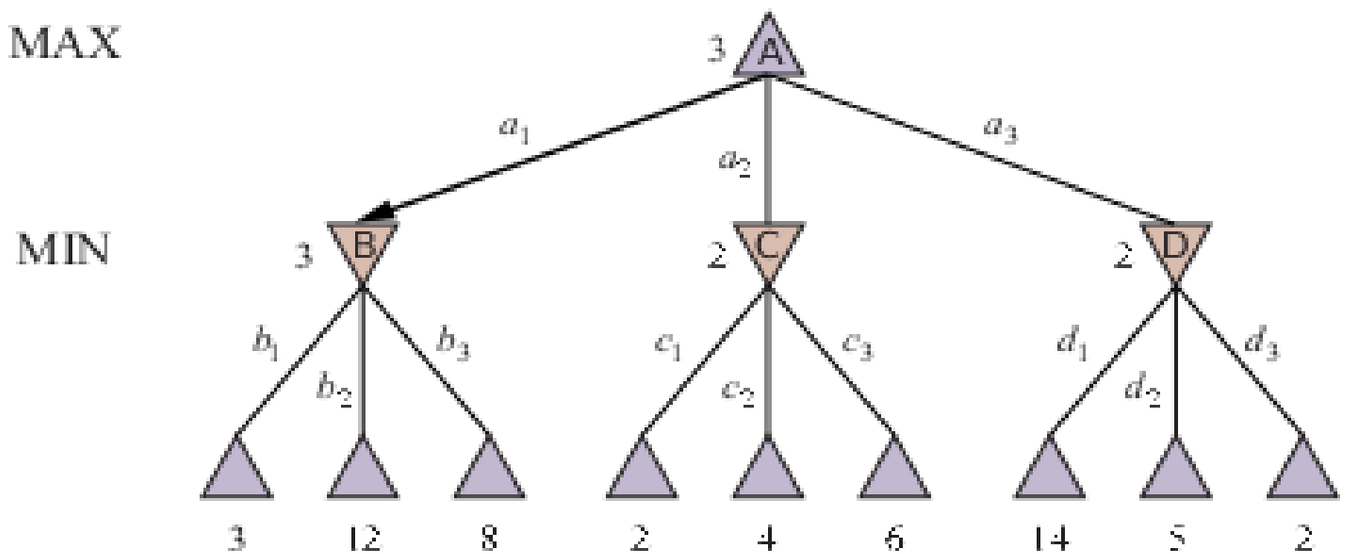

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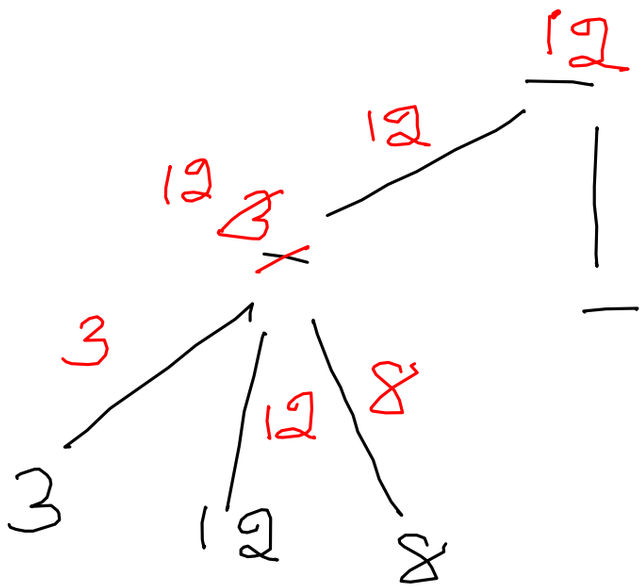

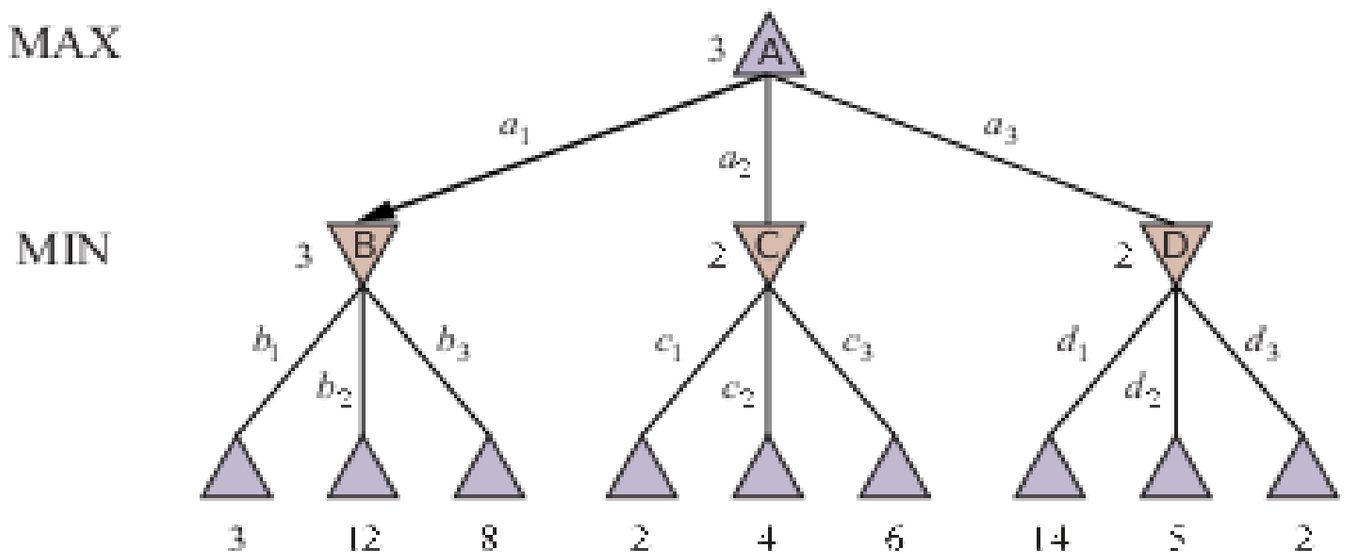

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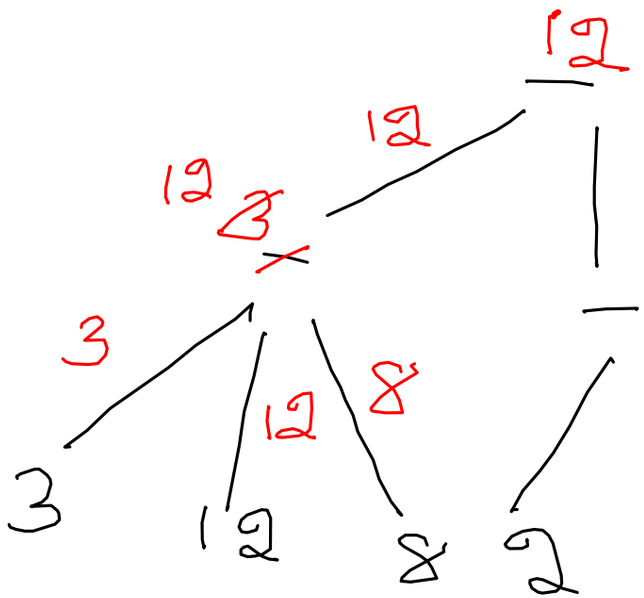

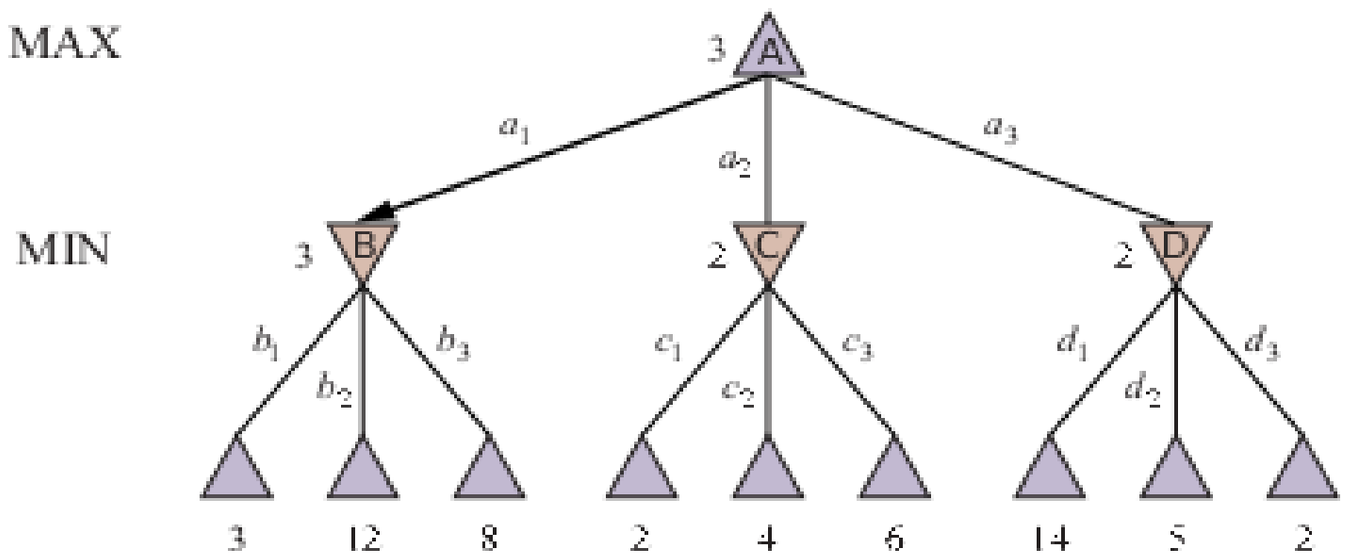

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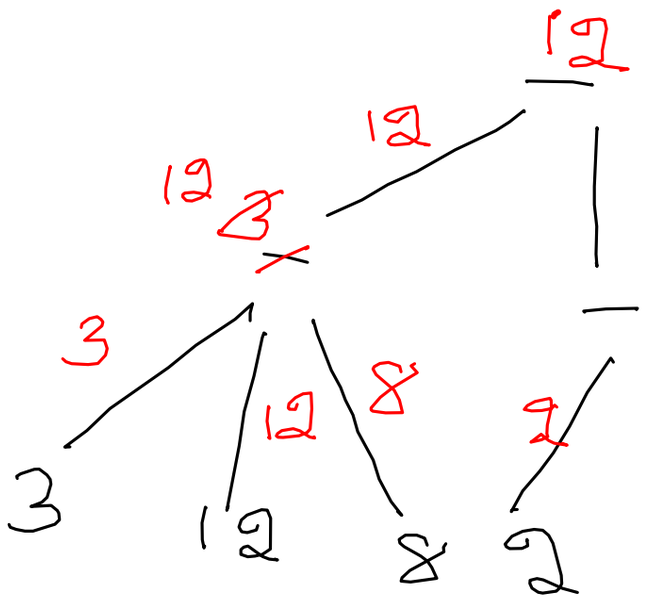

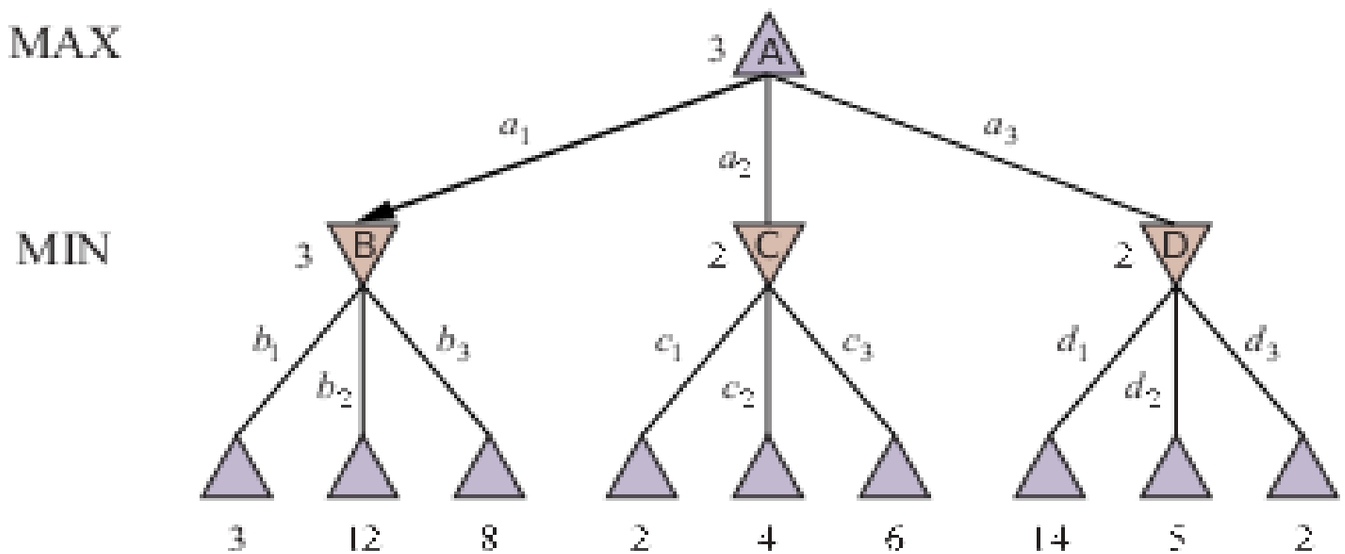

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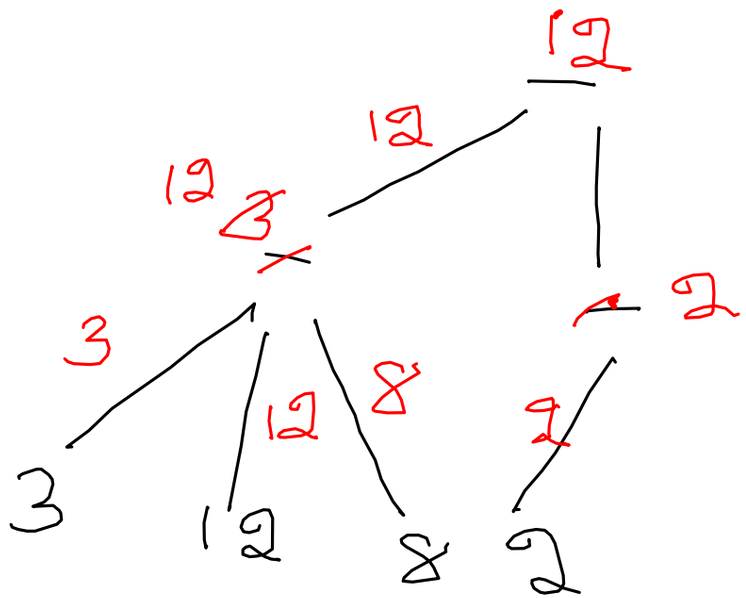

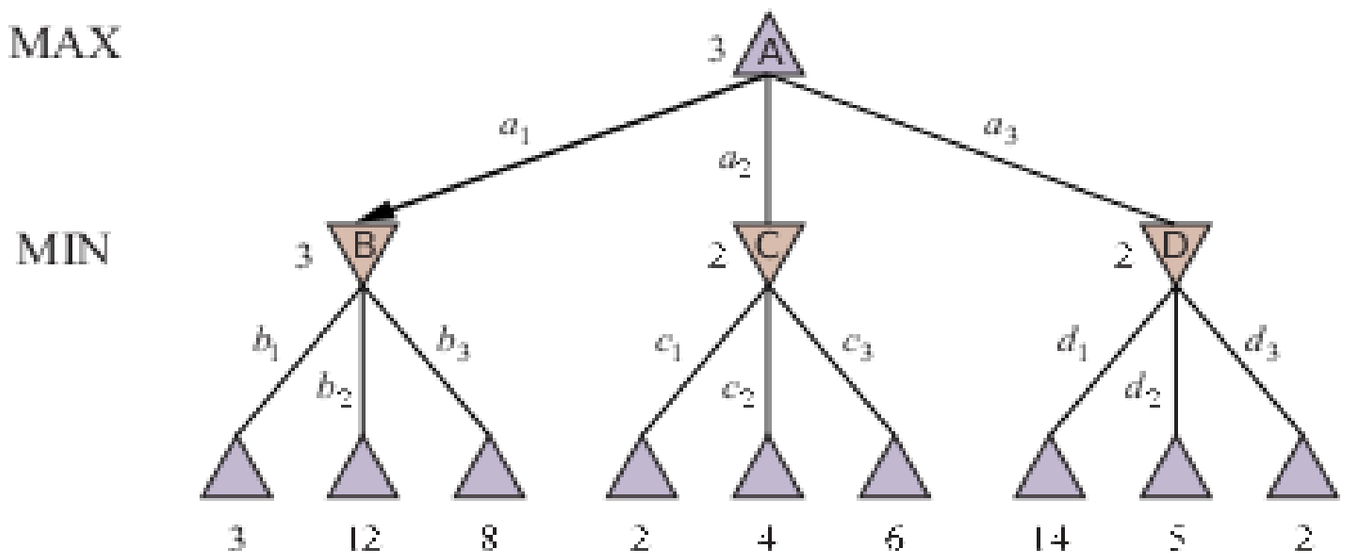

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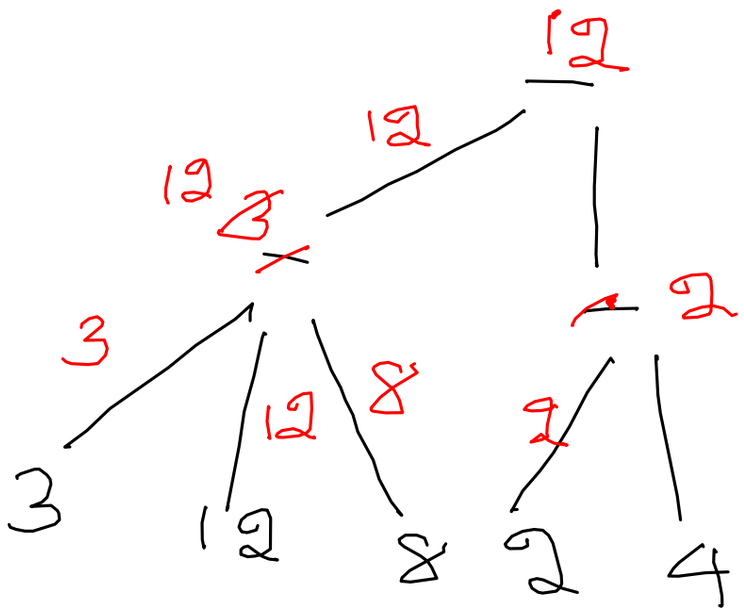

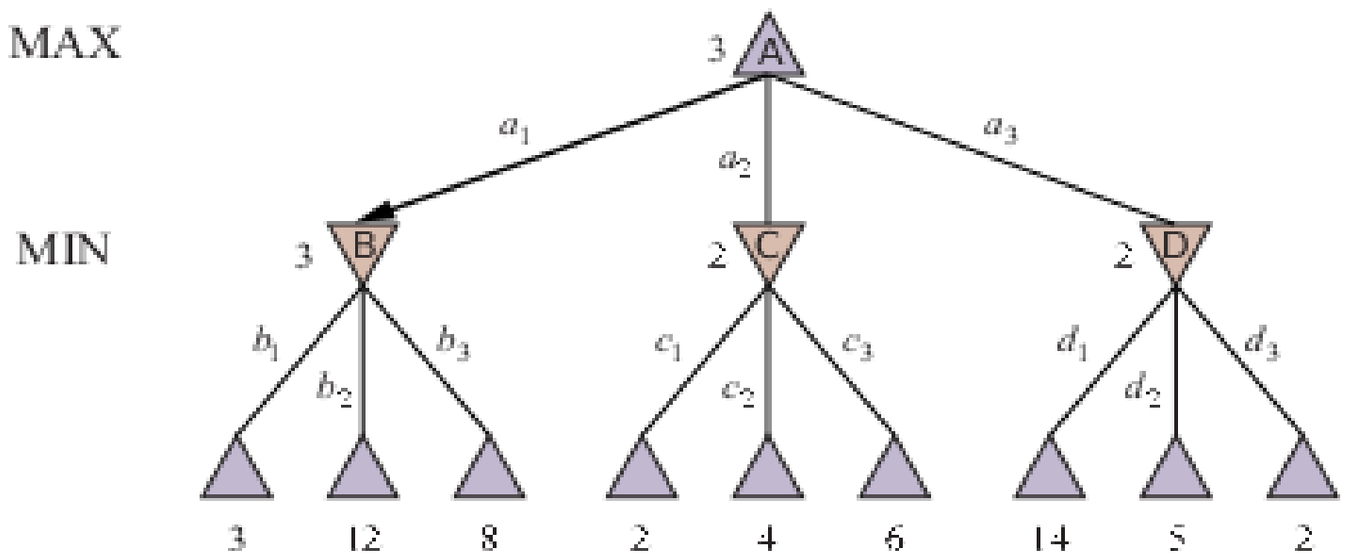

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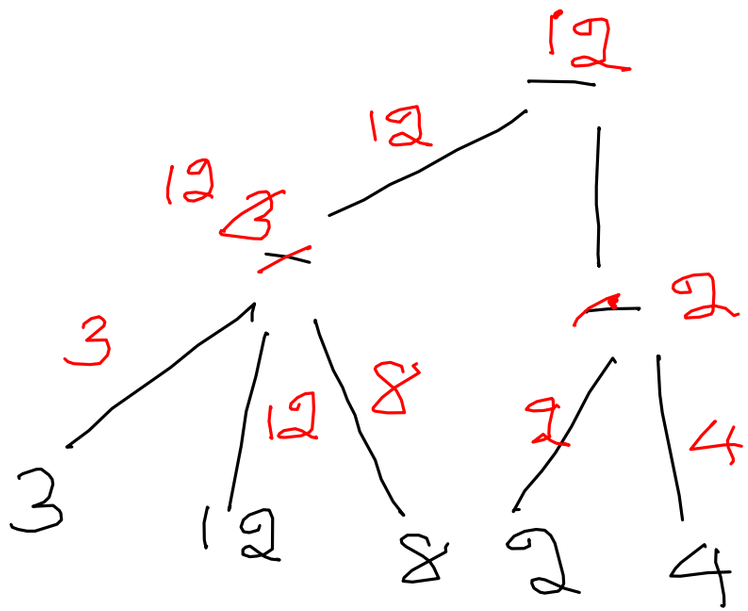

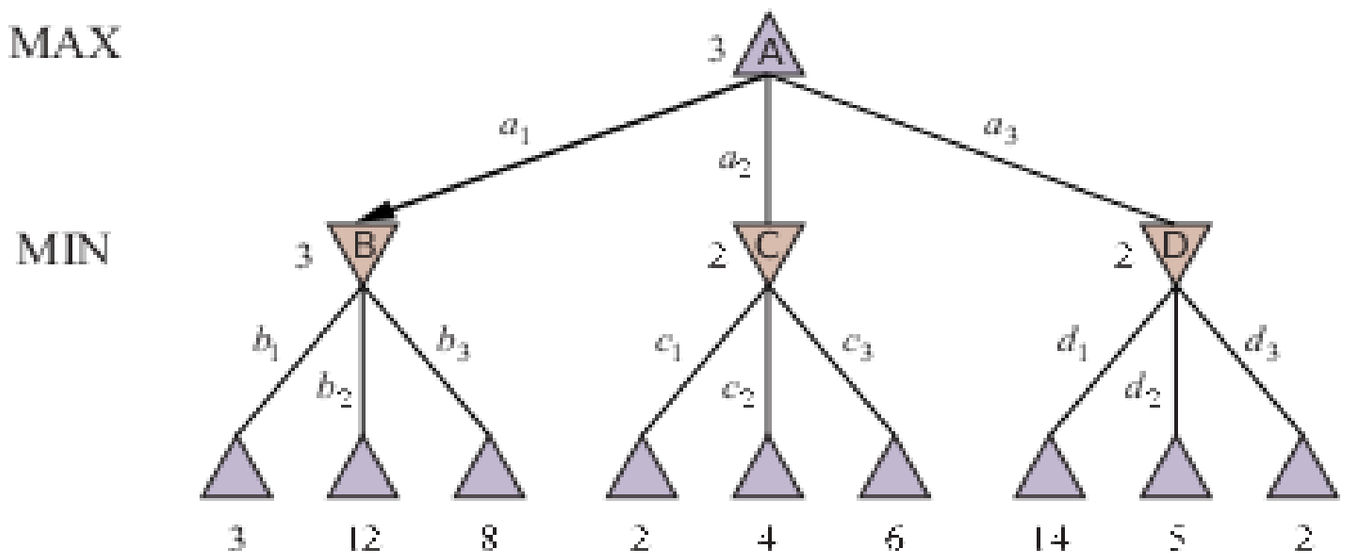

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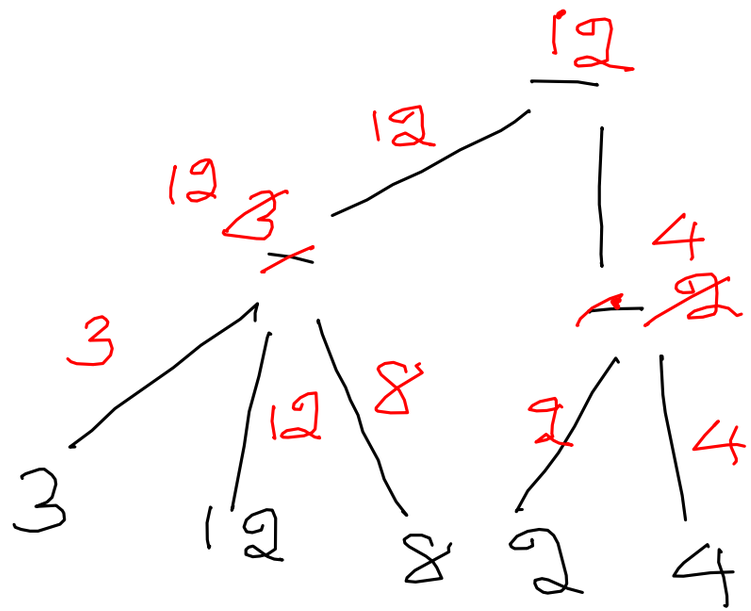

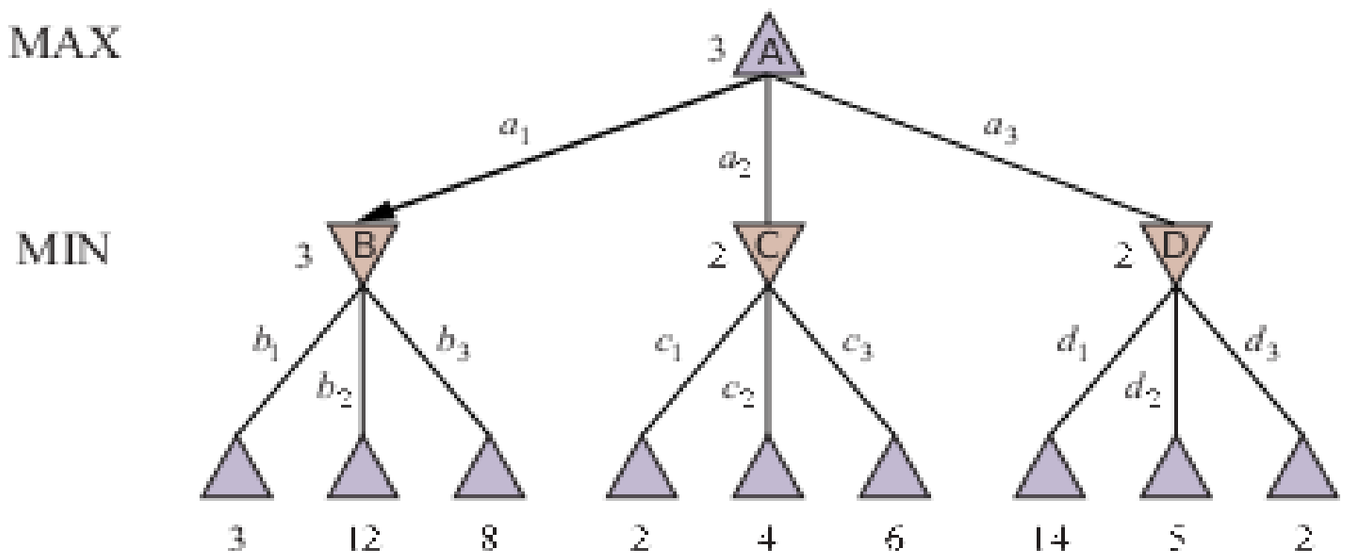

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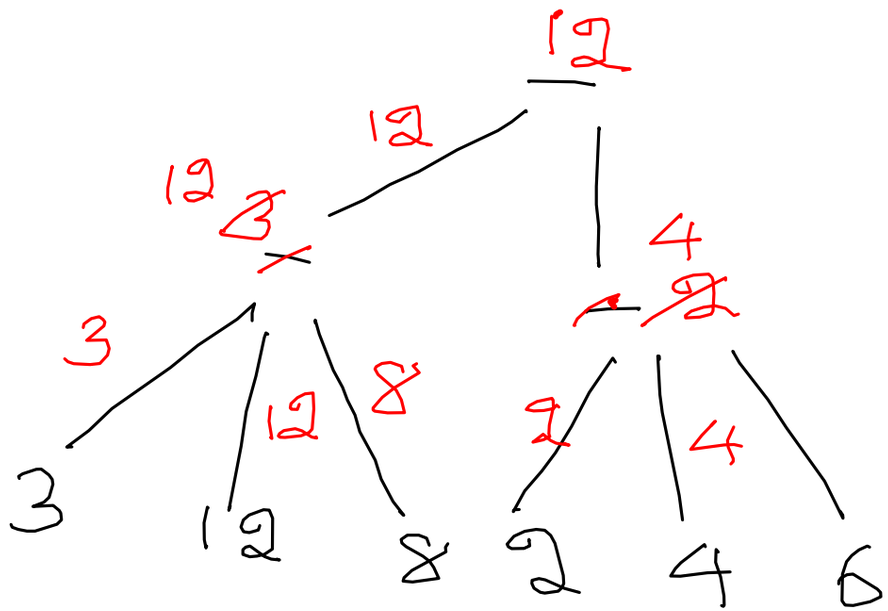

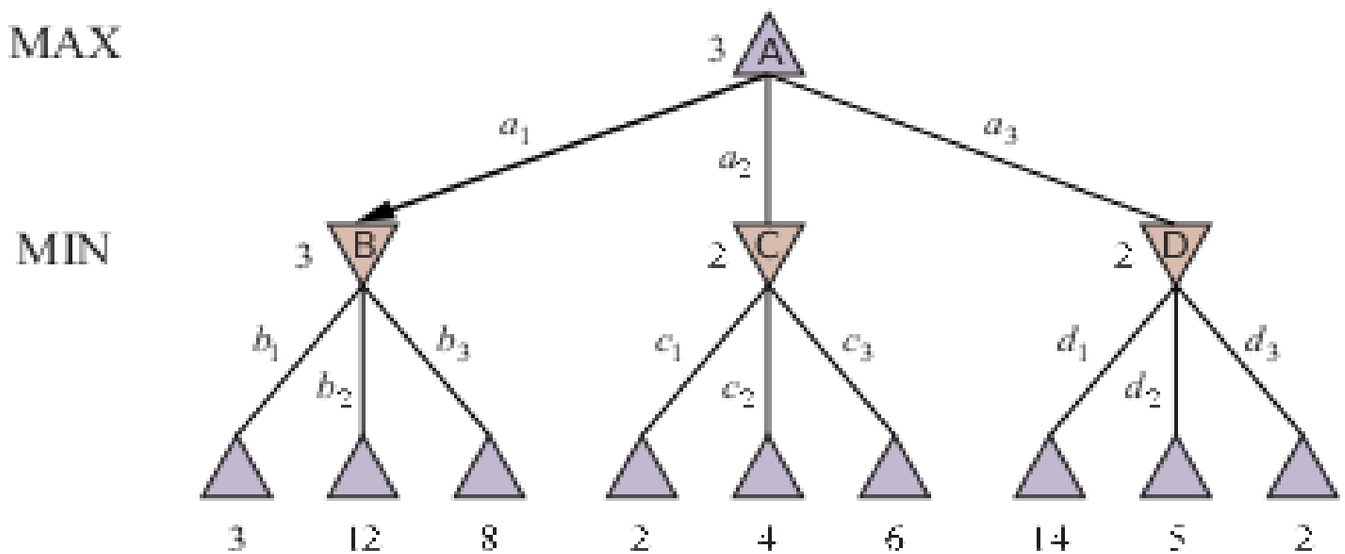

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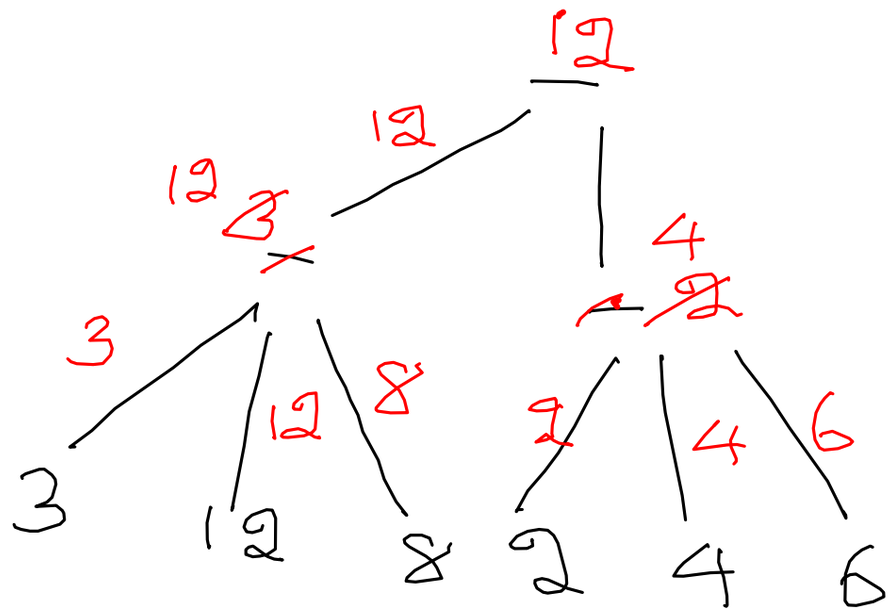

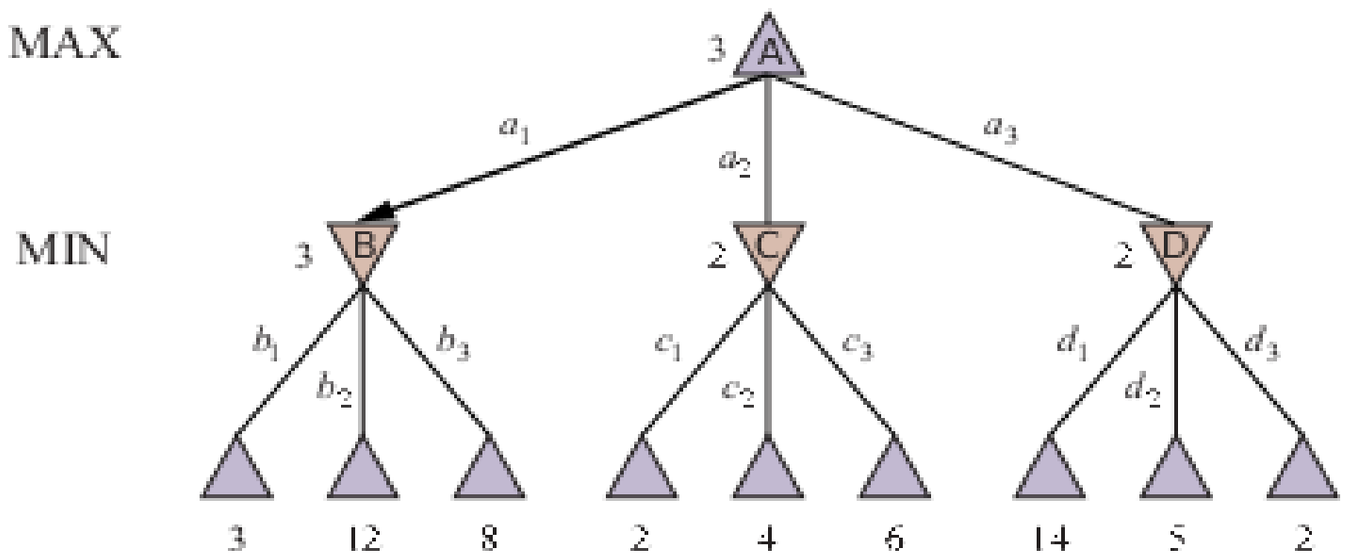

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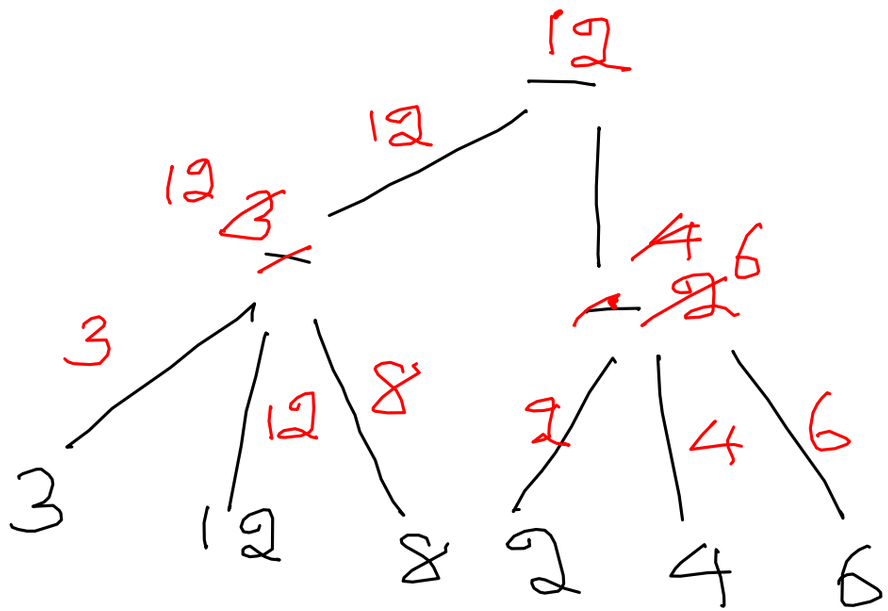

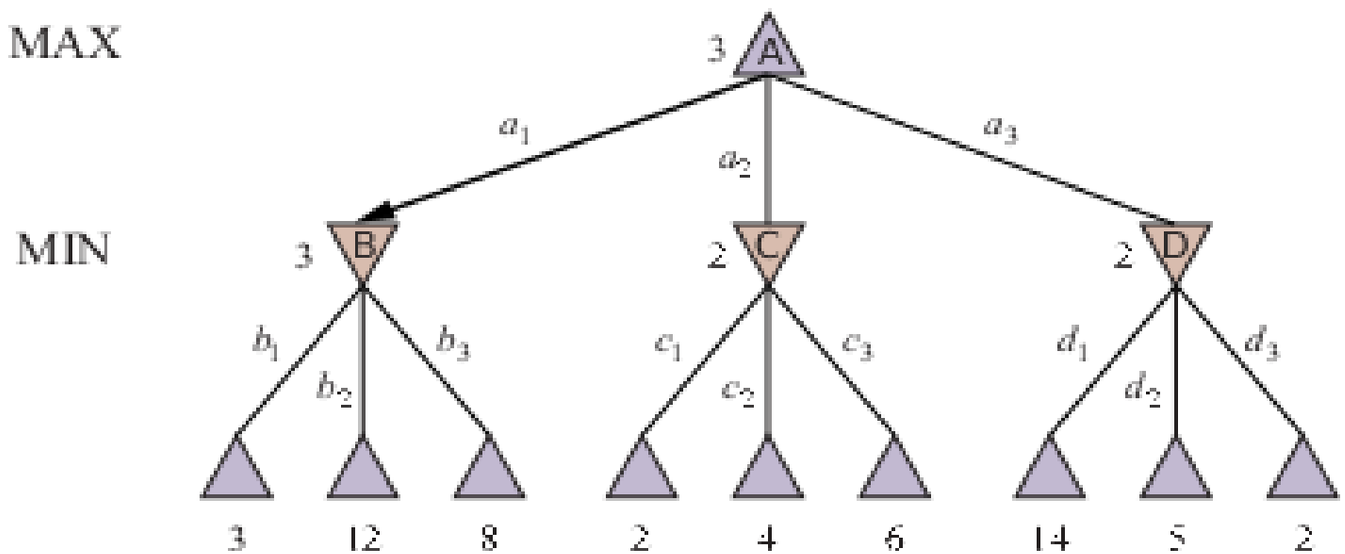

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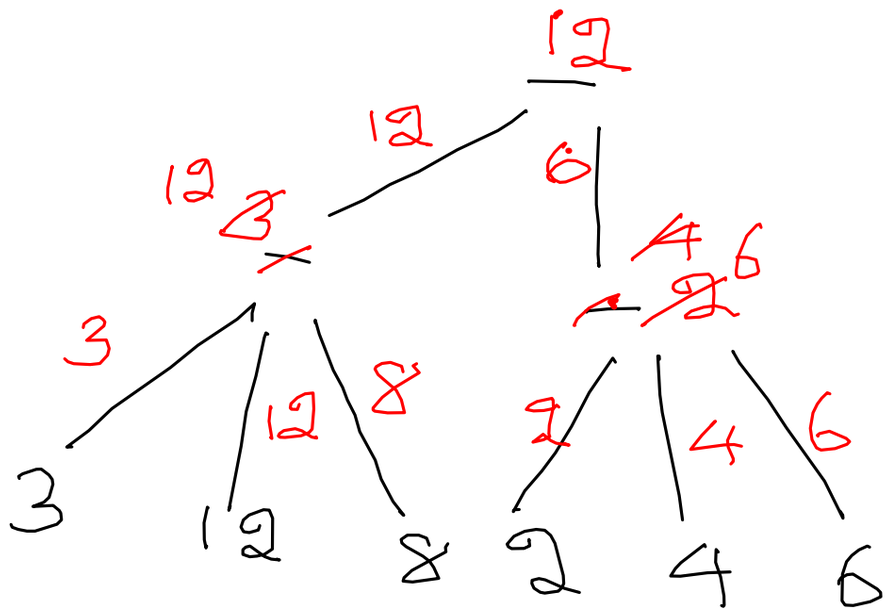

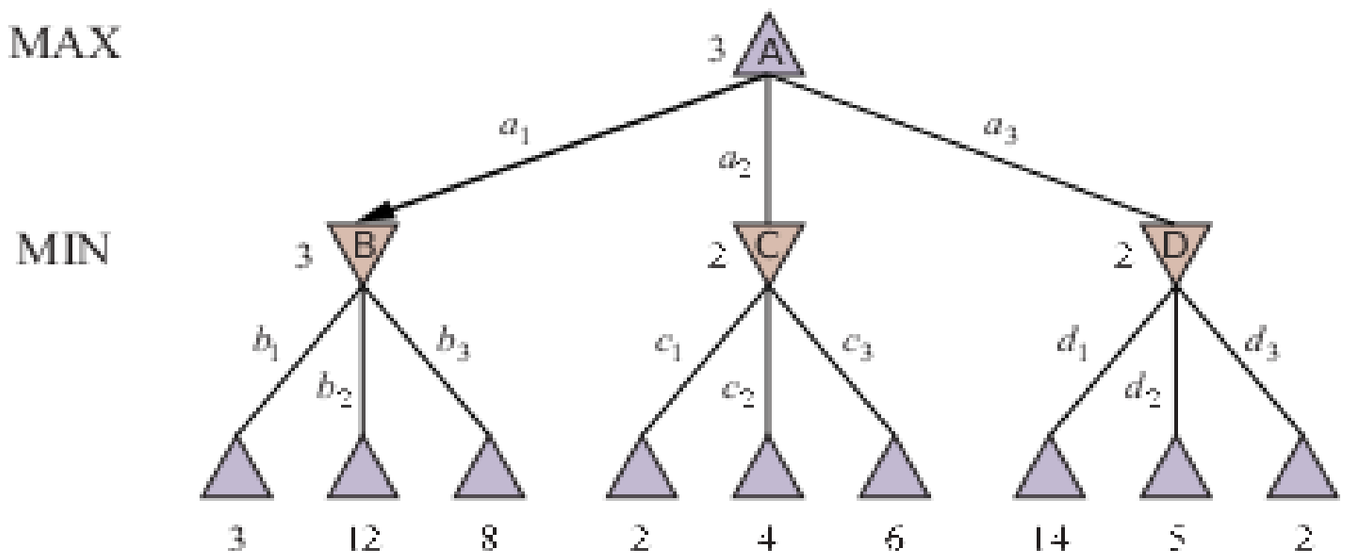

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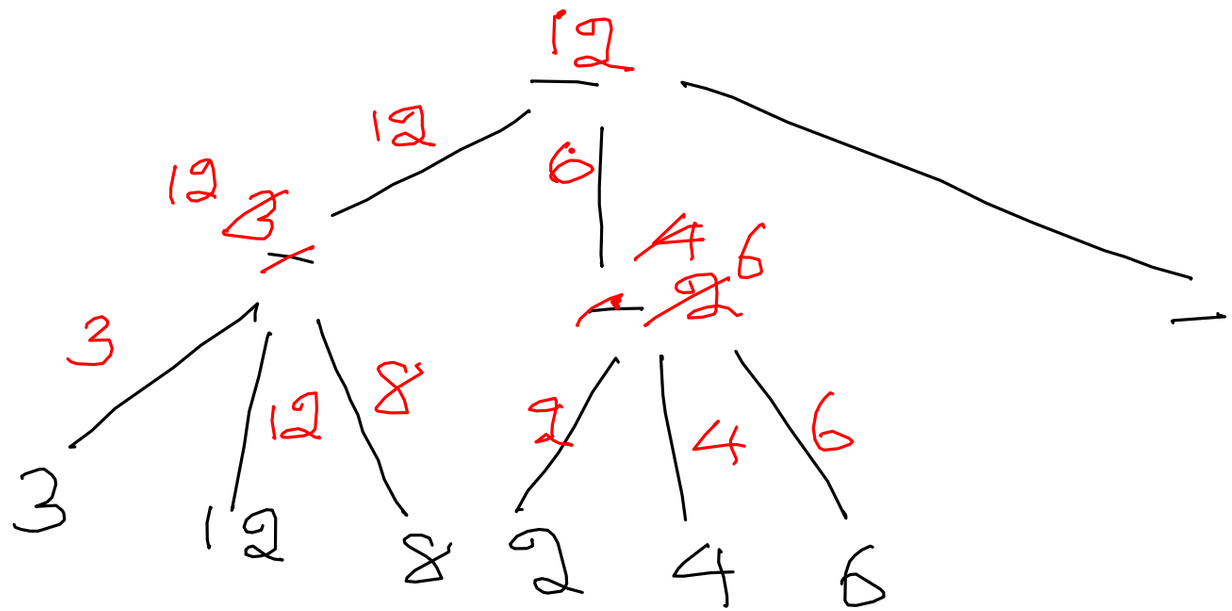
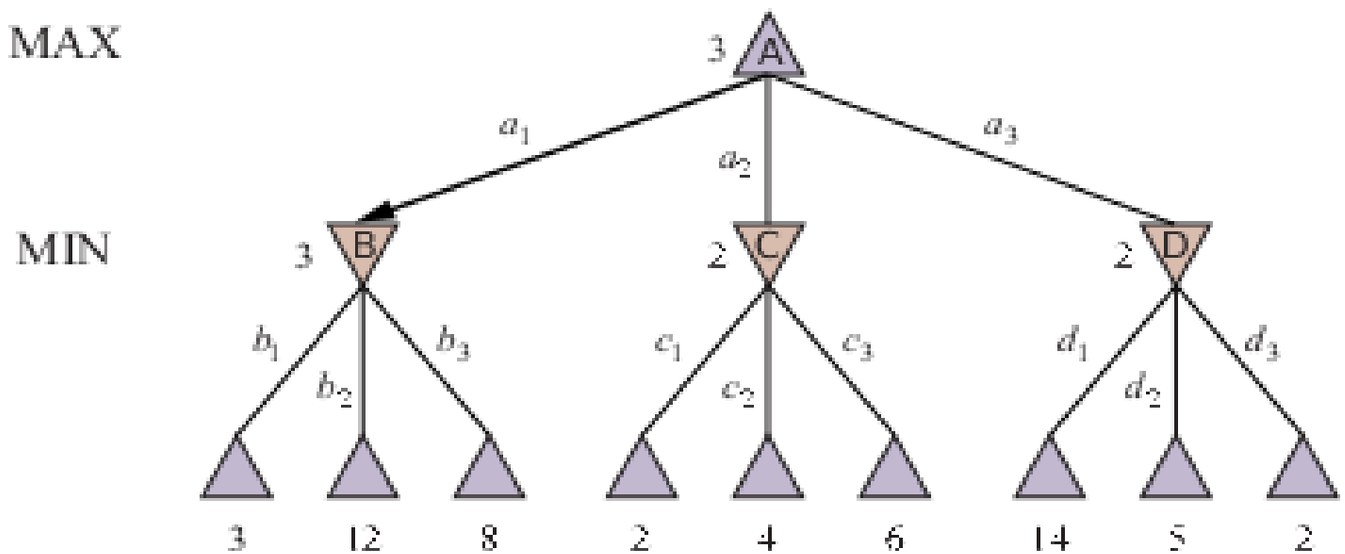
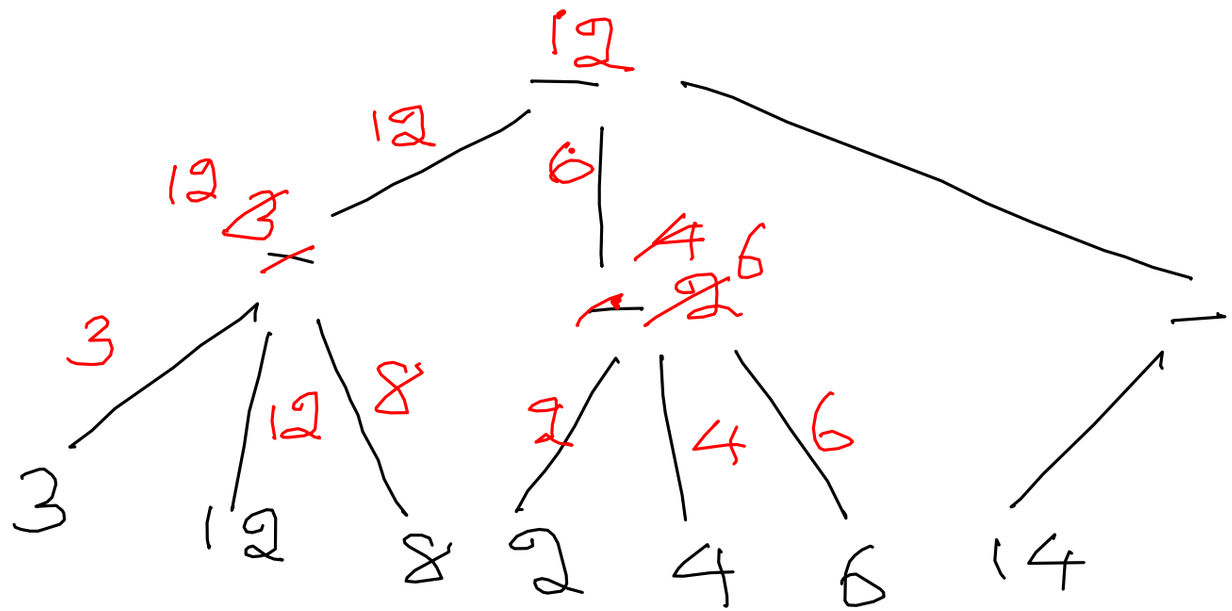
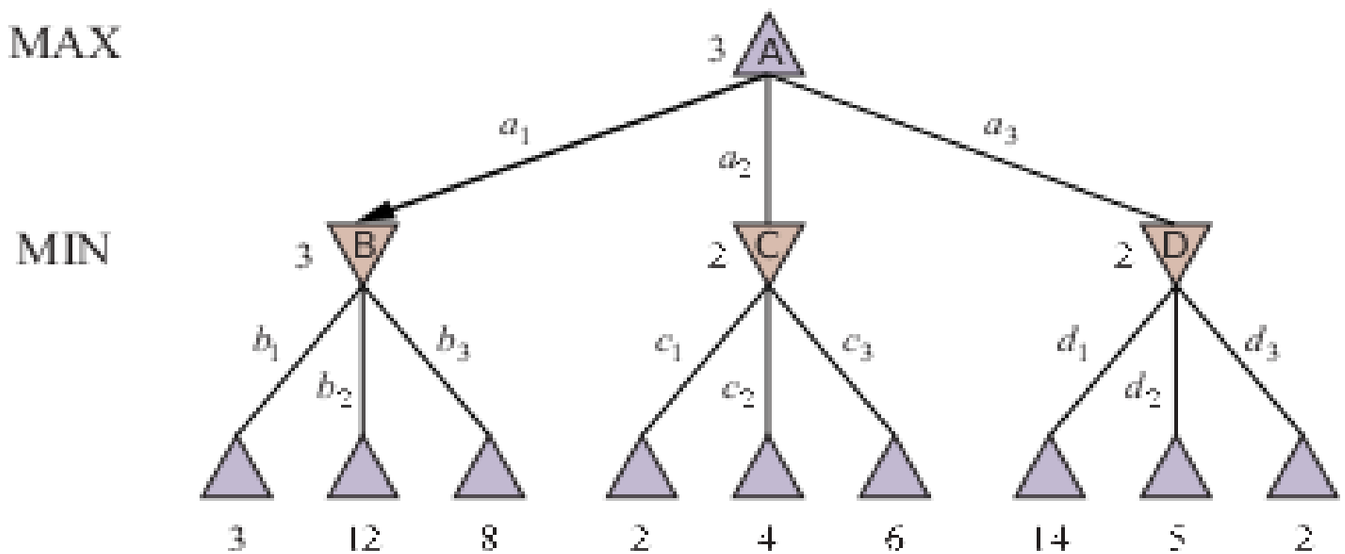


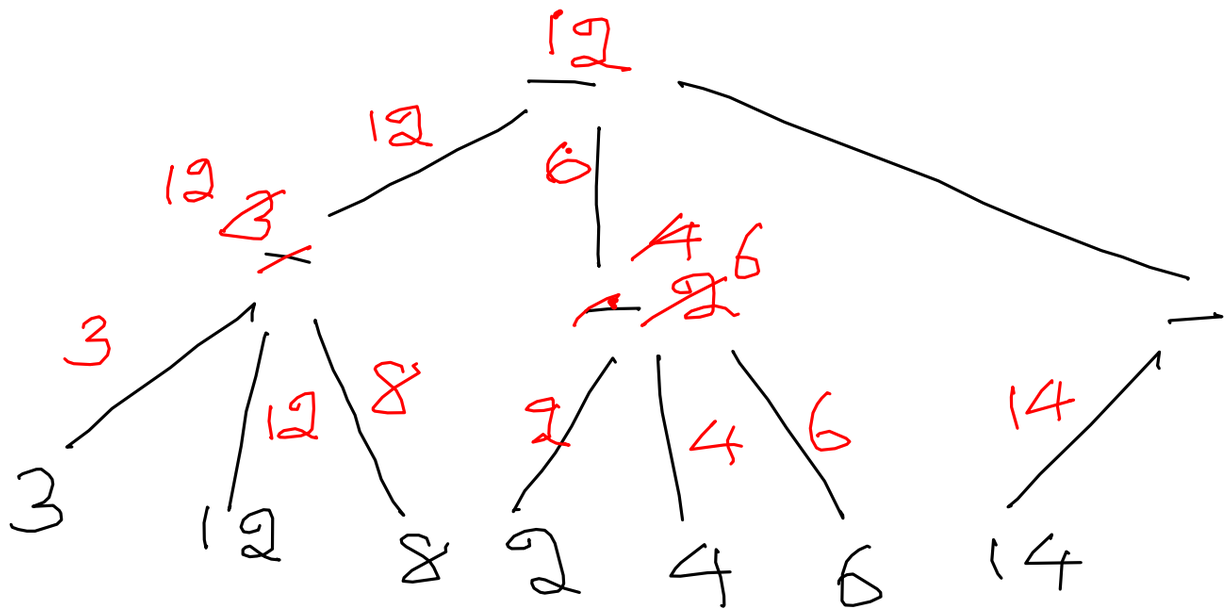


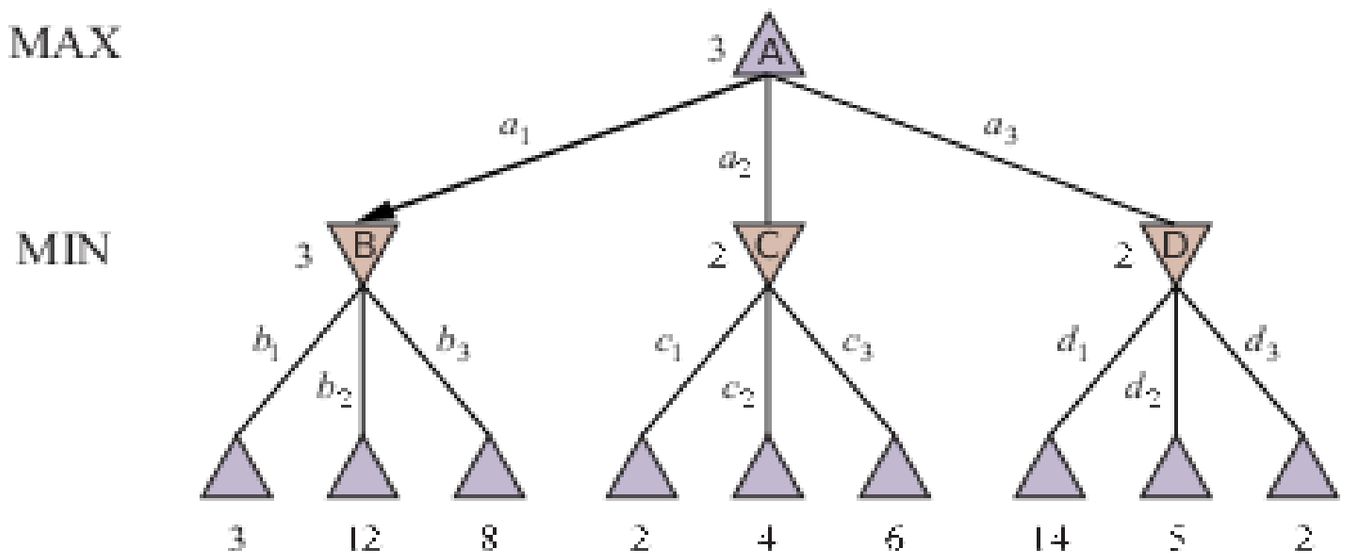
Maxvalue Example Trace



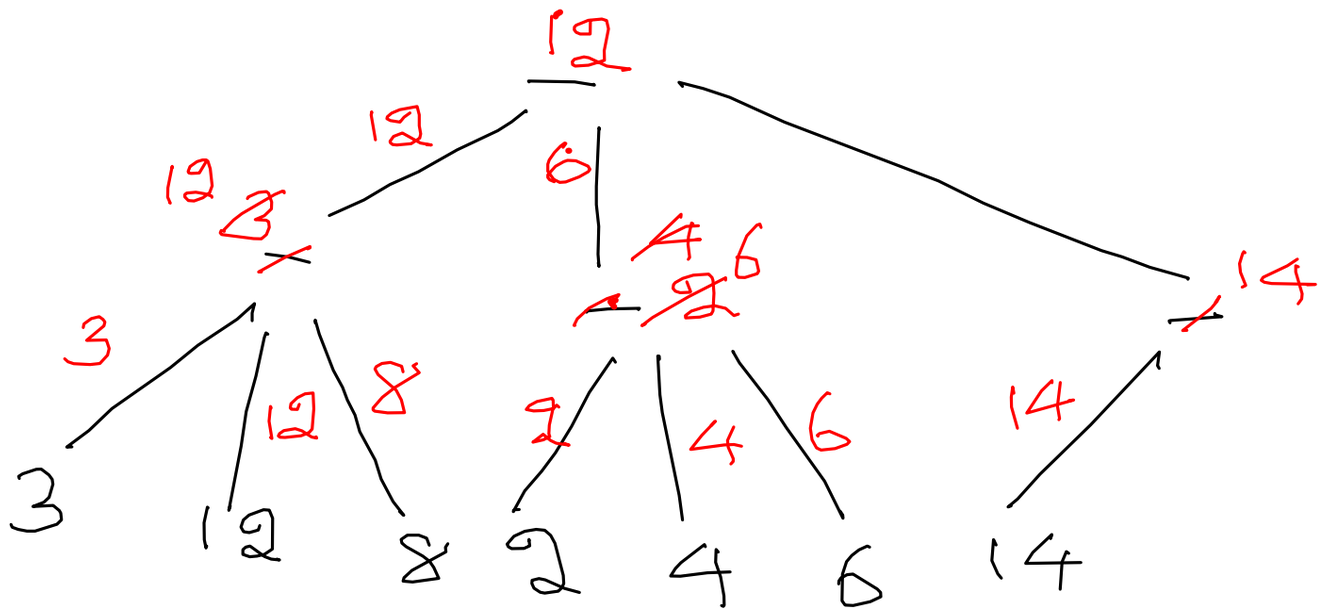


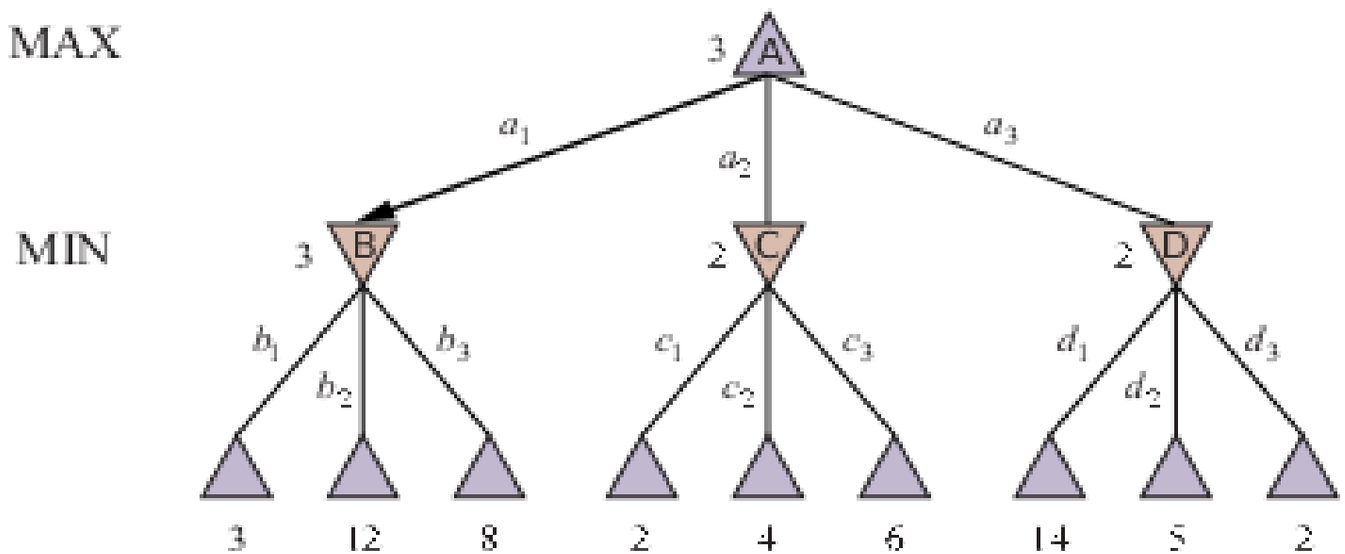
Maxvalue Example Trace



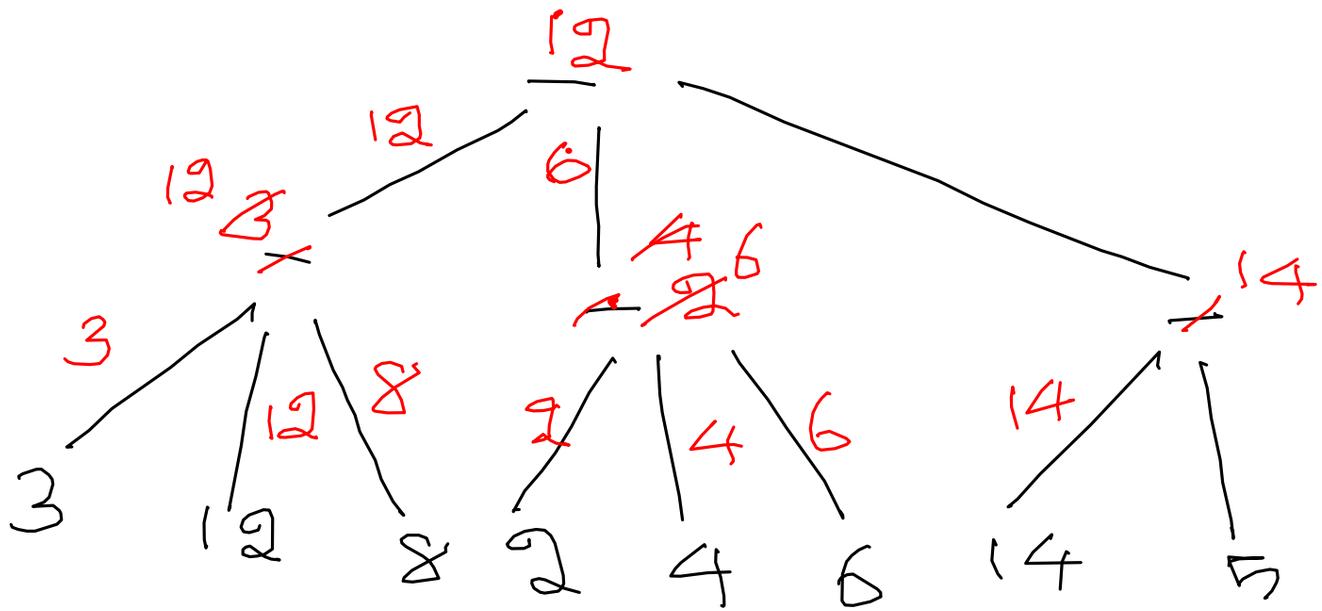


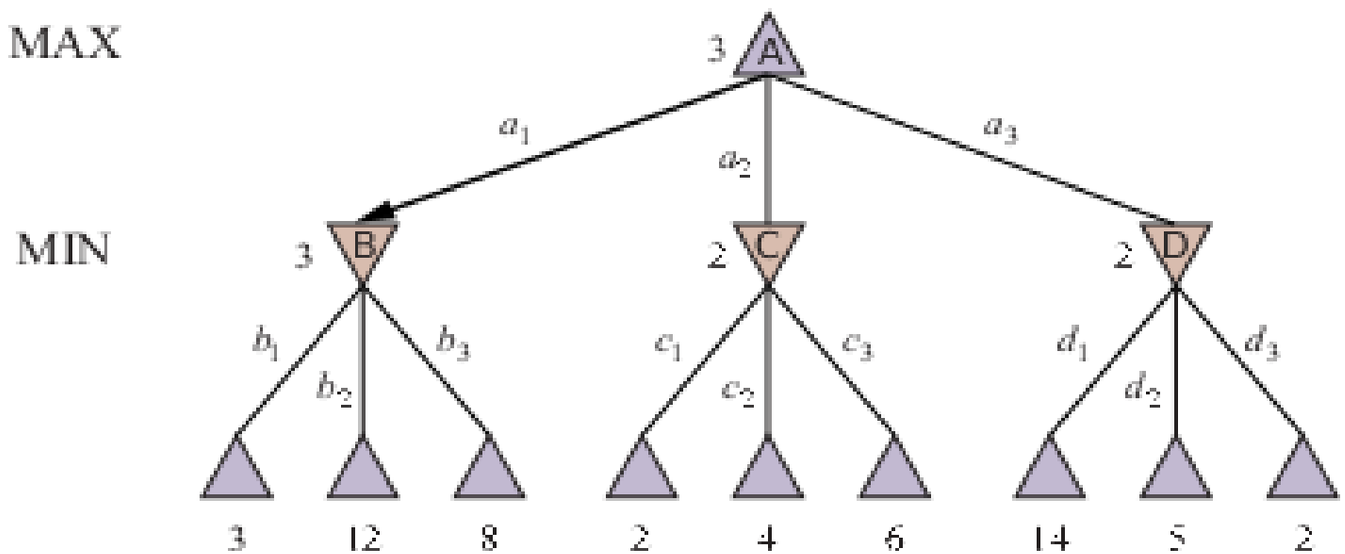
Maxvalue Example Trace





Maxvalue Example Trace





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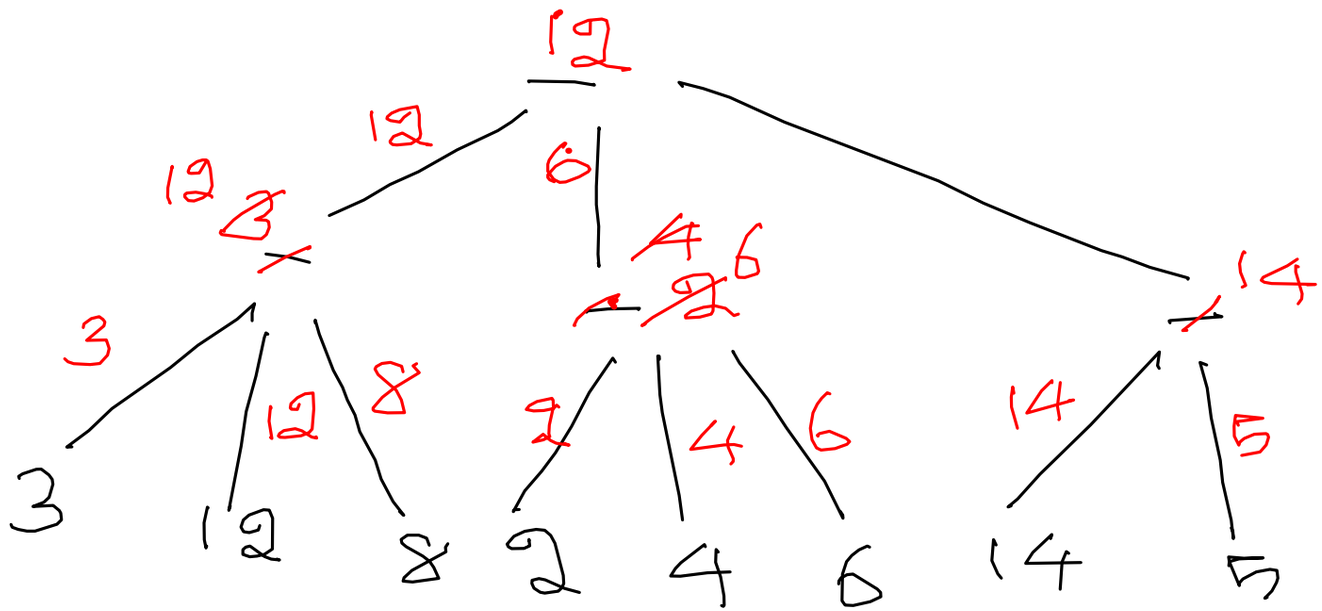
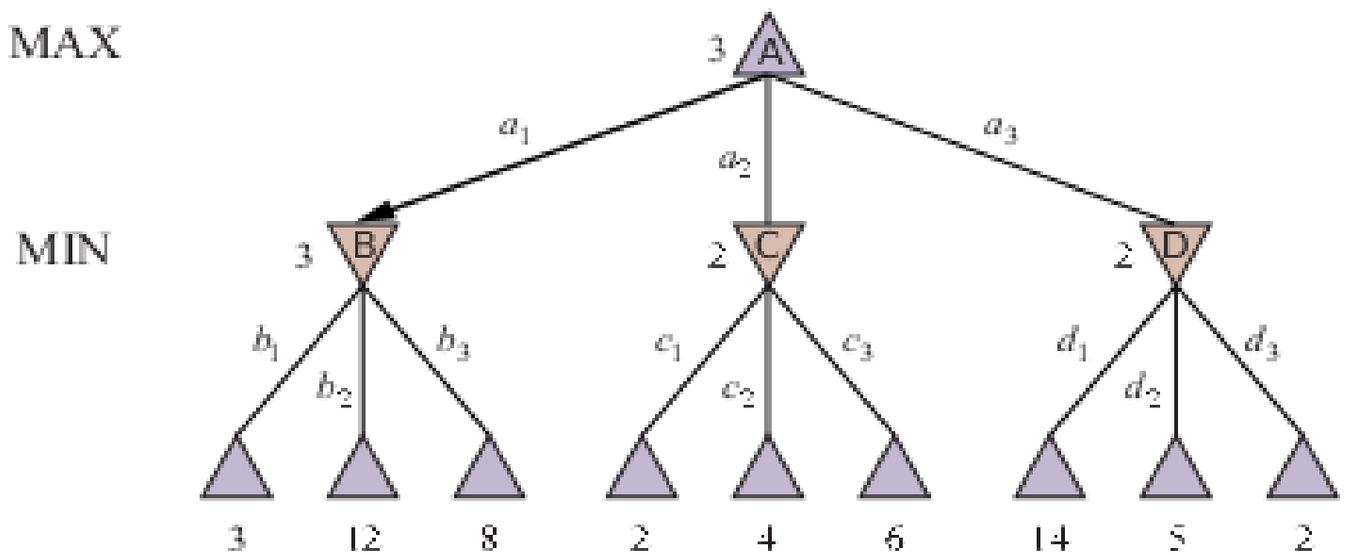
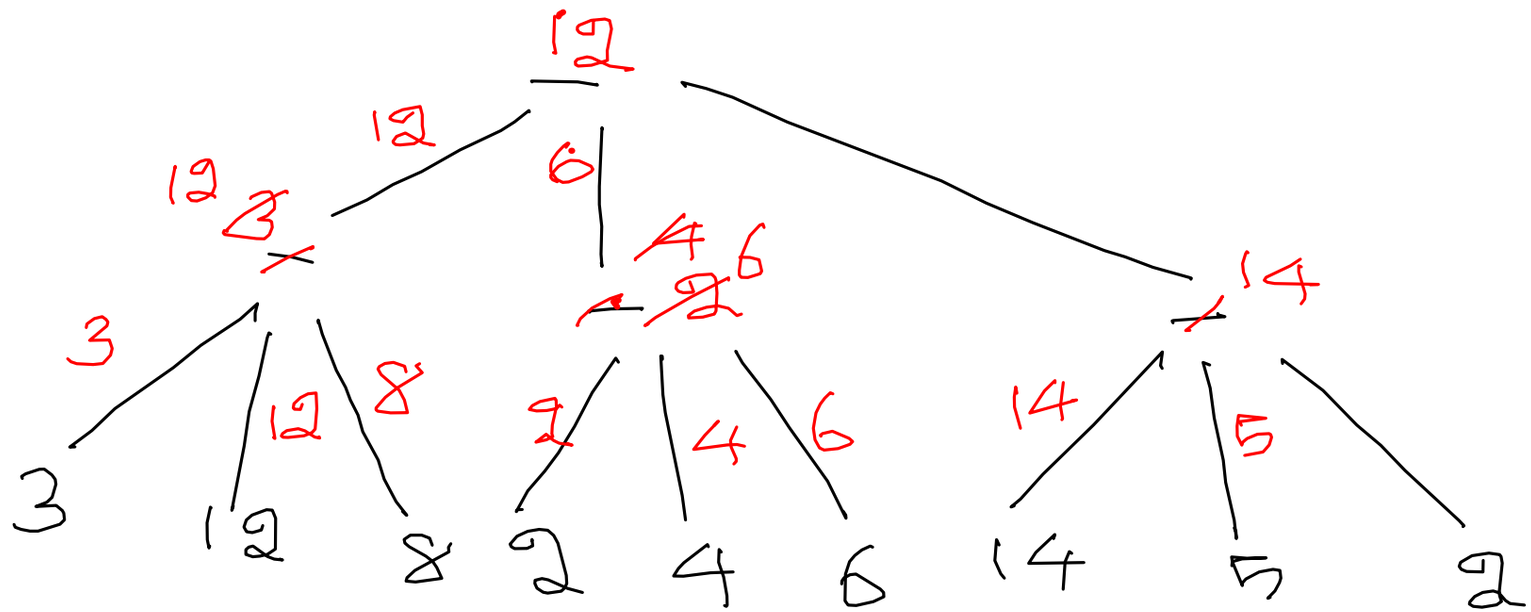
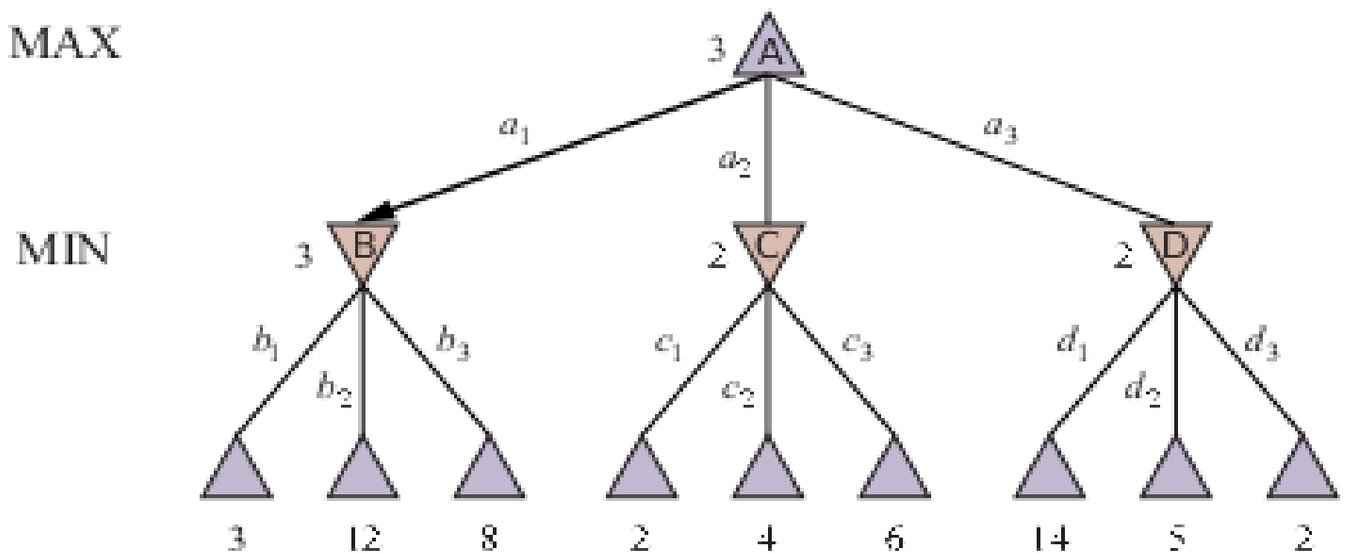




Maxvalue Example Trace





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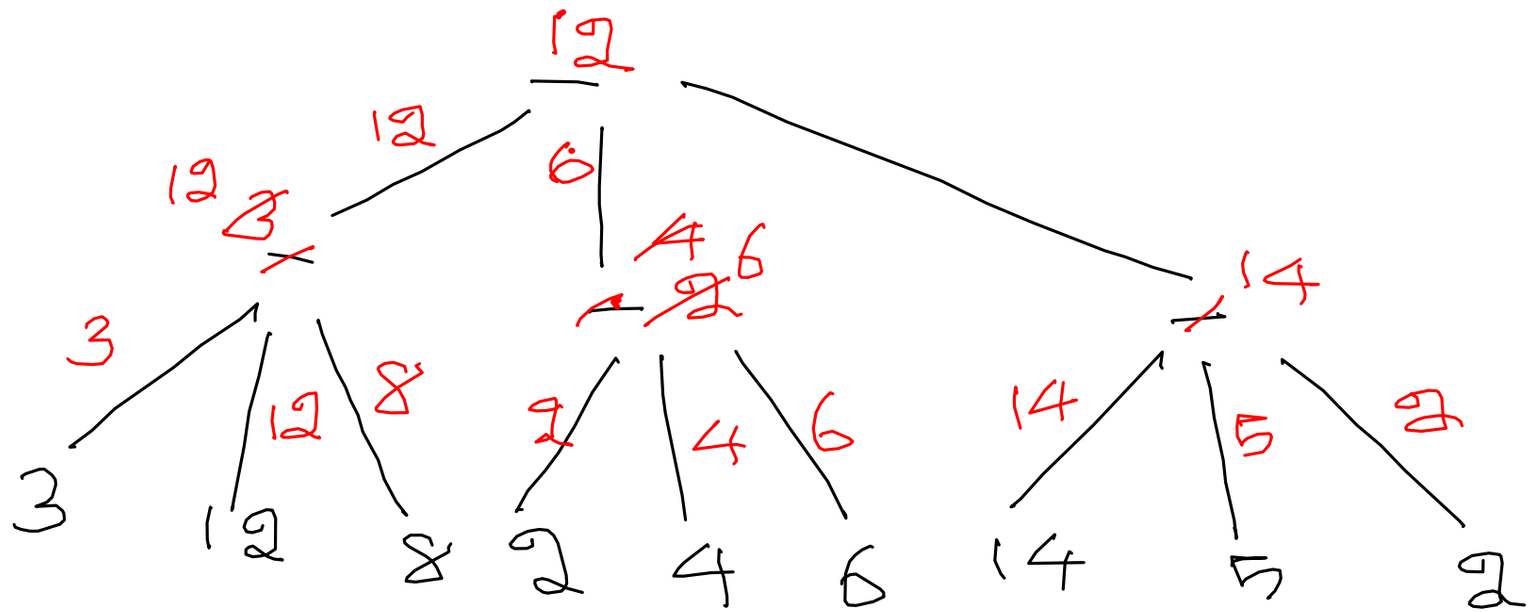

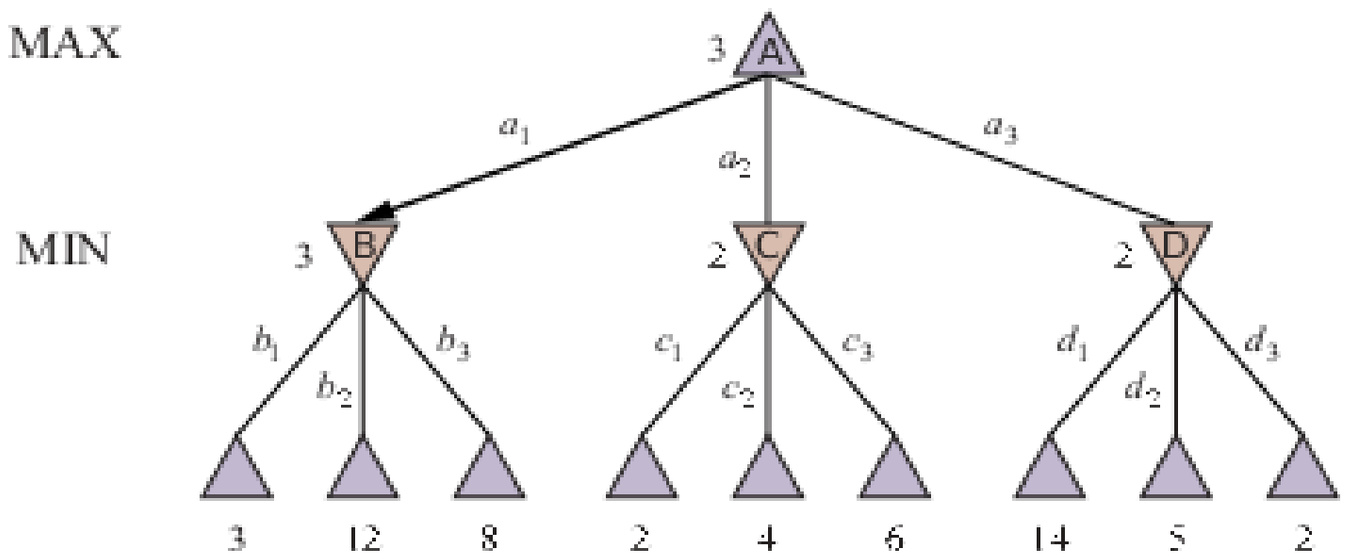

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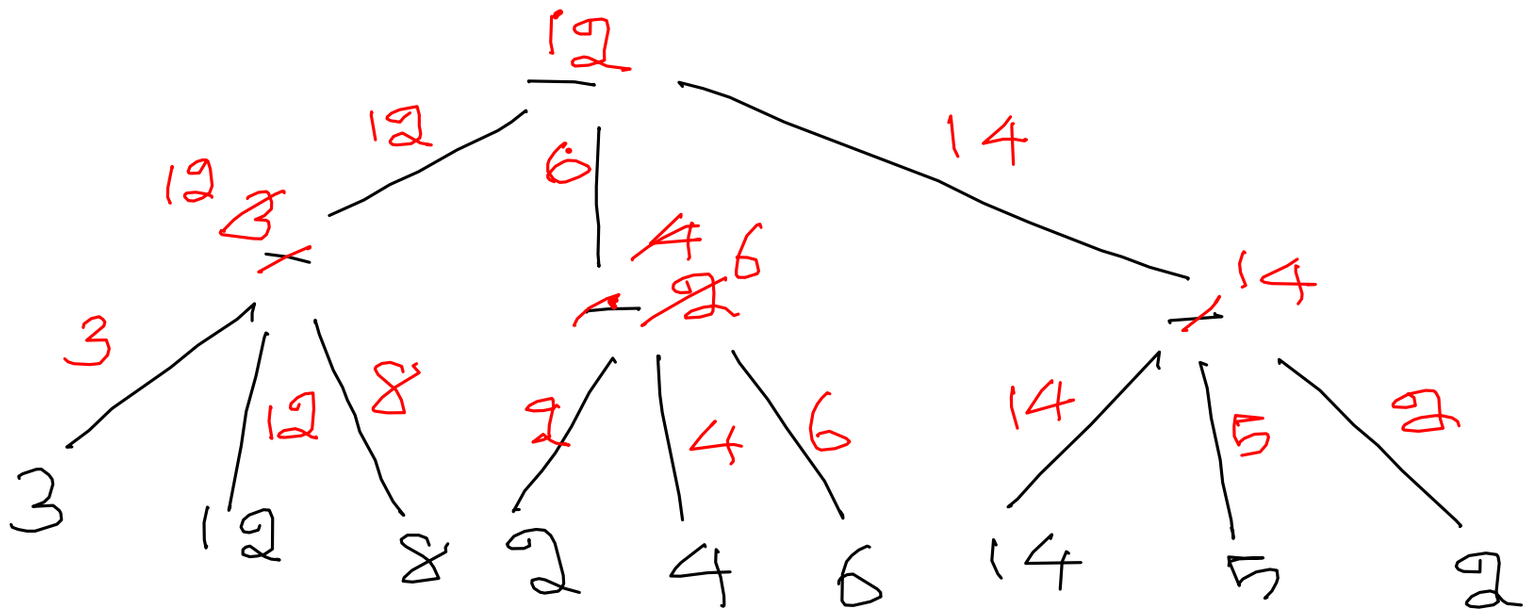
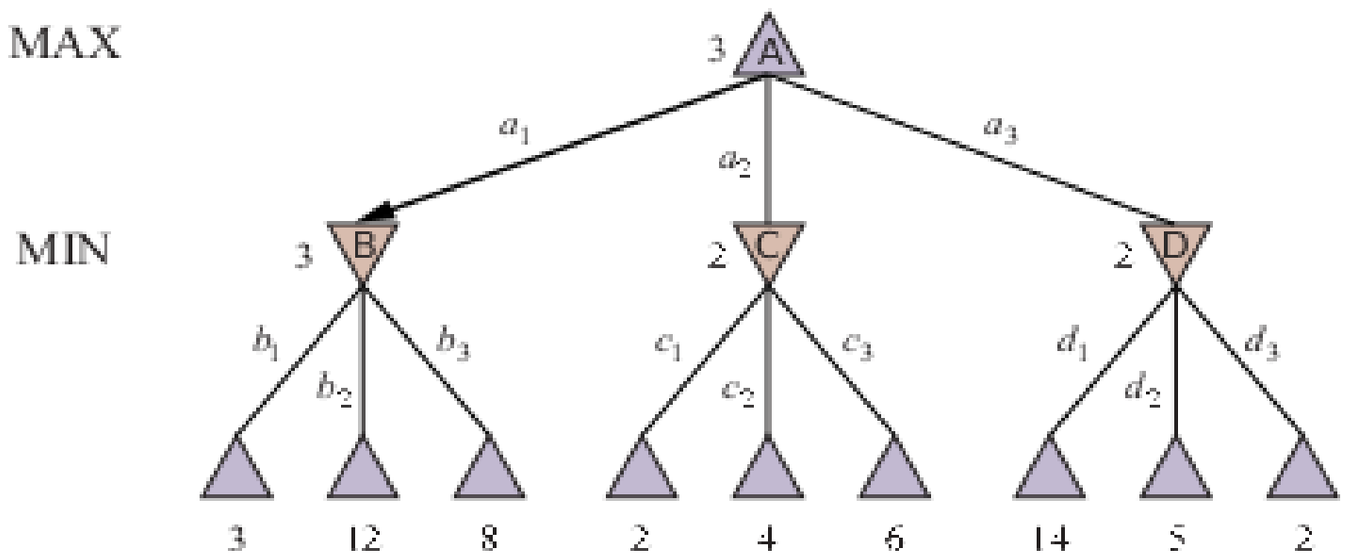
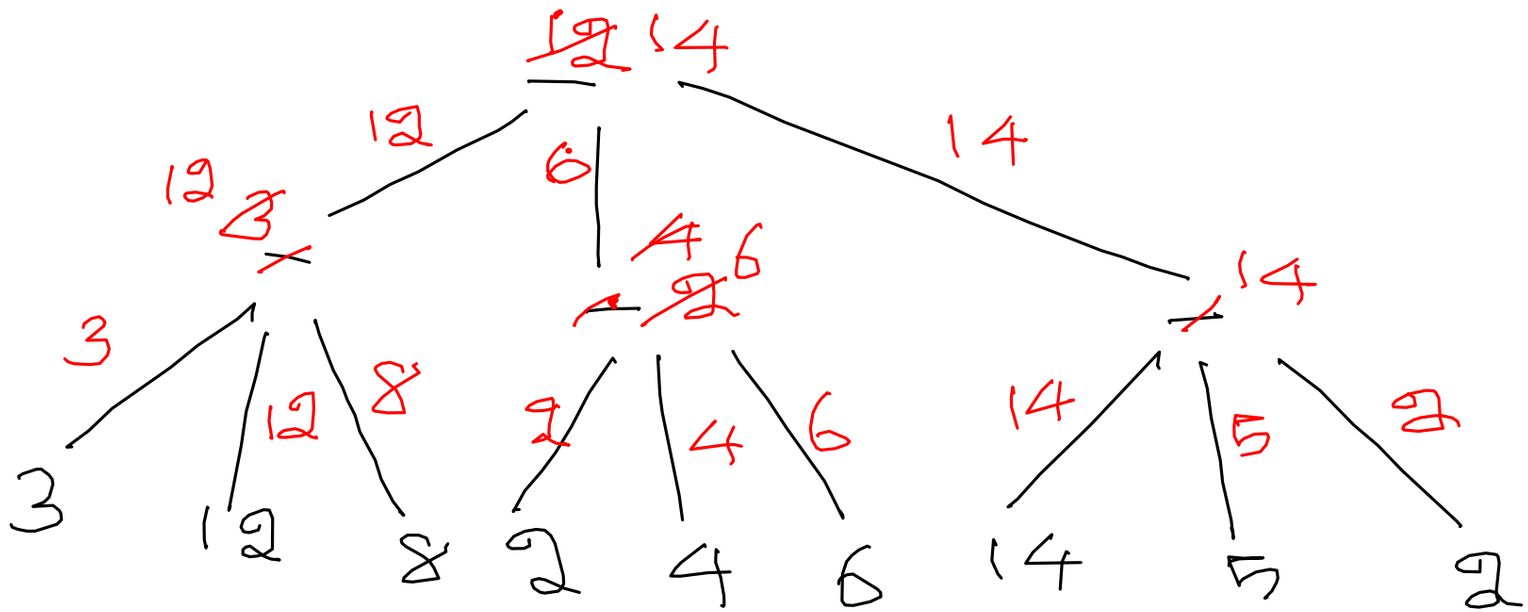
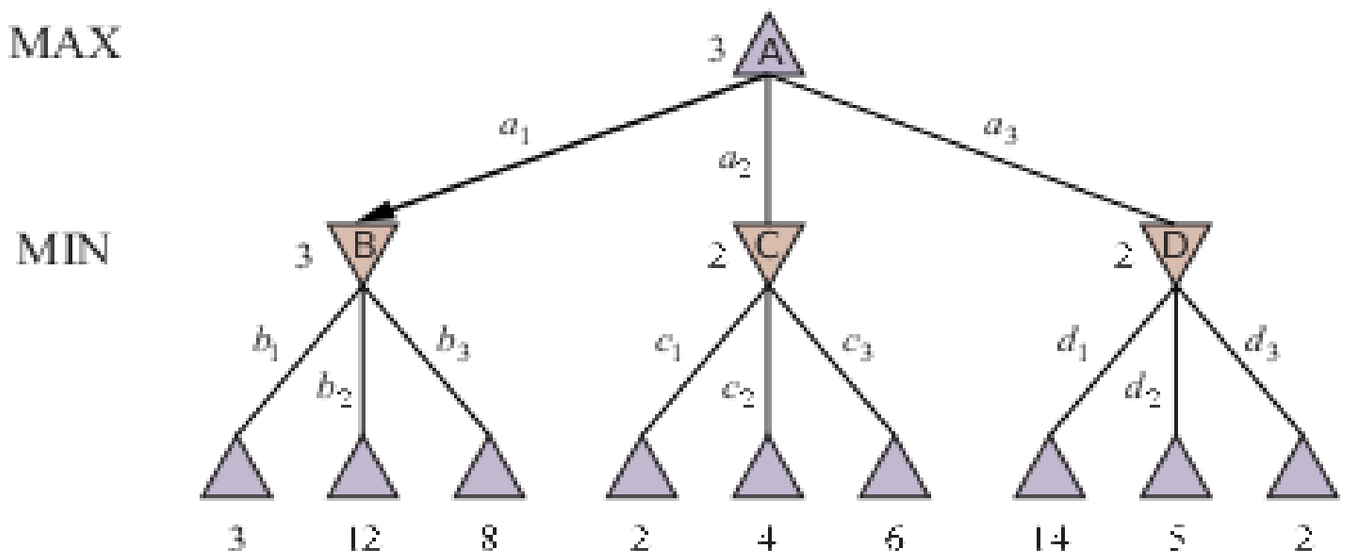


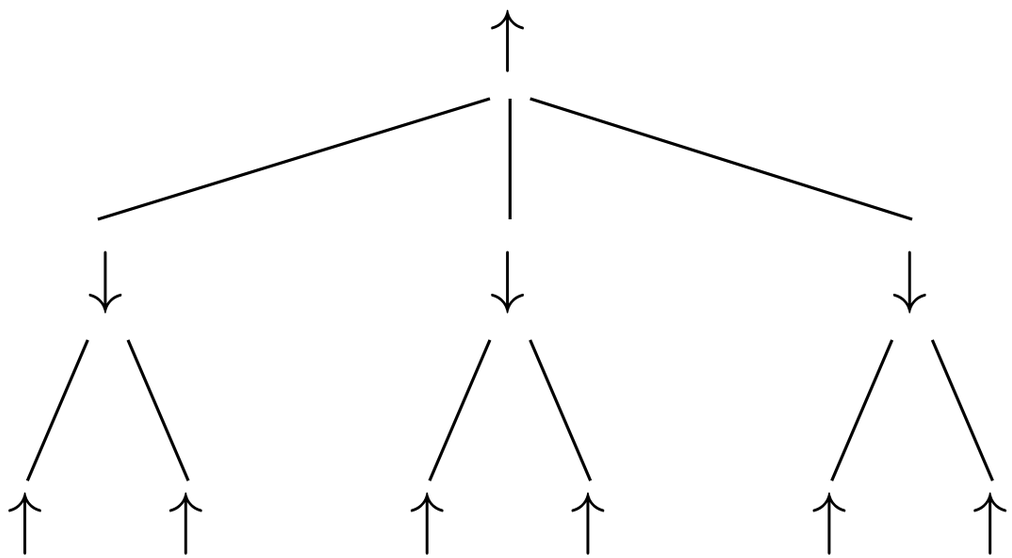


Maxvalue Example Trace





Minimax Value of States



MAX player ↑'s move in this state
$\text{Value}(n) = \max_{c \in \text{Successors}(n)} \text{Value(c)}$

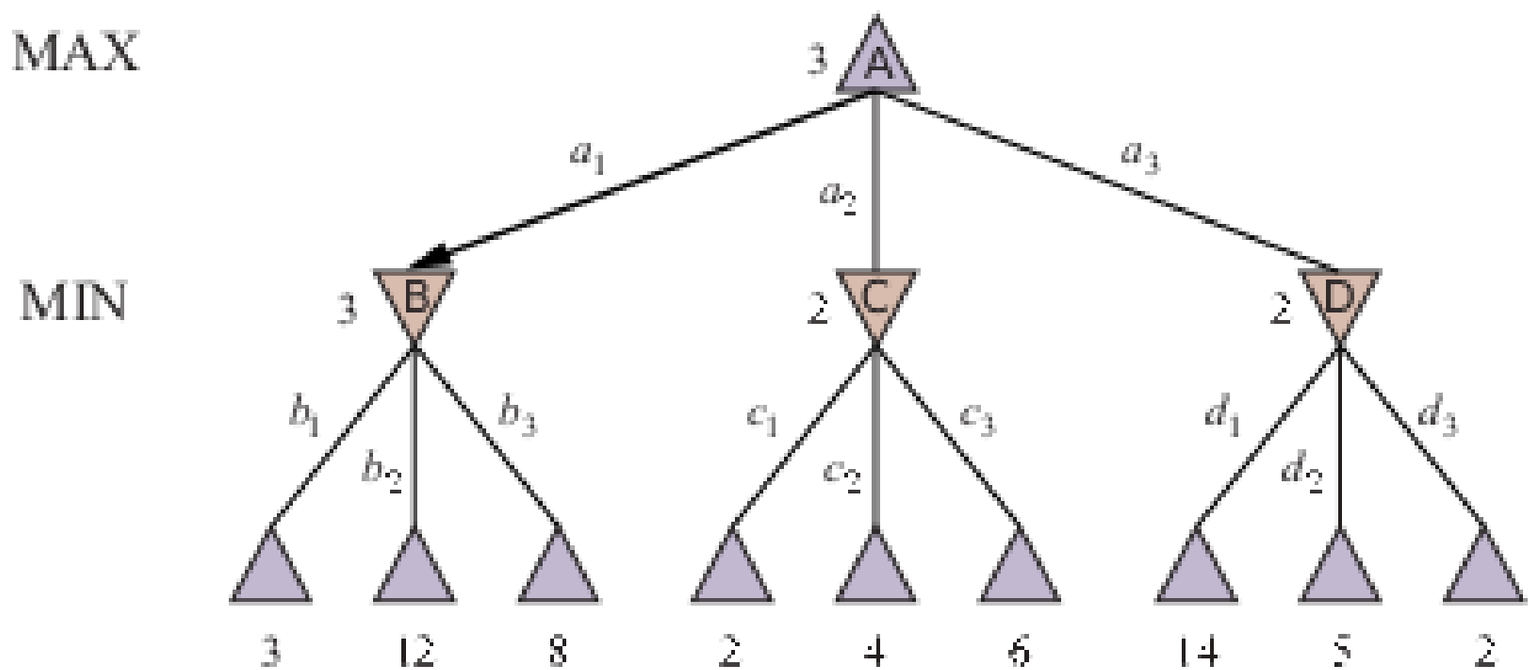
MIN player ↓'s move in this state
$\text{Value}(n) = \min_{c \in \text{Successors}(n)} \text{Value(c)}$

Terminal state
$\text{Value}(n) = \text{Utility}(n)$

Minimax Value and Minimax Action in a Node

- Utility of being in that state
- Assuming both the players play optimally from there to the end of the game

$$\text{Minimax}(n) = \begin{cases} \text{Utility}(n) & \text{if Leaf?}(n) \\ \max_{c \in \text{Successors}(n)} \text{Minimax(c)} & n \text{ is a MAX node} \\ \min_{c \in \text{Successors}(n)} \text{Minimax(c)} & n \text{ is a MIN node} \end{cases}$$



Two-ply Game Tree

- A two-ply game tree.
  - One move deep
  - One move = two half-moves (half-move = ply)
- "MAX nodes" – MAX's turn to move
- "MIN nodes" – MIN's turn to move
- Terminal nodes show the utility values for MAX
- Other nodes labeled with their minimax values
- MAX's best move at the root is a_1
  - leads to the state with the highest minimax value
- MIN's best reply move is b_1
  - leads to the state with the lowest minimax value.

Minimax Idea

- Minimax value of a state from the minimax values of its successors, recursively
- Base case: minimax values of terminal states (leaves)
- As the recursion unwinds, the minimax value are backed up from leaves to the root

Algorithm: Minimax(n)

Input: n a node in a game tree
Output: A pair, (value for the agent in node n, path that gives this value)

```
if n is a leaf node then  return evaluate(n)
else if n is a MAX node then
    max_val ← −∞
    foreach child c of n do
        val ← Minimax(c)
        if max_val < val  then
            max_val ← val
    return max_val
else
    min_val ← ∞
    foreach child c of n do
        val ← Minimax(c)
        if val < min_val then
            min_val ← val
    return min_val
```

Algorithm: Minimax(n)

Input: n a node in a game tree
Output: A pair, (value for the agent in node n, path that gives this value)

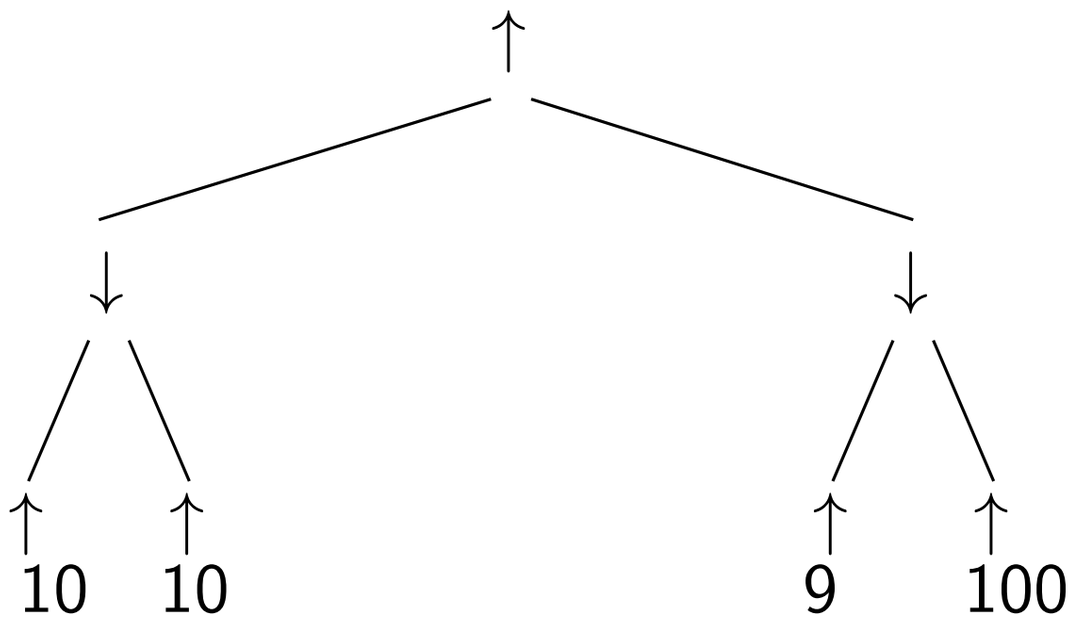
```
if n is a leaf node then  return evaluate(n), [ ]
else if n is a MAX node then
    max_val ← −∞
    foreach child c of n do
        val, path ← Minimax(c)
        if max_val < val  then
            max_val, max_path ← val, [c] + path
    return max_val, max_path
else
    min_val ← ∞
    foreach child c of n do
        val, path ← Minimax(c)
        if val < min_val  then
            min_val, min_path ← val, [c] + path
    return min_val, min_path
```

Minimax Algorithm Analysis

- DFS
- Height of the tree $= m$, Legal moves $= b$
- Time complexity $= O(b^m)$, impractical
- Space complexity $= O(bm)$ or $O(m)$

Minimax Performance

- Optimal play for MAX assumes MIN also plays optimally
- Maximize the worst-case outcome for MAX
- If MIN does not play optimally?
- There are better strategies against suboptimal opponents
  - Do worse against optimal opponents.



Resource Limits

- Problem: In realistic games, cannot search to leaves!
- Solution: Depth-limited search
  - Search only to a limited depth in the tree
  - Replace terminal utilities with an evaluation function for non-terminal positions
- Guarantee of optimal play is gone
- More plies makes a BIG difference
- Use iterative deepening for an anytime algorithm

Evaluation Functions

- Evaluation functions are always imperfect
- Evaluation functions score non-terminals in depth-limited search
- Ideal function: returns the actual minimax value of the position
- In practice: typically weighted linear sum of features:

$$\text{eval}(n) = w_1 f_1(n) + w_2 f_2(n) + \ldots + w_k f_k(n)$$

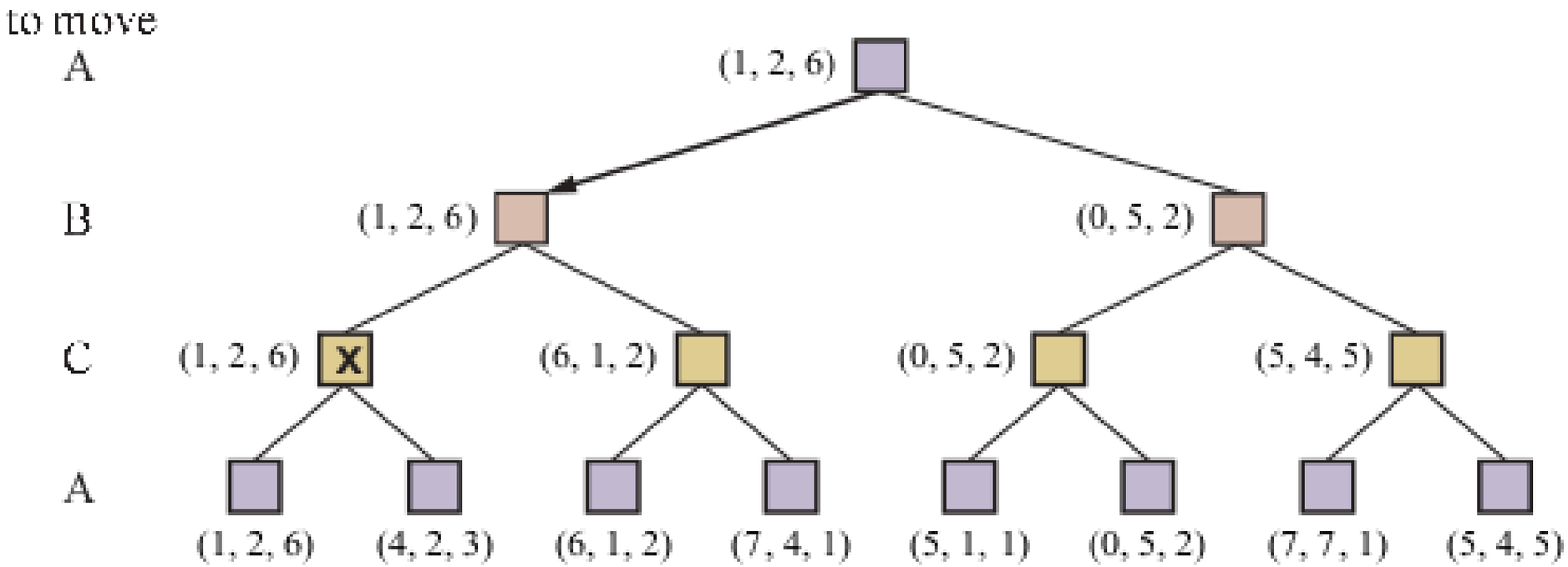
- e.g. $f_1(n) =$(num white queens $-$ num black queens), etc.

Optimal Decisions in Multiplayer Games

- Each node labeled with a vector of n values, from the viewpoint of n players

$$(v_A, v_B, v_C)$$

- Terminal nodes – `Utility` returns n - vector
- Backed-up value of node n = utility vector of the successor state with the highest value for the player choosing at n.



Minimax with Alpha-Beta Pruning

Alpha-Beta Pruning

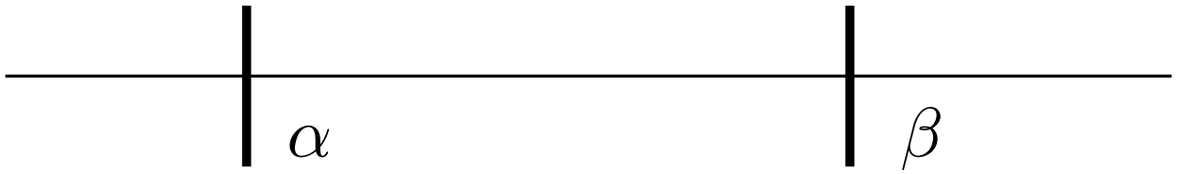
- Game tree: number of states exponential in the depth of the tree
- Is it possible to compute the correct minimax decision without looking at every node
- Pruning: eliminate subtrees from consideration
- Alpha-beta pruning: eliminate branches that cannot influence the final decision

α and β

- At any node, a subtree of the game tree has already been traversed.
- In the traversed subtree, the MAX player can achieve a maximum value of max_sofar (α) and there is a path to get it.
- The MIN player can achieve a minimum value of min_sofar (β) and there is a path to get it.
- As more of the game tree is traversed, max_sofar (α) can increase and min_sofar (β) can decrease.
- It is impossible that max_sofar (α) for the MAX player is greater than the min_sofar (β) for the MIN player.

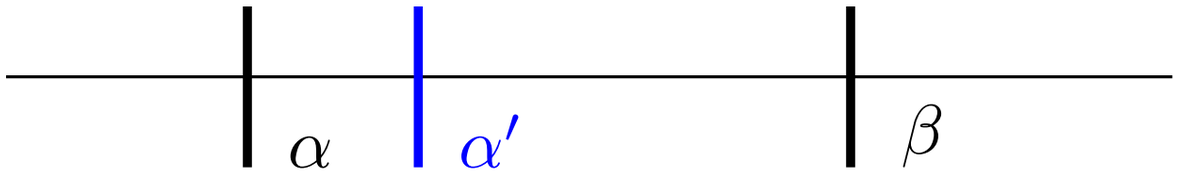
$$\text{max_sofar} \leq \text{min_sofar}$$

$$\alpha \leq \beta$$

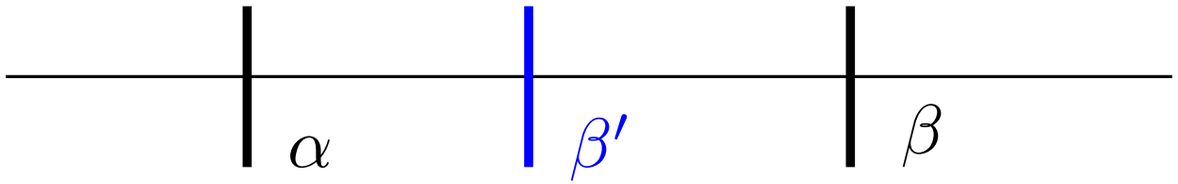


Improving α and β

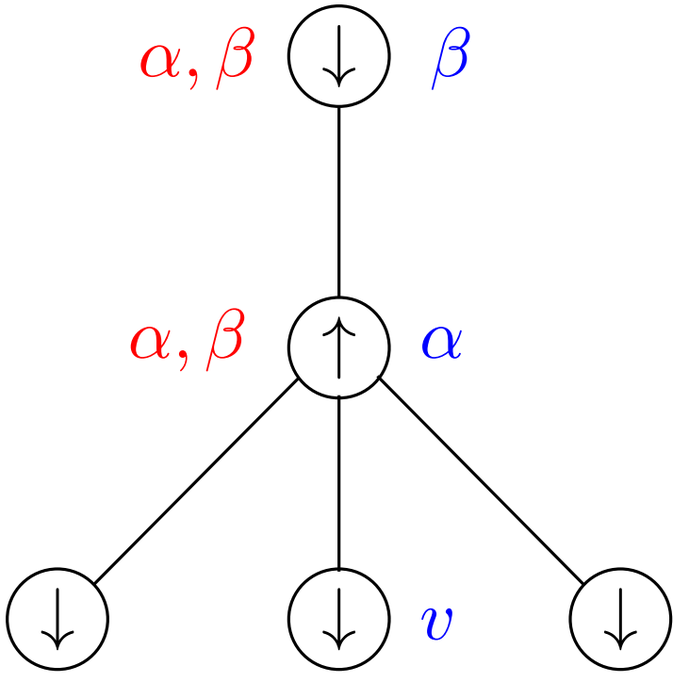
- Each node receives as arguments α and β of the subtree traversed so far.
- A MAX node may find a path with better maximum value through its subtrees and returns an updated α'.
- The returned α' should be in the range $[\alpha, \beta]$, inclusive, where α and β are the arguments received at the node.

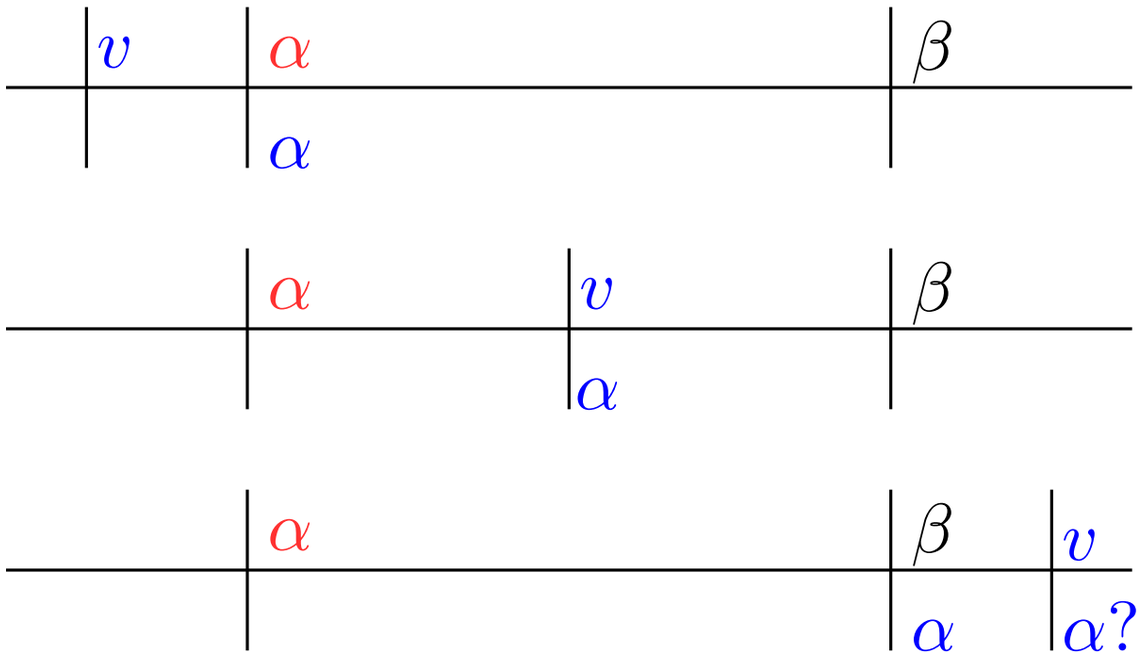


- A MIN node, after traversing its subtrees, returns an updated β.
- The returned β' should be in the range $[\alpha, \beta]$, inclusive, where α and β are the arguments received at the node.



Updating α in a MAX Node



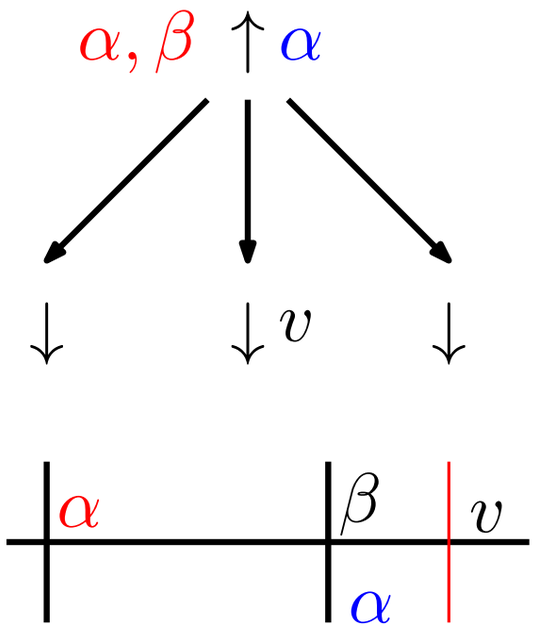


if $v < \alpha$ **then** // $v < \alpha$
  $\alpha \leftarrow \alpha$
else if $v < \beta$ **then** // $\alpha \leq v < \beta$
  $\alpha \leftarrow v$
else // $\beta \leq v$
  $\alpha \leftarrow \beta$

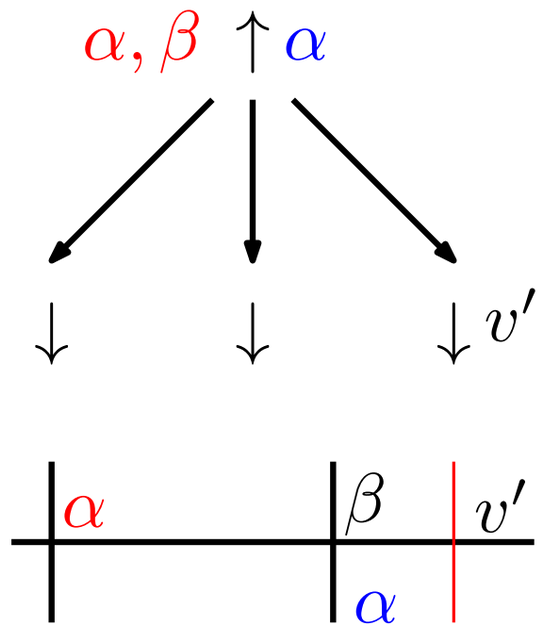
$\equiv$

if $\alpha < v$ **then** // $\alpha < v$
  $\alpha \leftarrow v$
if $\beta \leq v$ **then** // $\beta \leq v$
  $\alpha \leftarrow \beta$

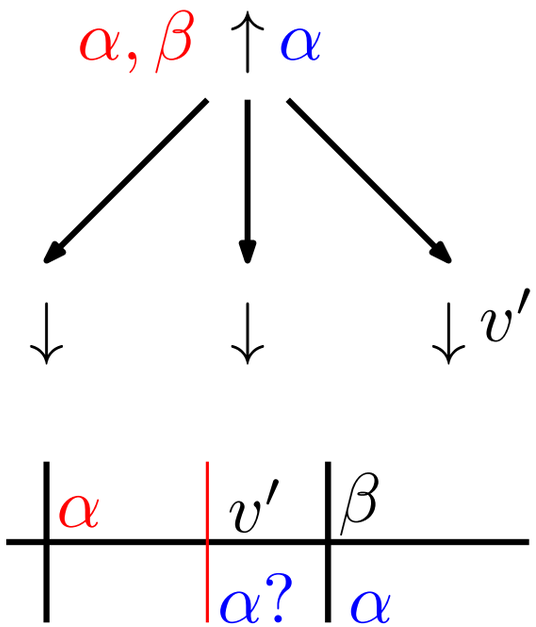
Pruning



$\beta \leq v$



$\beta \leq v'$



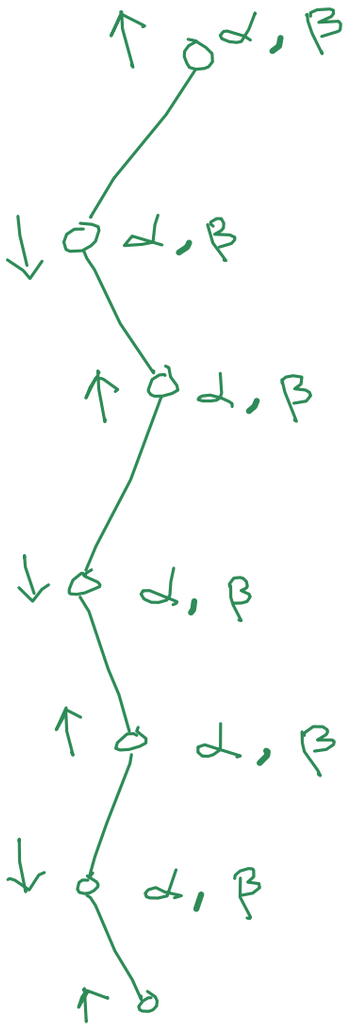
$\alpha \leq v' < \beta$

Pruning Logic

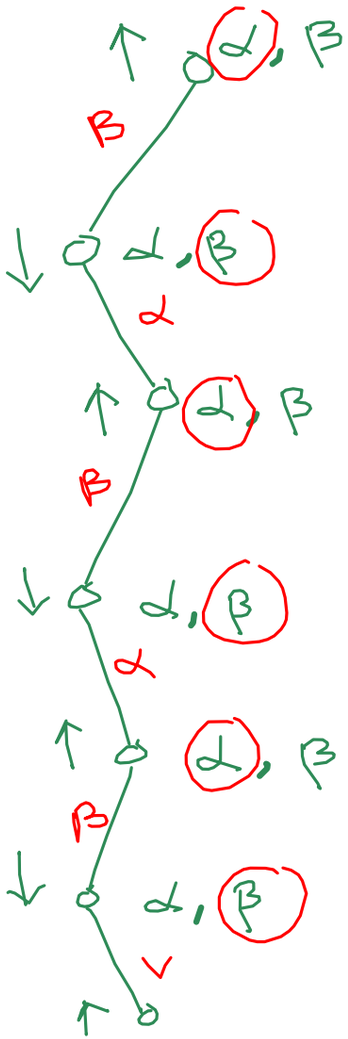
```
// update α
if α < val then          // α < val
    α ← val
if β ≤ val then          // β ≤ val
    α ← β
    return α
```

```
// update β
if val < β then          // val < β
    β ← val
if val ≤ α then          // val ≤ α
    β ← α
    return β
```

DFS: Descending and Ascending



DFS: Descending and Ascending



Minimax with α-β Pruning

Algorithm: Alpha_Beta(n, α, β)

Input: n a node in a game tree; α, β real numbers
Output: A pair of a value for each agent for node n, path that gives this value

```
if n is a leaf node then  return evaluate(n), None
else if n is a MAX node then
    foreach child c of n do
        α ← α ↑ Alpha_Beta(c, α, β)
        if β ≤ α then
            α ← β
            return α
    return α
else
    foreach child c of n do
        β ← β ↓ Alpha_Beta(c, α, β)
        if β ≤ α then
            β ← α
            return β
    return β
```

Minimax with $\alpha\beta$ Pruning

Algorithm: Alpha_Beta(n, α, β)

Input: n a node in a game tree; α, β real numbers
Output: A pair of a value for each agent for node n, path that gives this value

```
if n is a leaf node then  return evaluate(n), None
else if n is a MAX node then
    foreach child c of n do
        val, path ← Alpha_Beta(c, α, β)
        if α < val  then
            α ← val
            best_path ← [c] + path
        if β ≤ val then  return val, best_path
    return α, best_path
else
    foreach child c of n do
        val, path ← Alpha_Beta(c, α, β)
        if val < β then
            β ← val
            best_path ← [c] + path
        if val ≤ α then  return val, best_path
```

Alpha-Beta Pruning Trace

α — ∞, ∞

β

α

β

α

Alpha-Beta Pruning Trace

α $\quad -\infty, \infty$

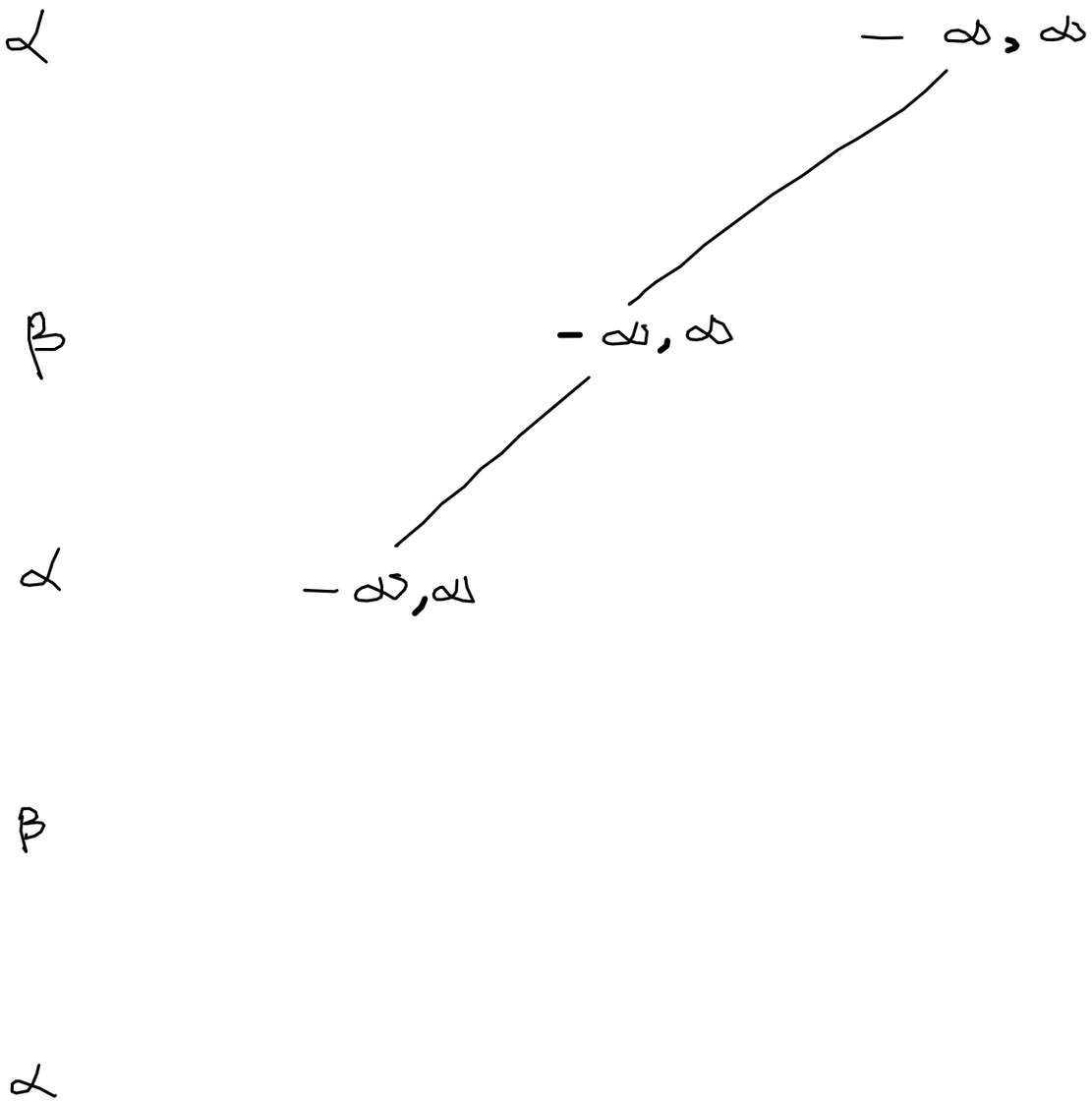
β $\quad -\infty, \infty$

α

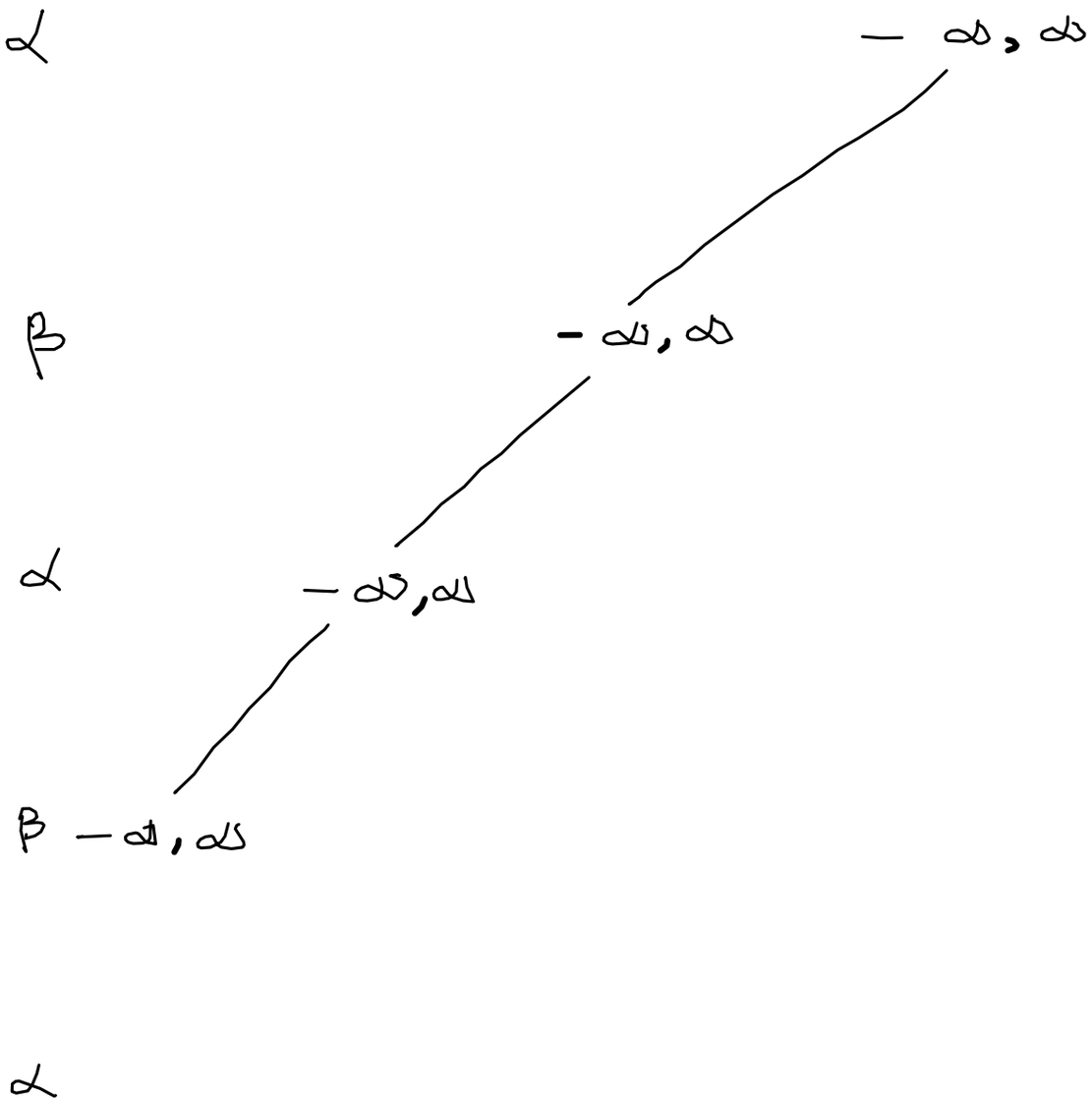
β

α

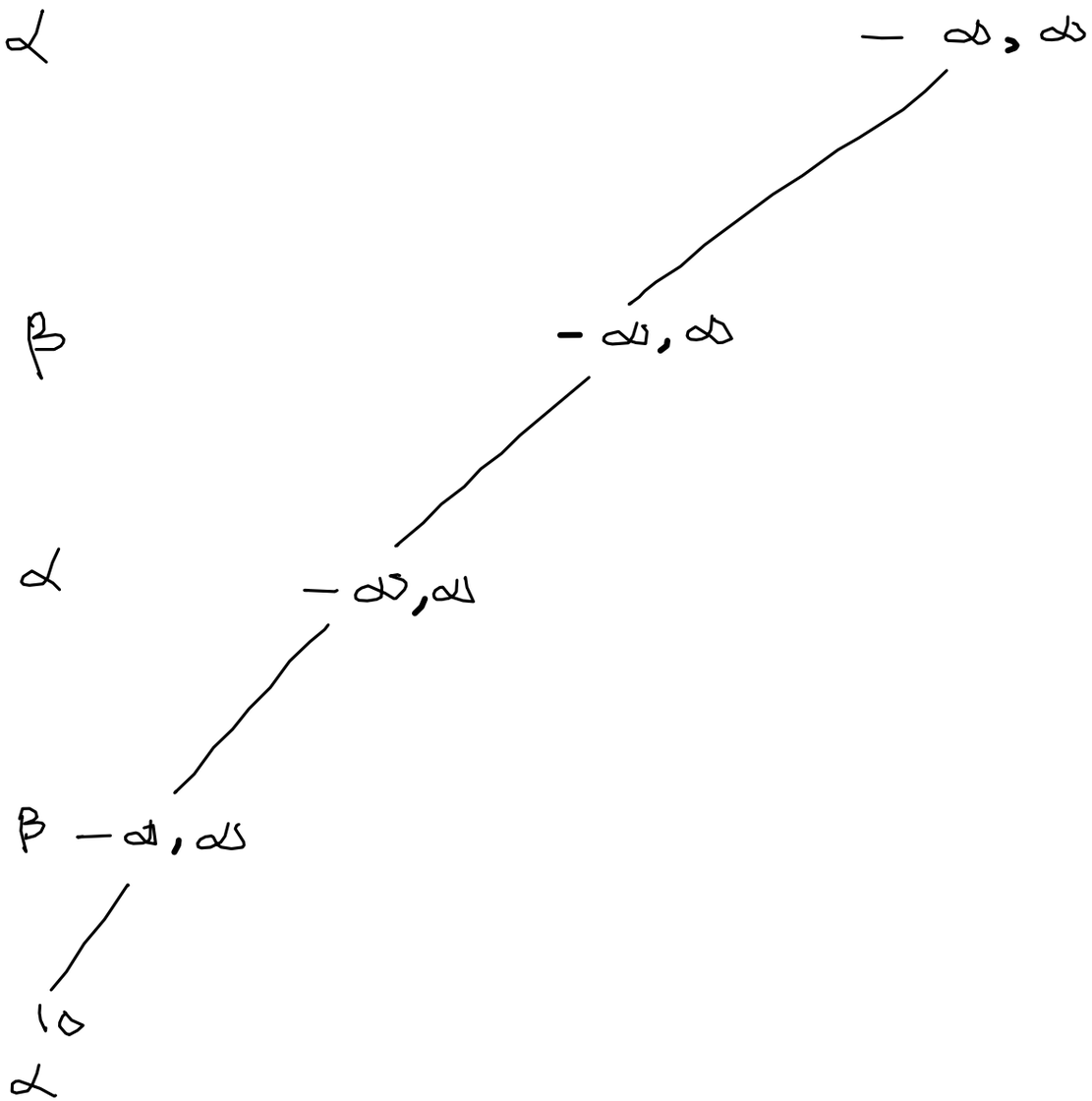
Alpha-Beta Pruning Trace



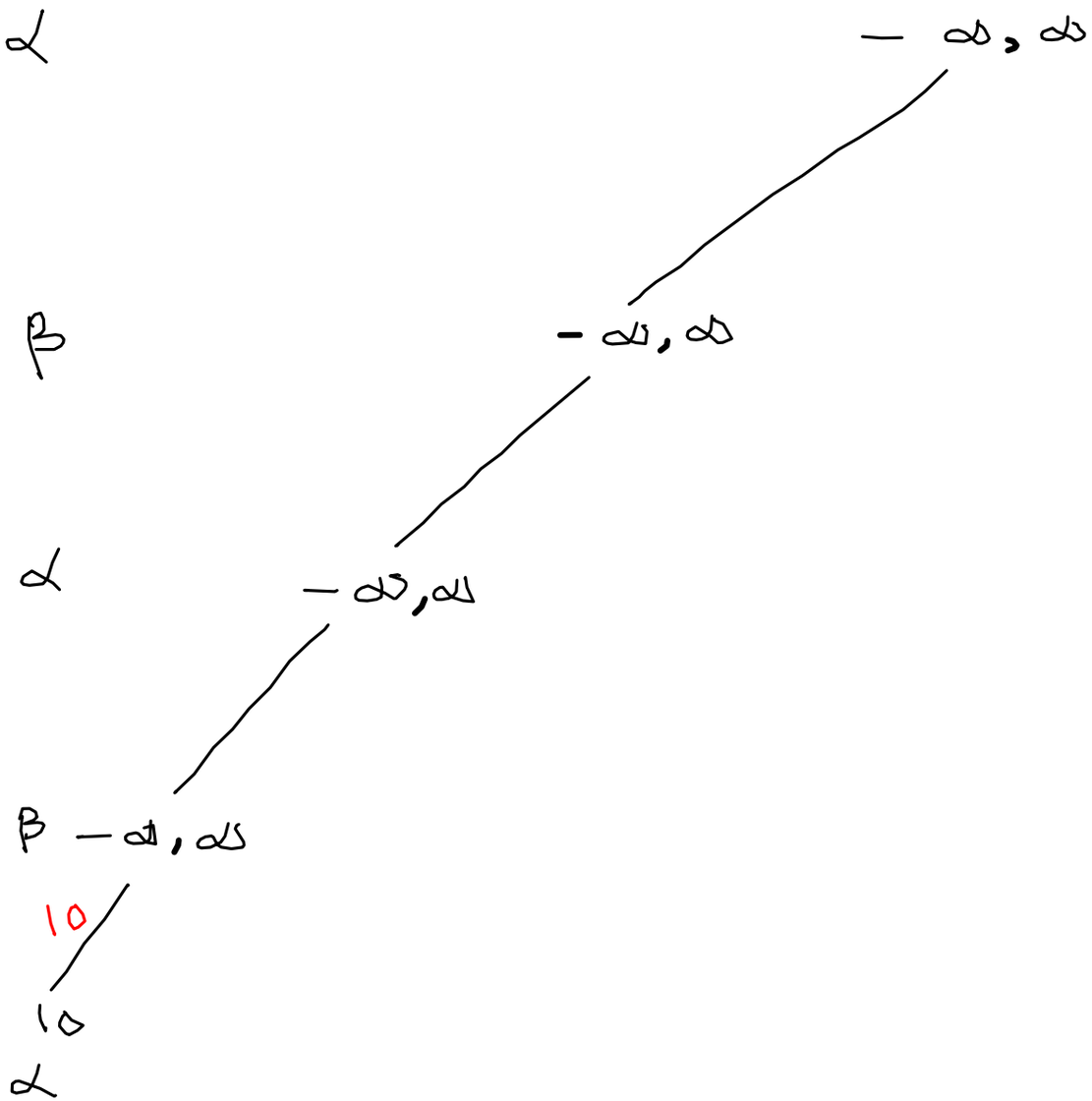
Alpha-Beta Pruning Trace



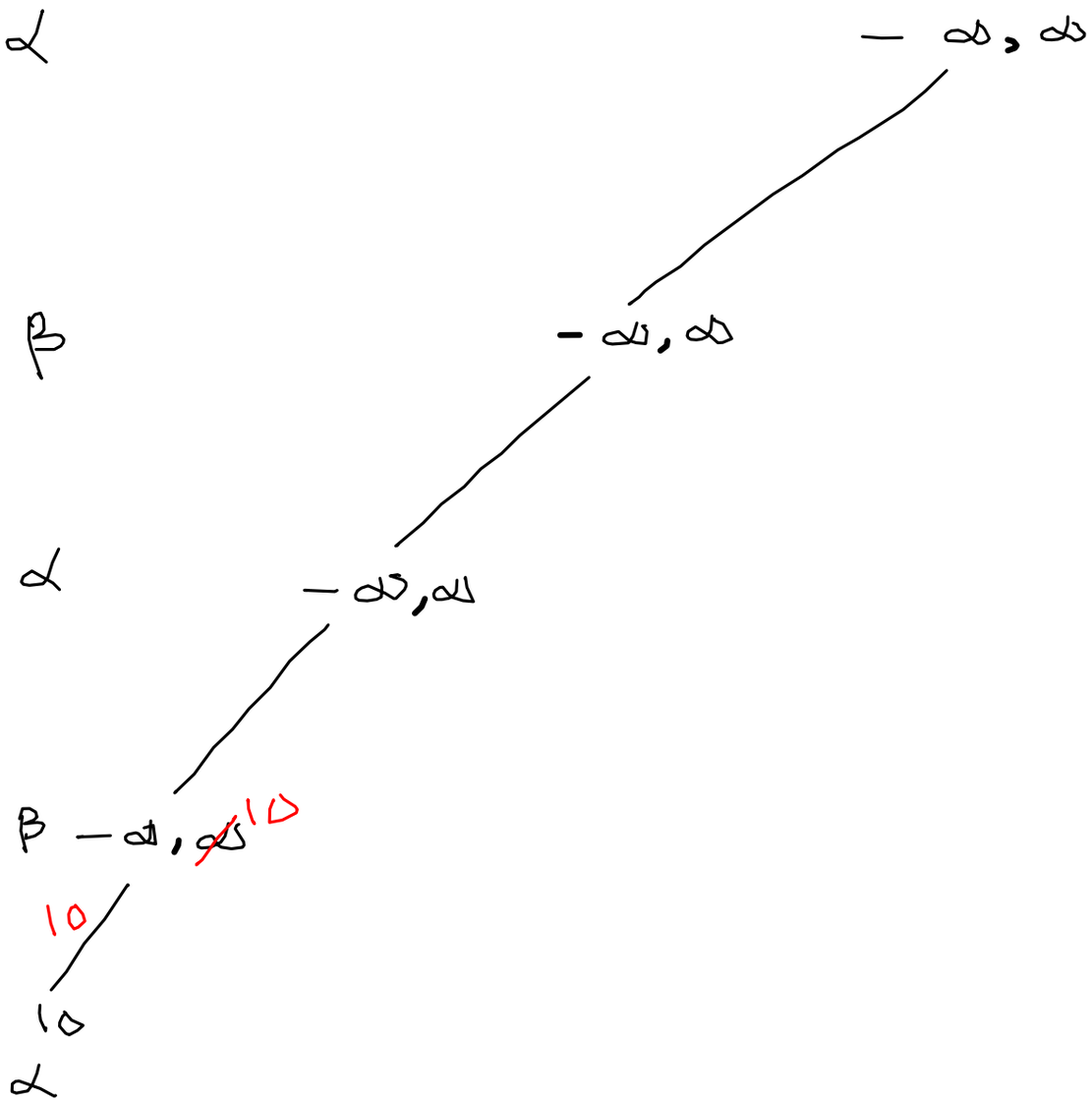
Alpha-Beta Pruning Trace



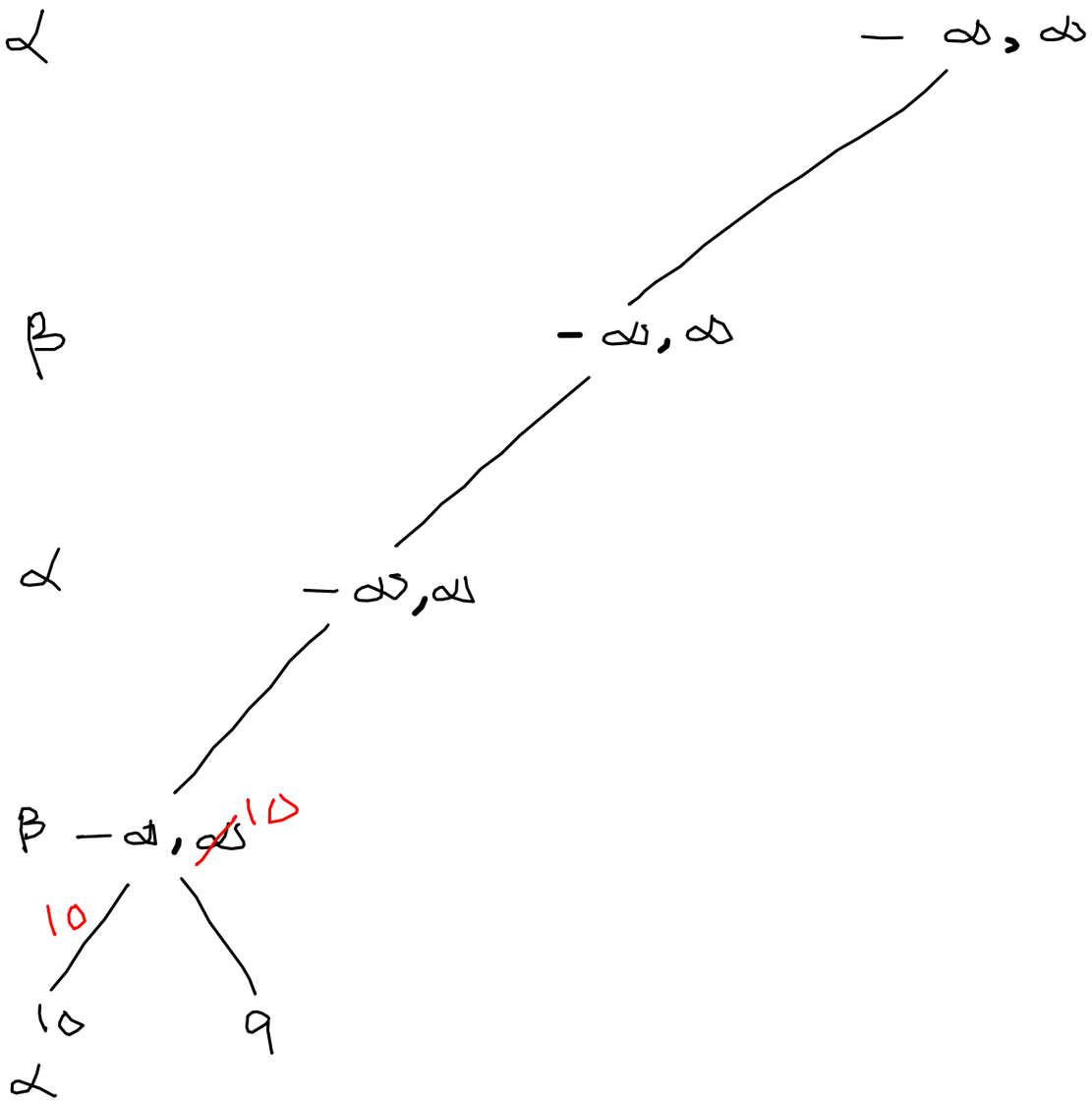
Alpha-Beta Pruning Trace



Alpha-Beta Pruning Trace



Alpha-Beta Pruning Trace



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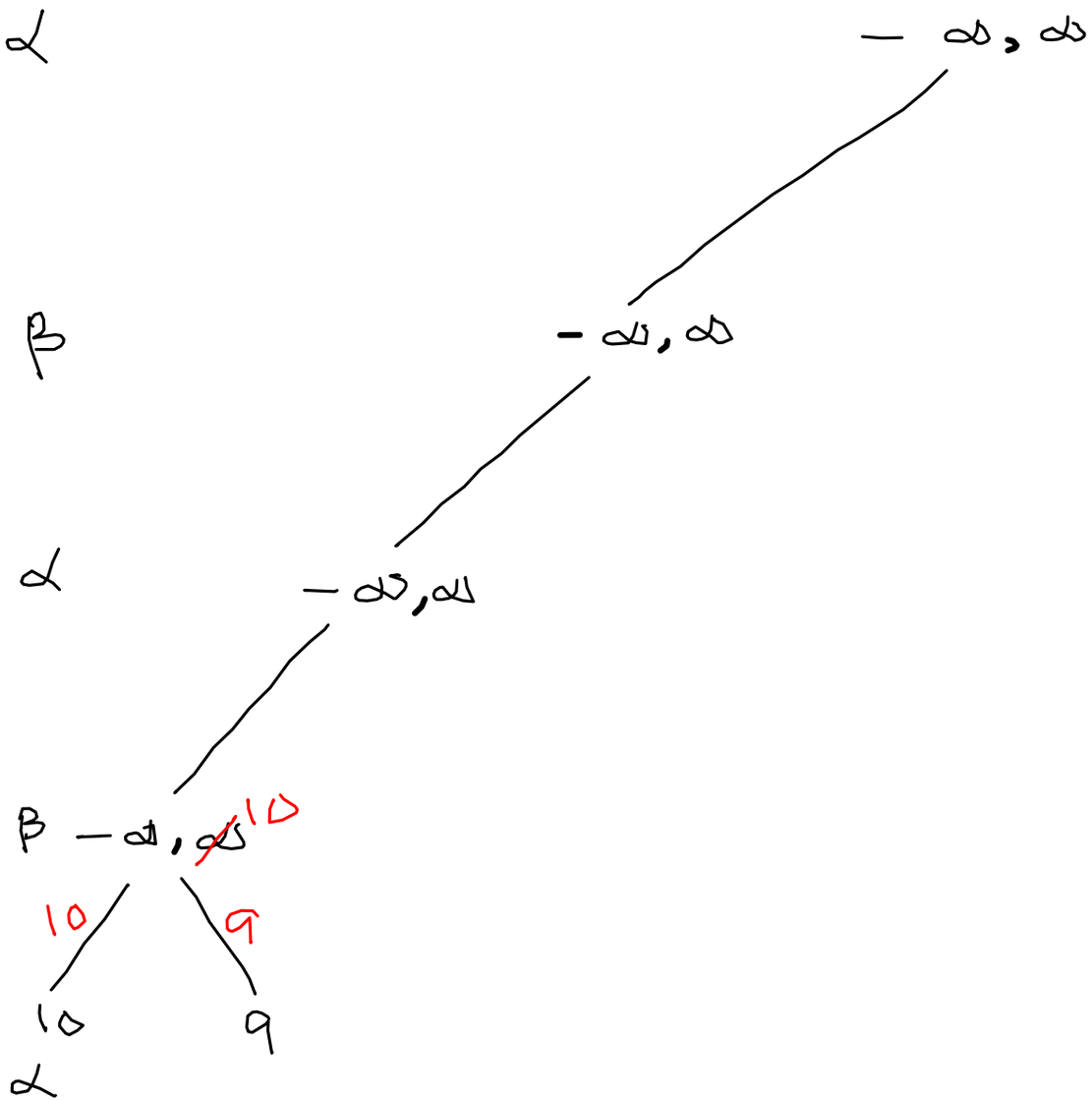

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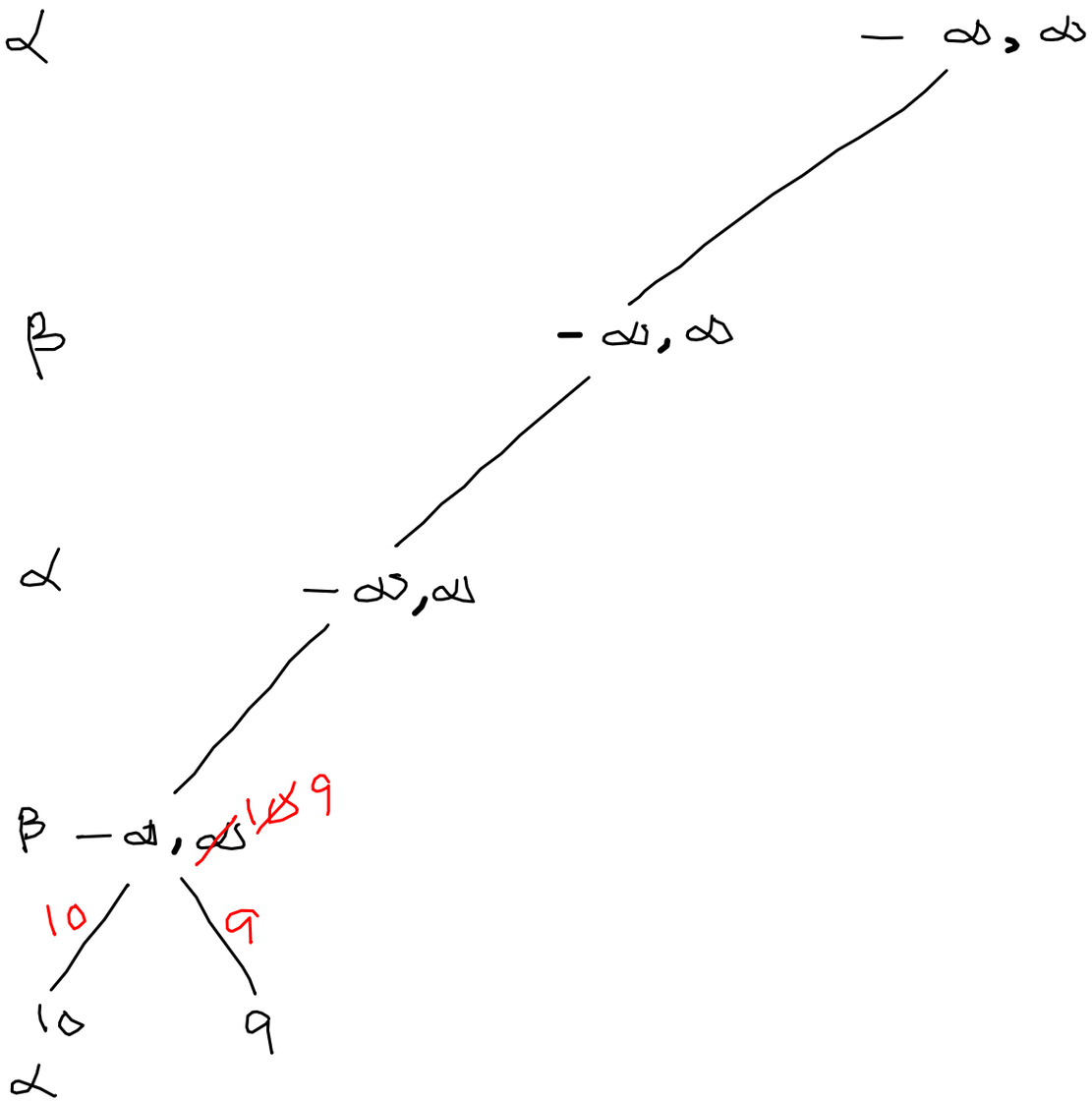

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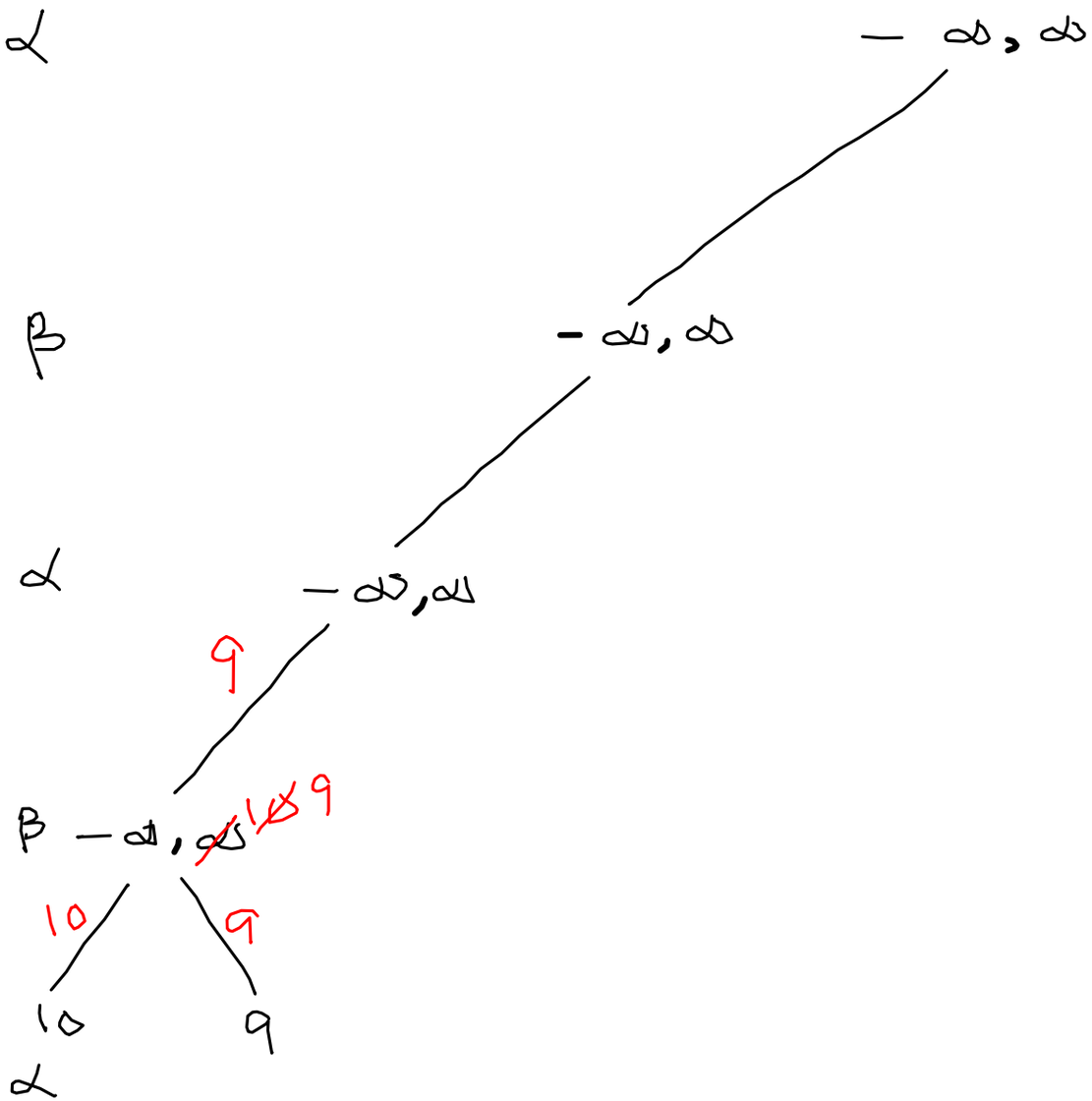

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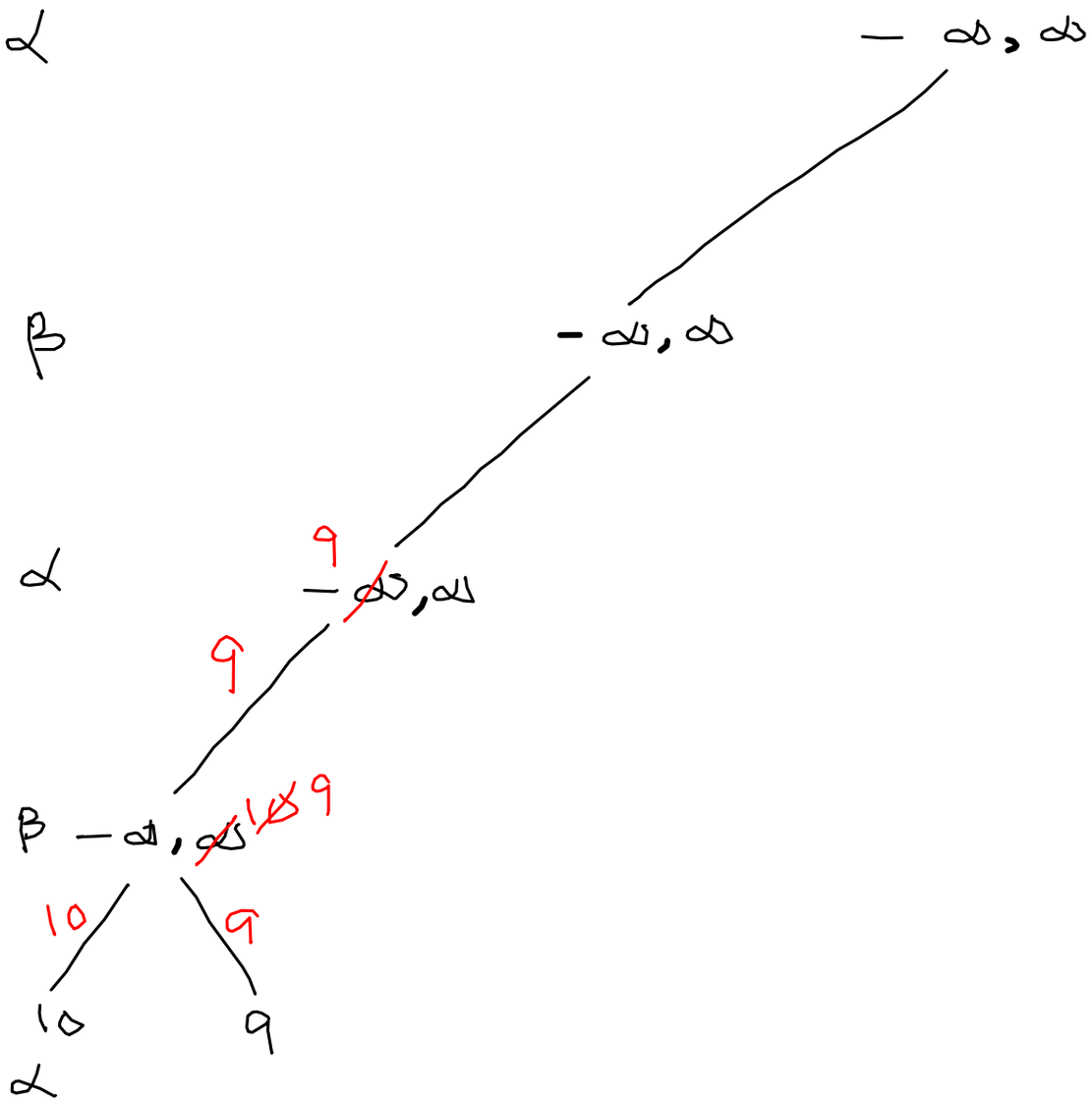

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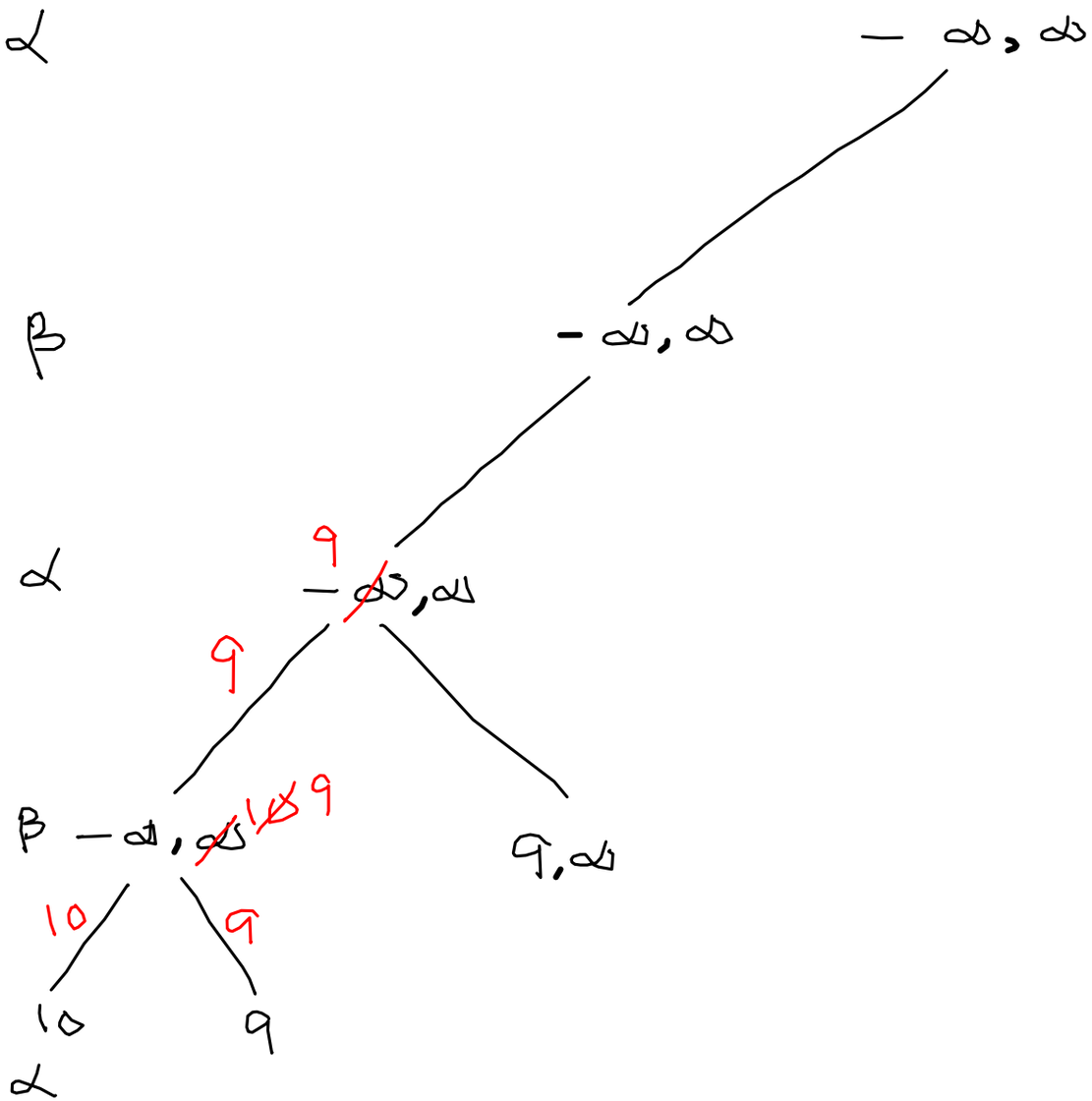

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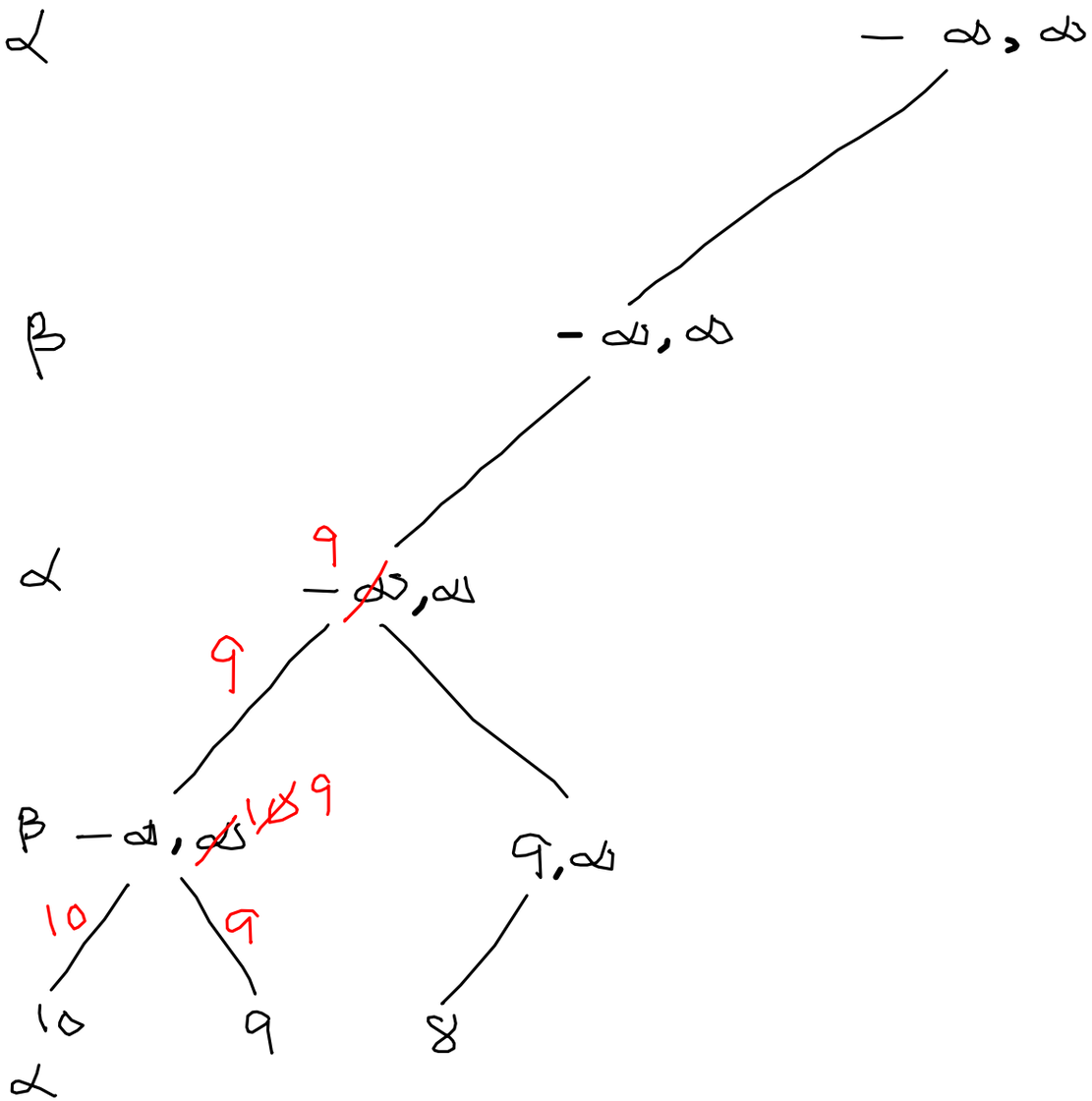

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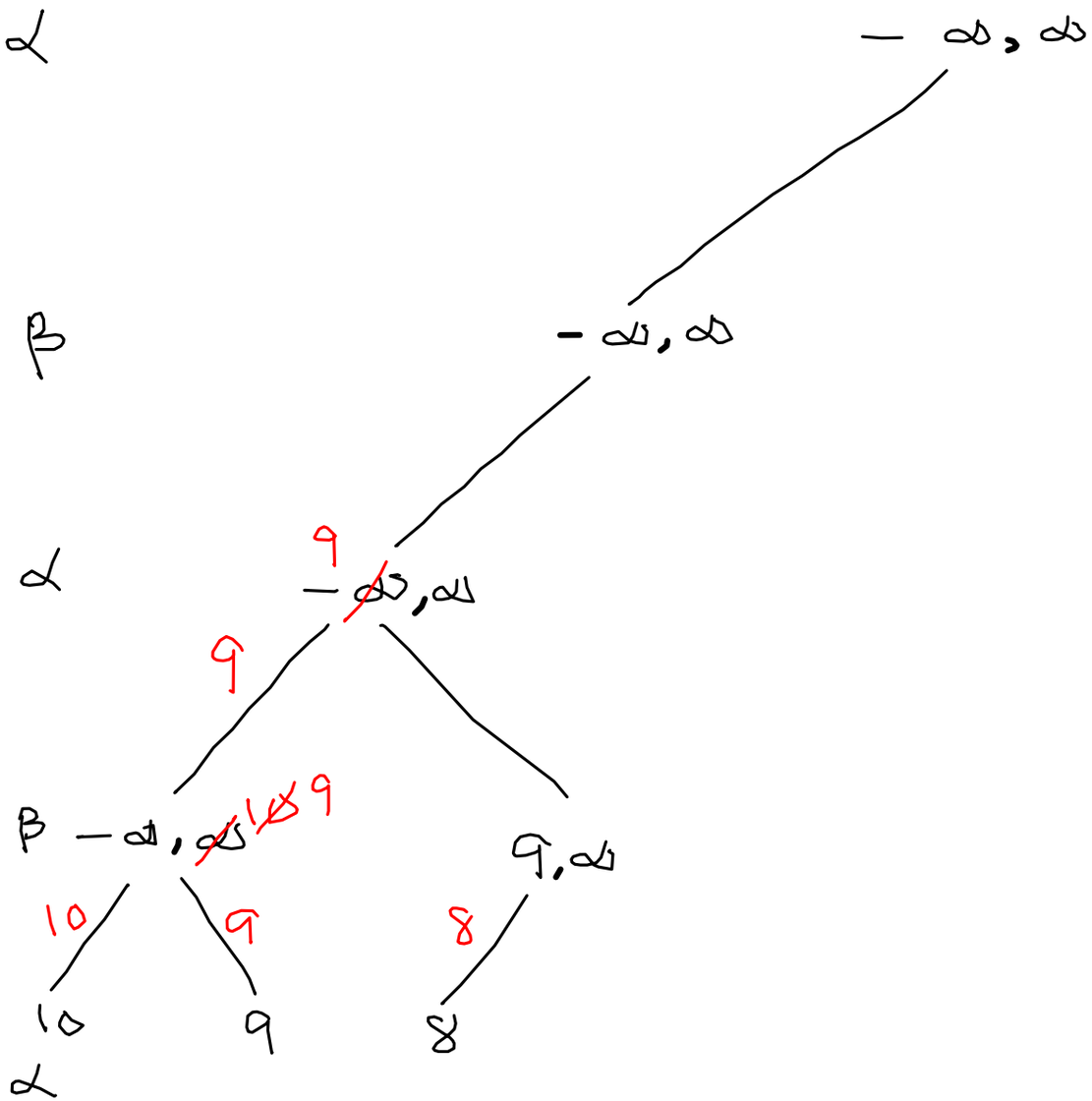

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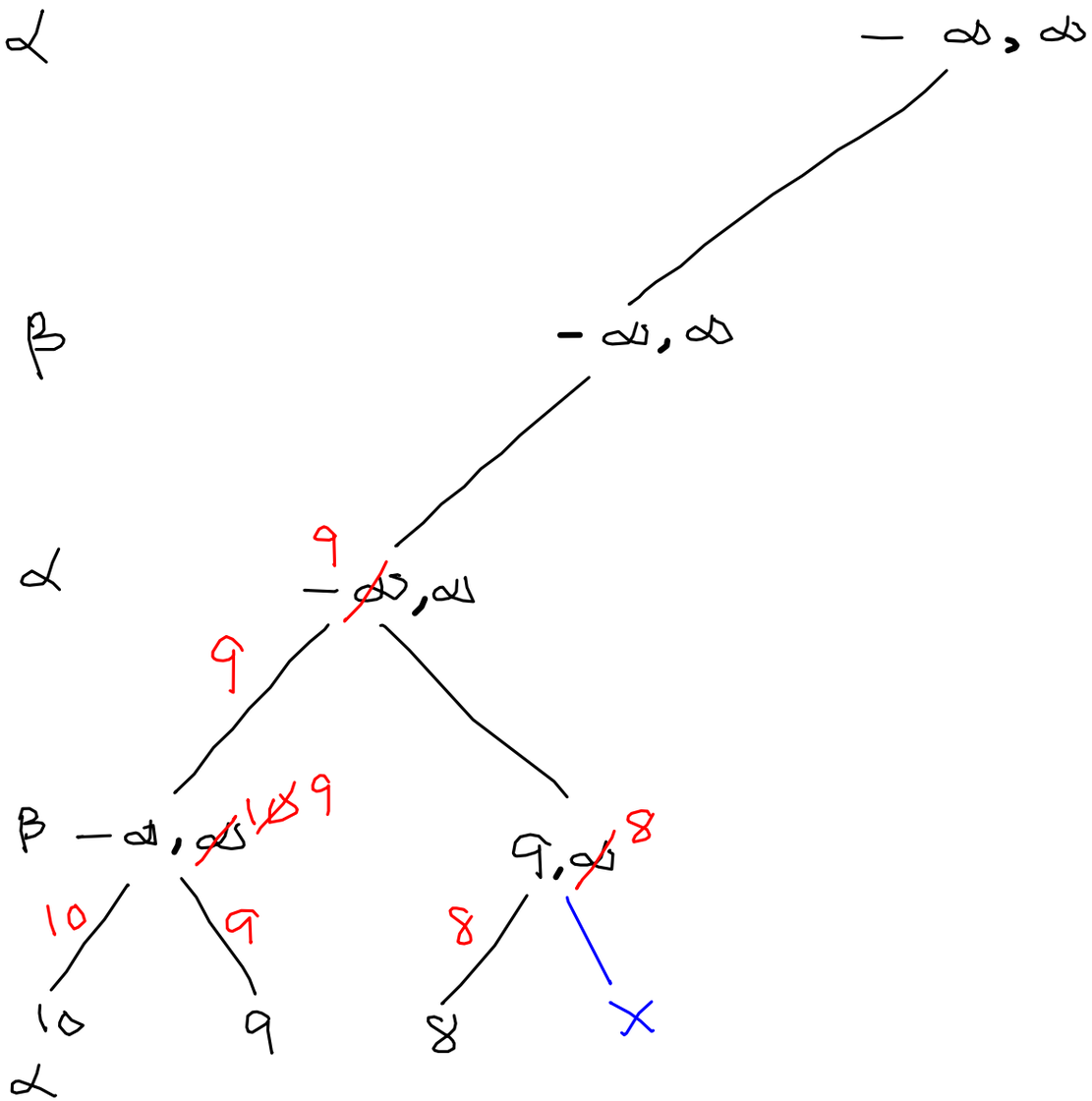

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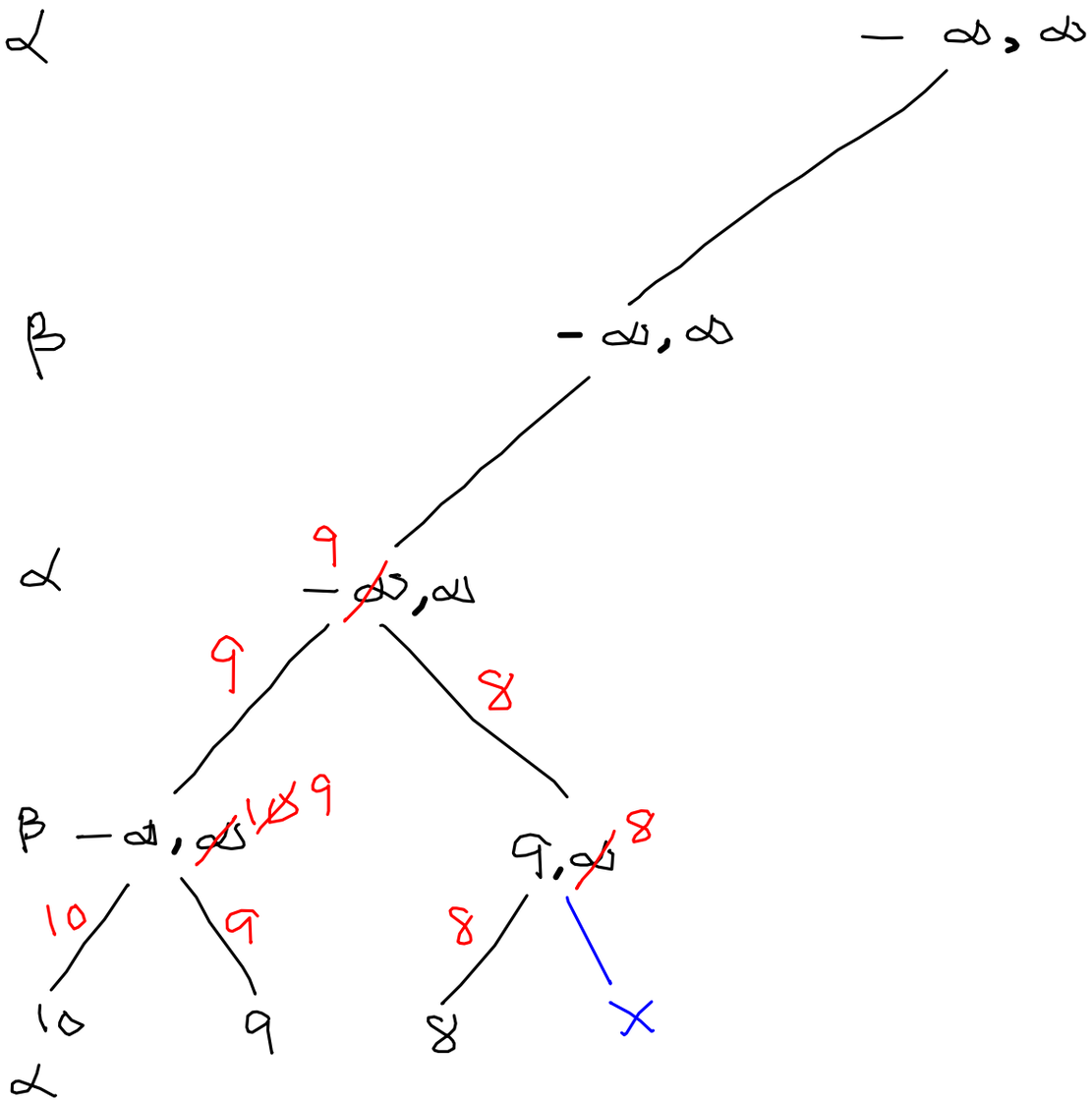

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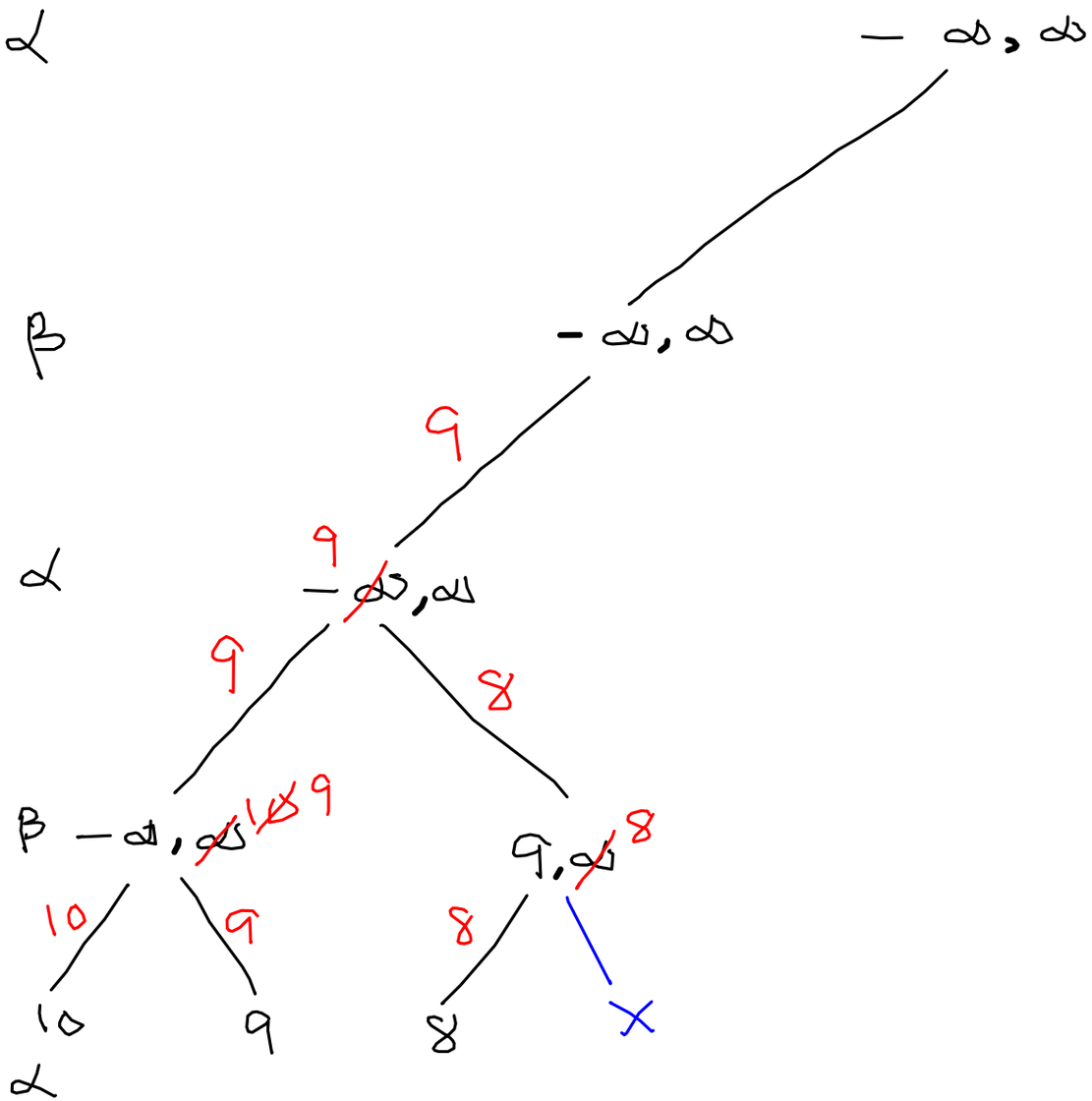

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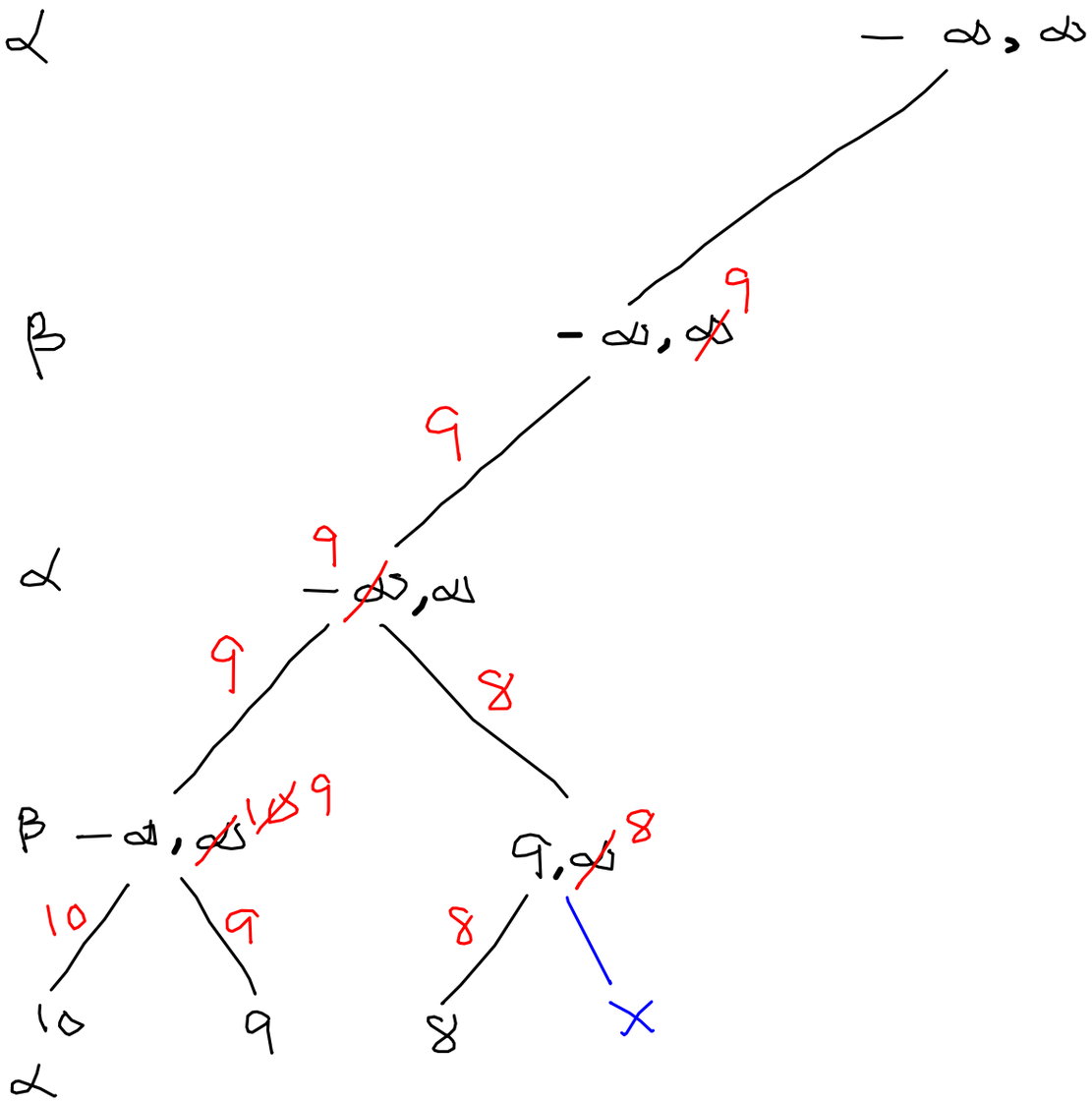

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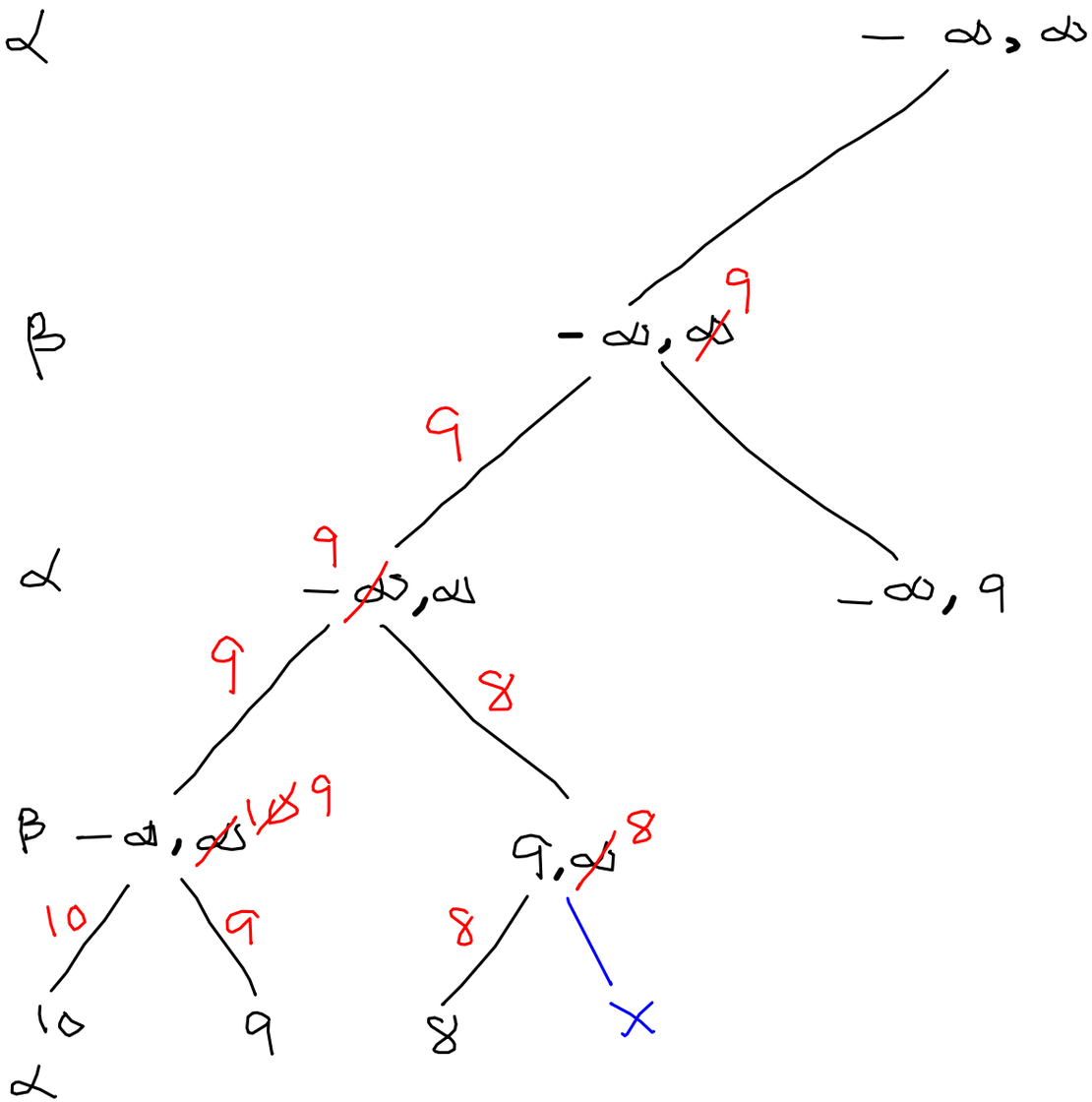

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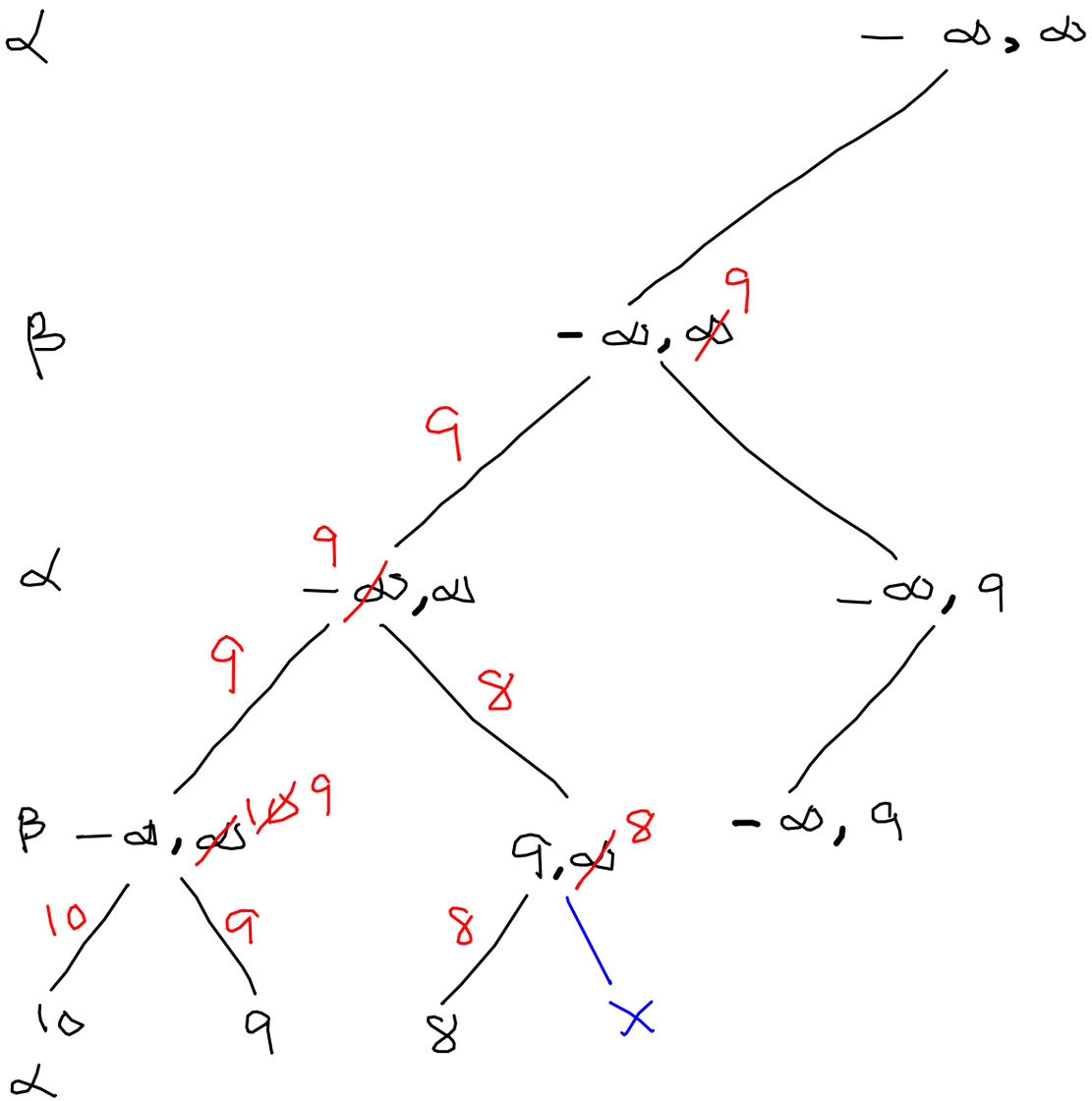

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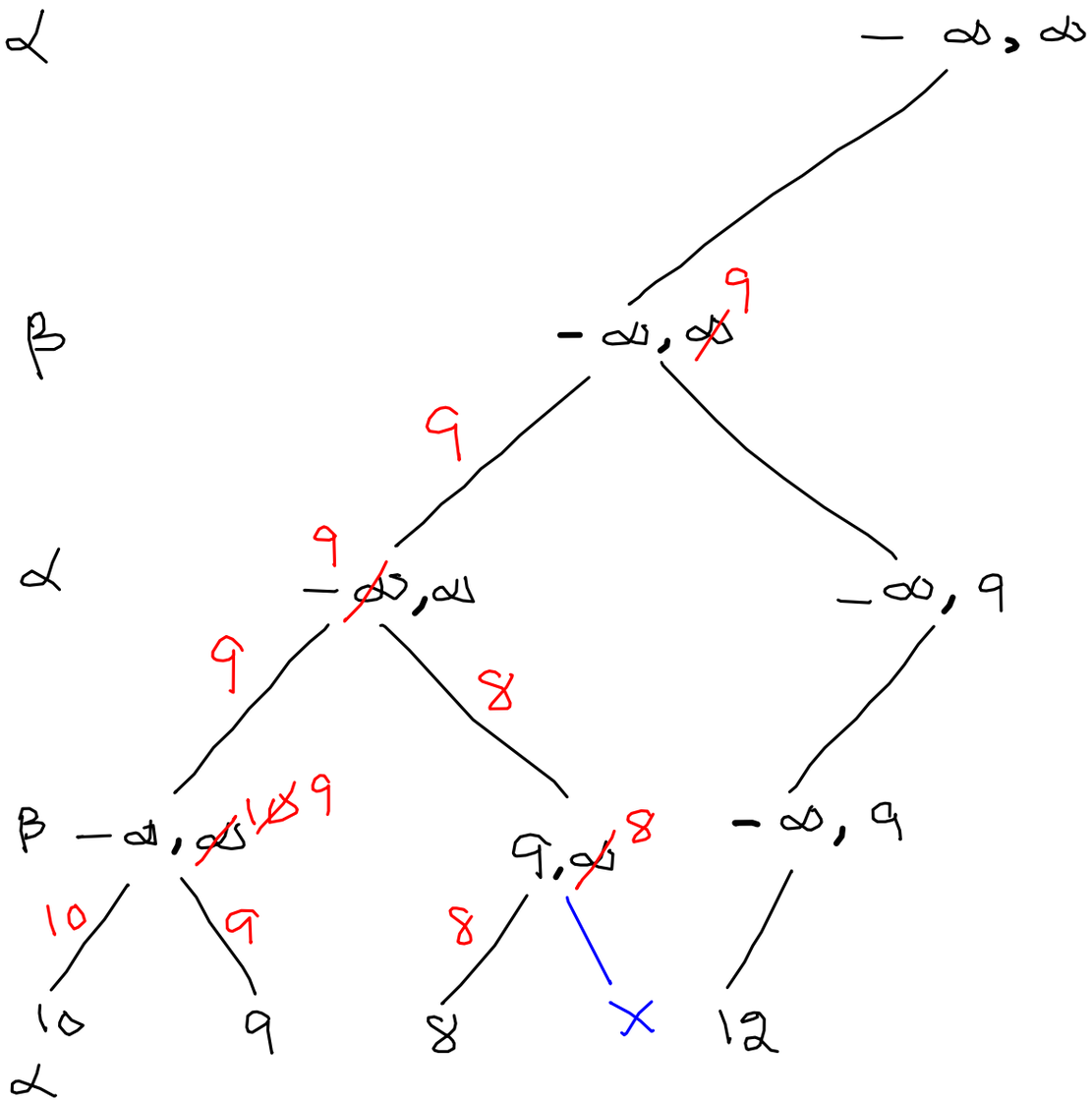

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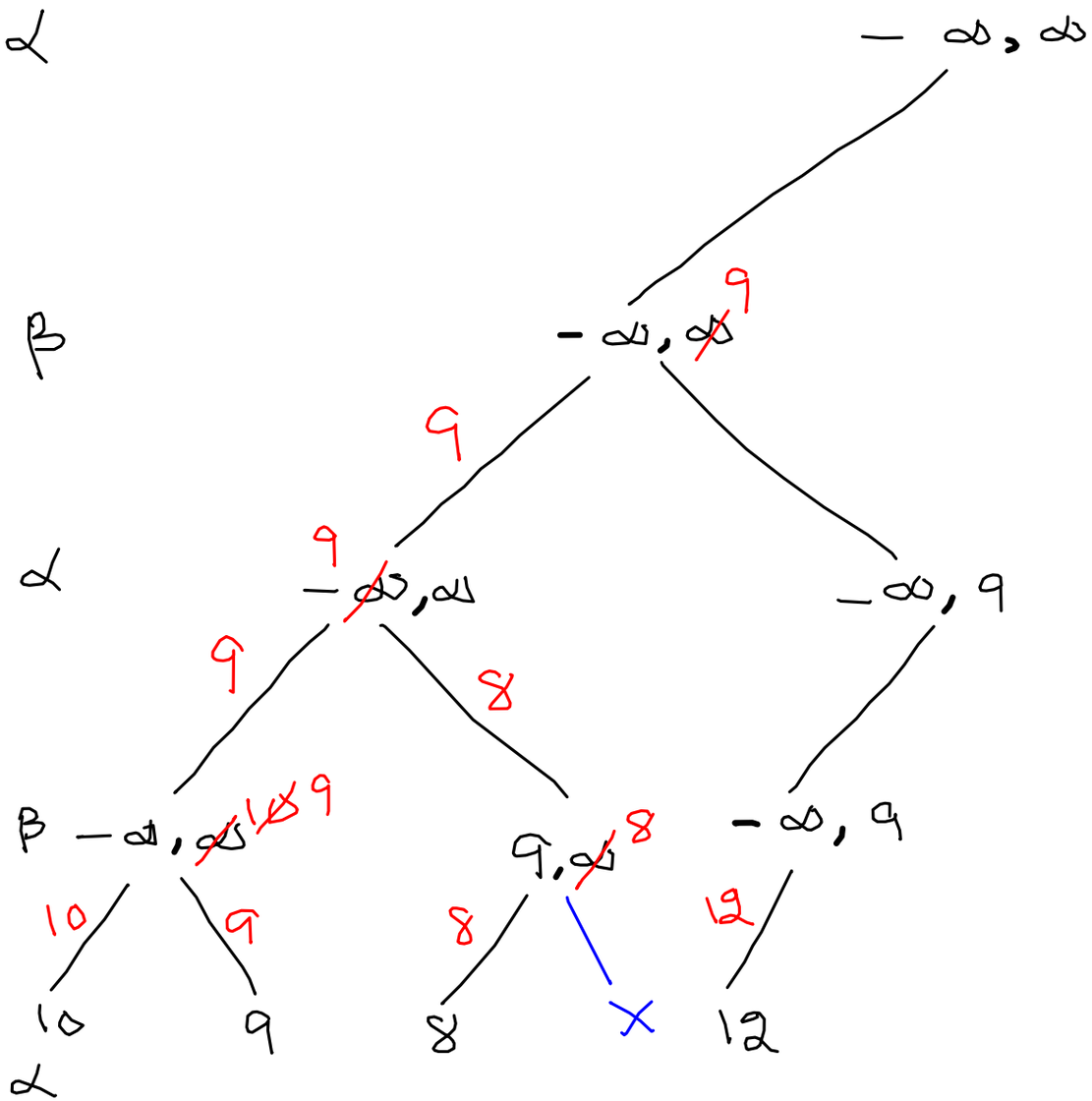

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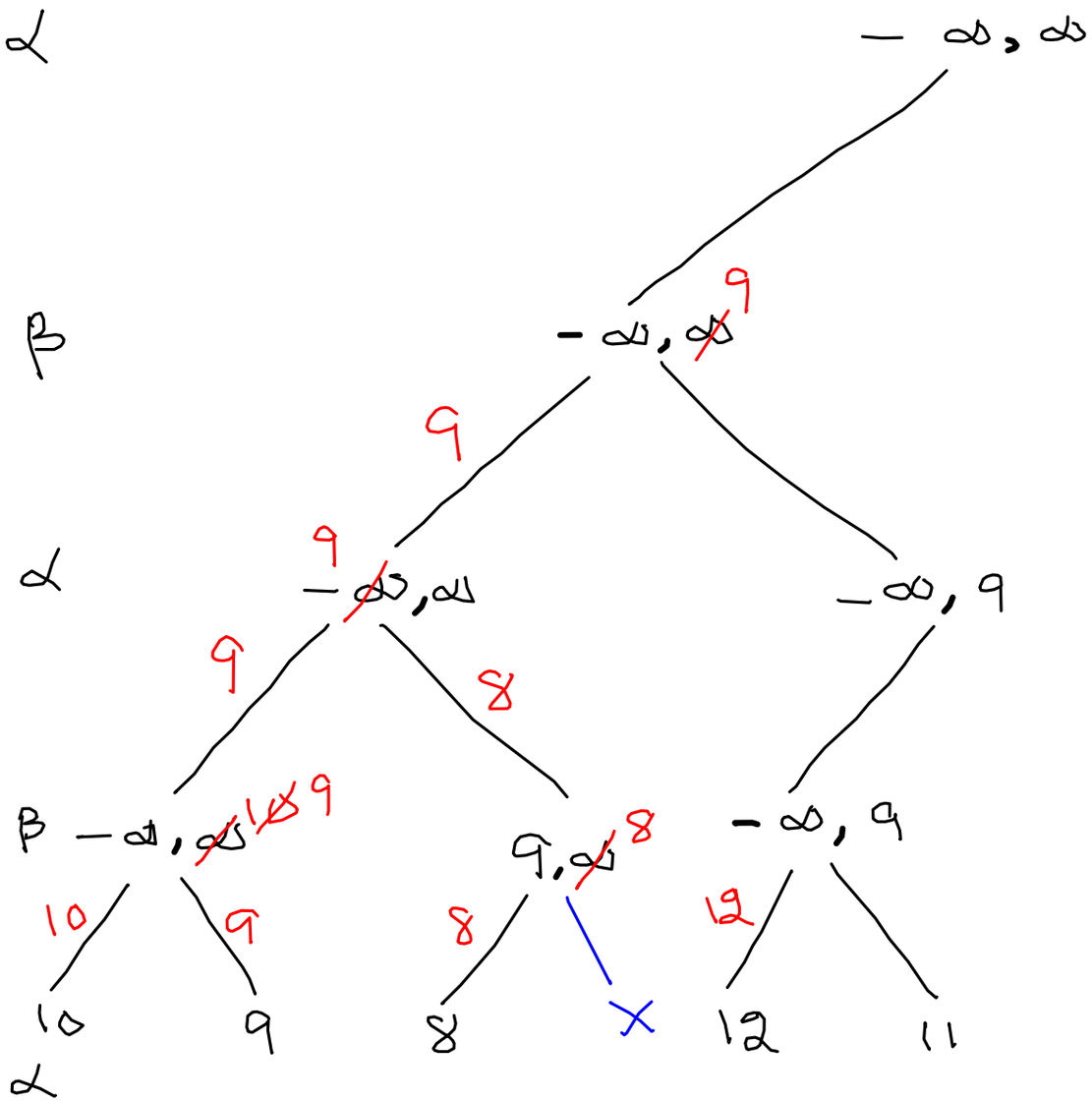

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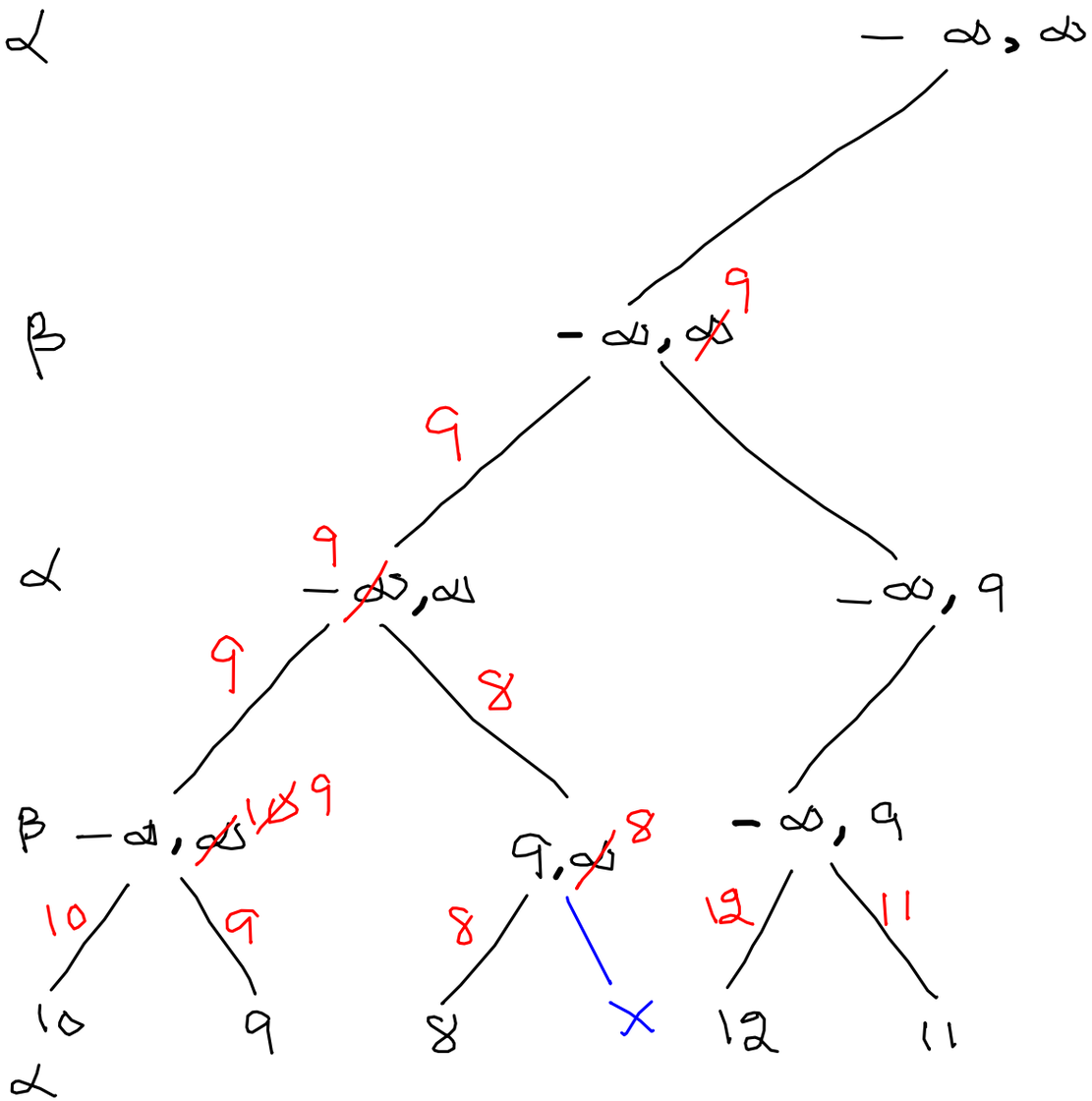

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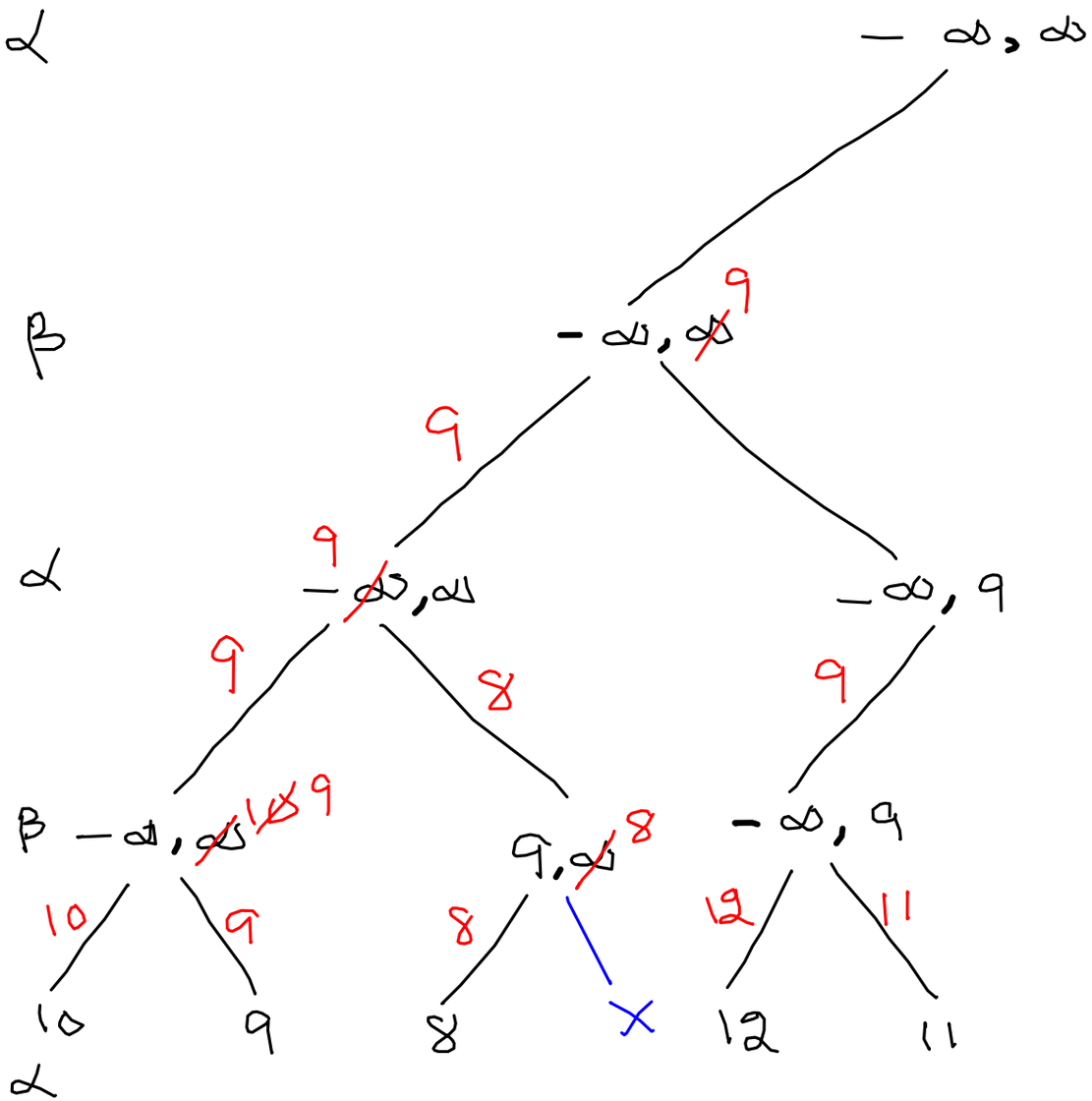

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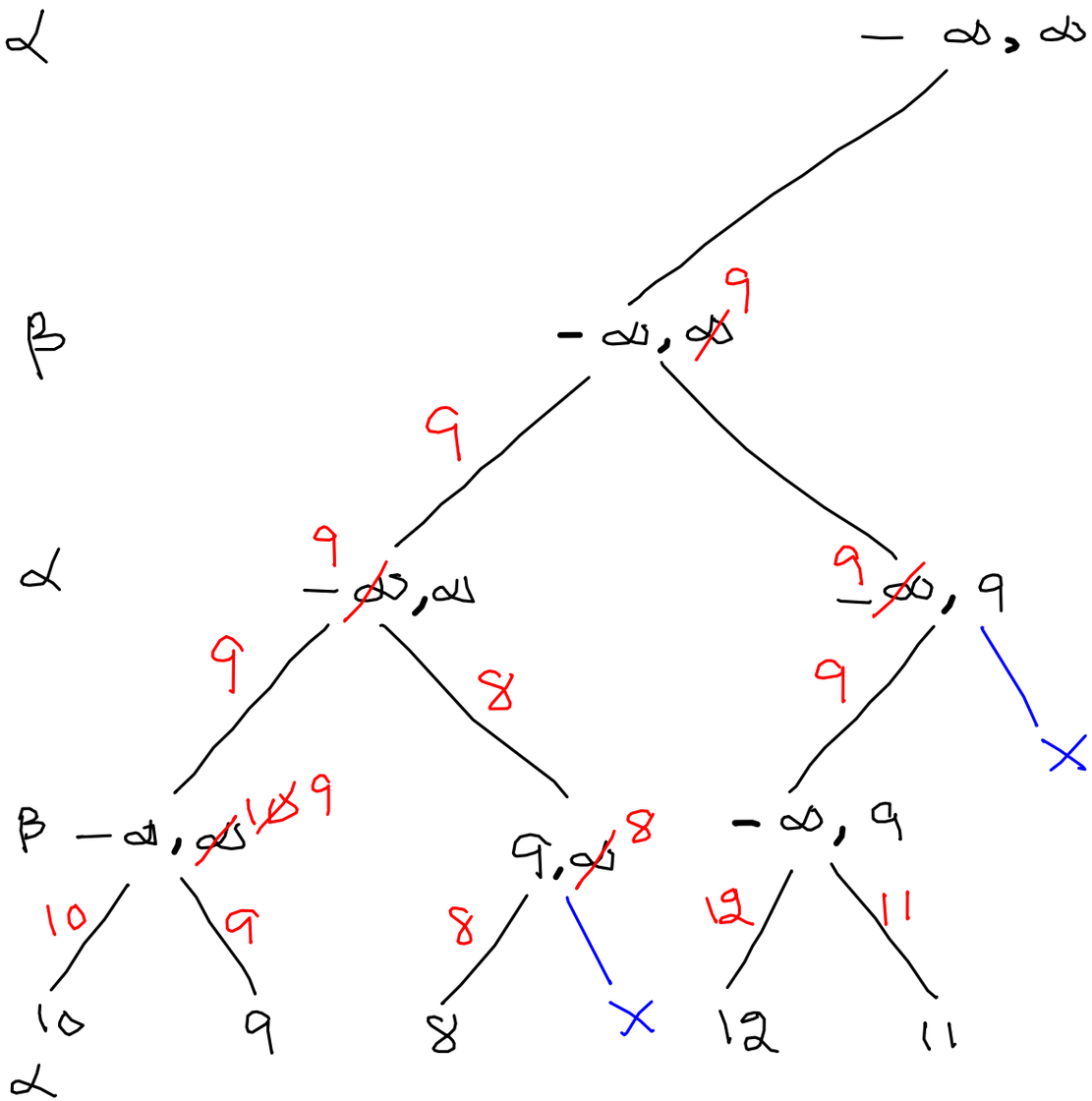

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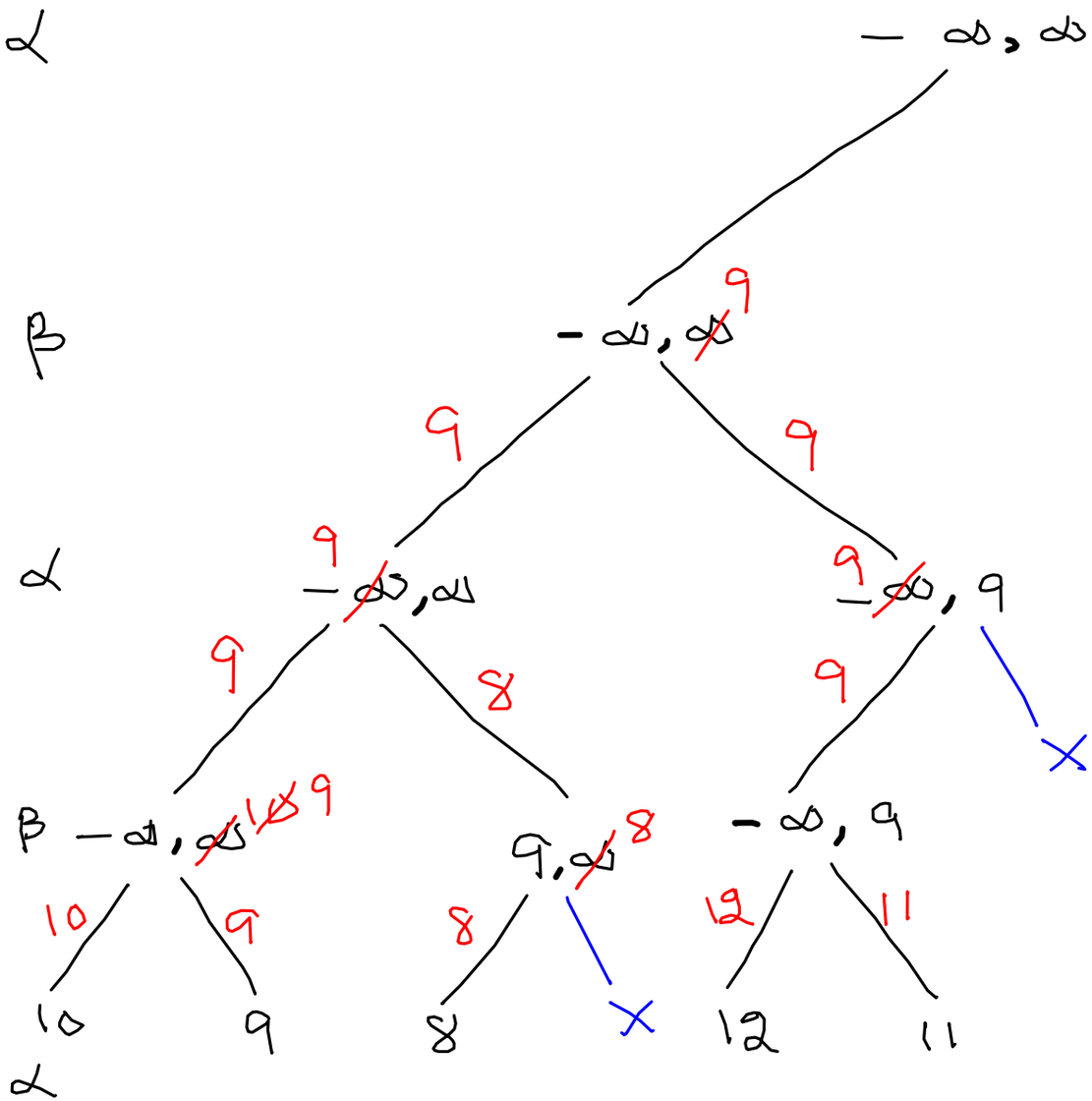

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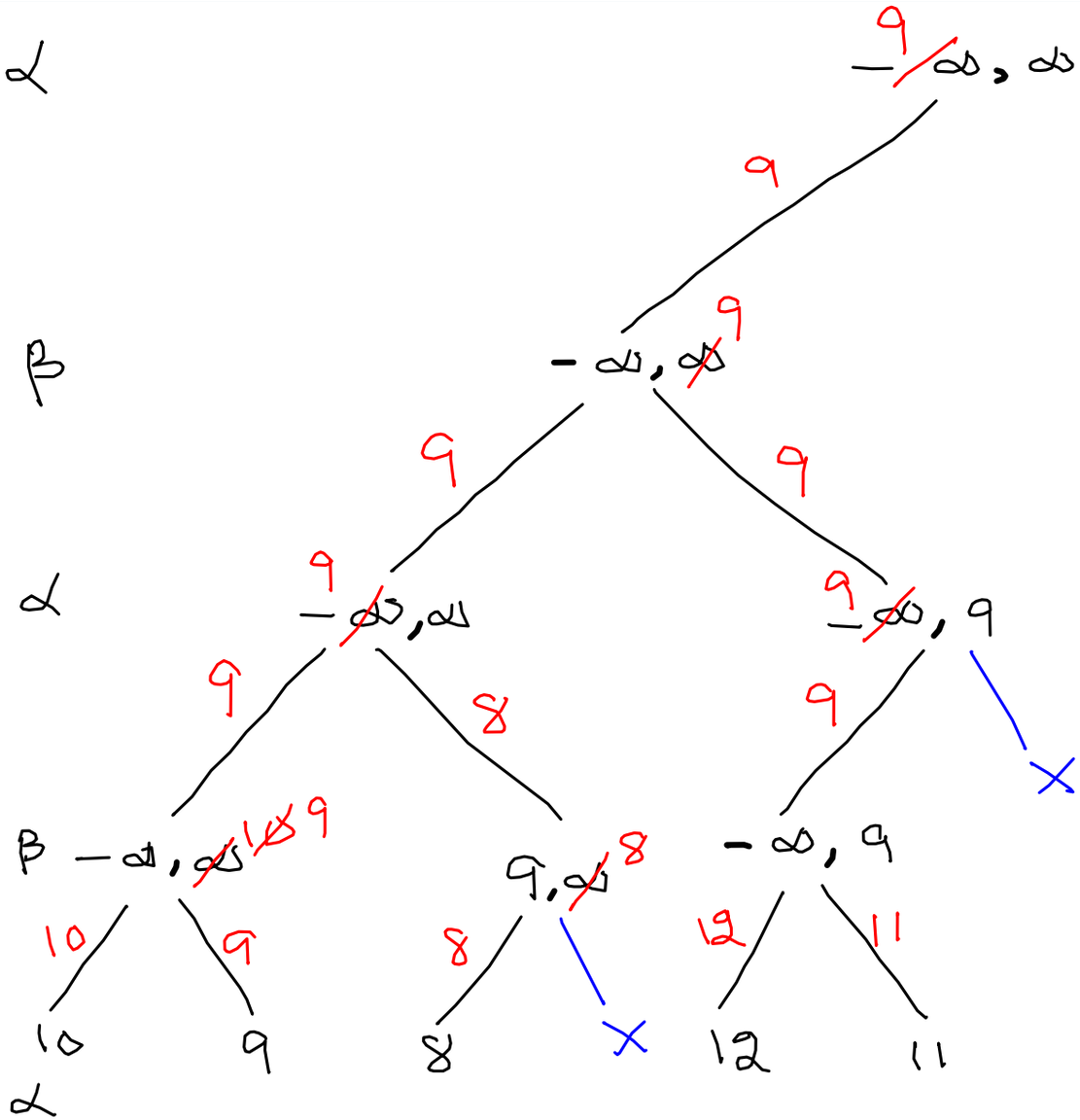

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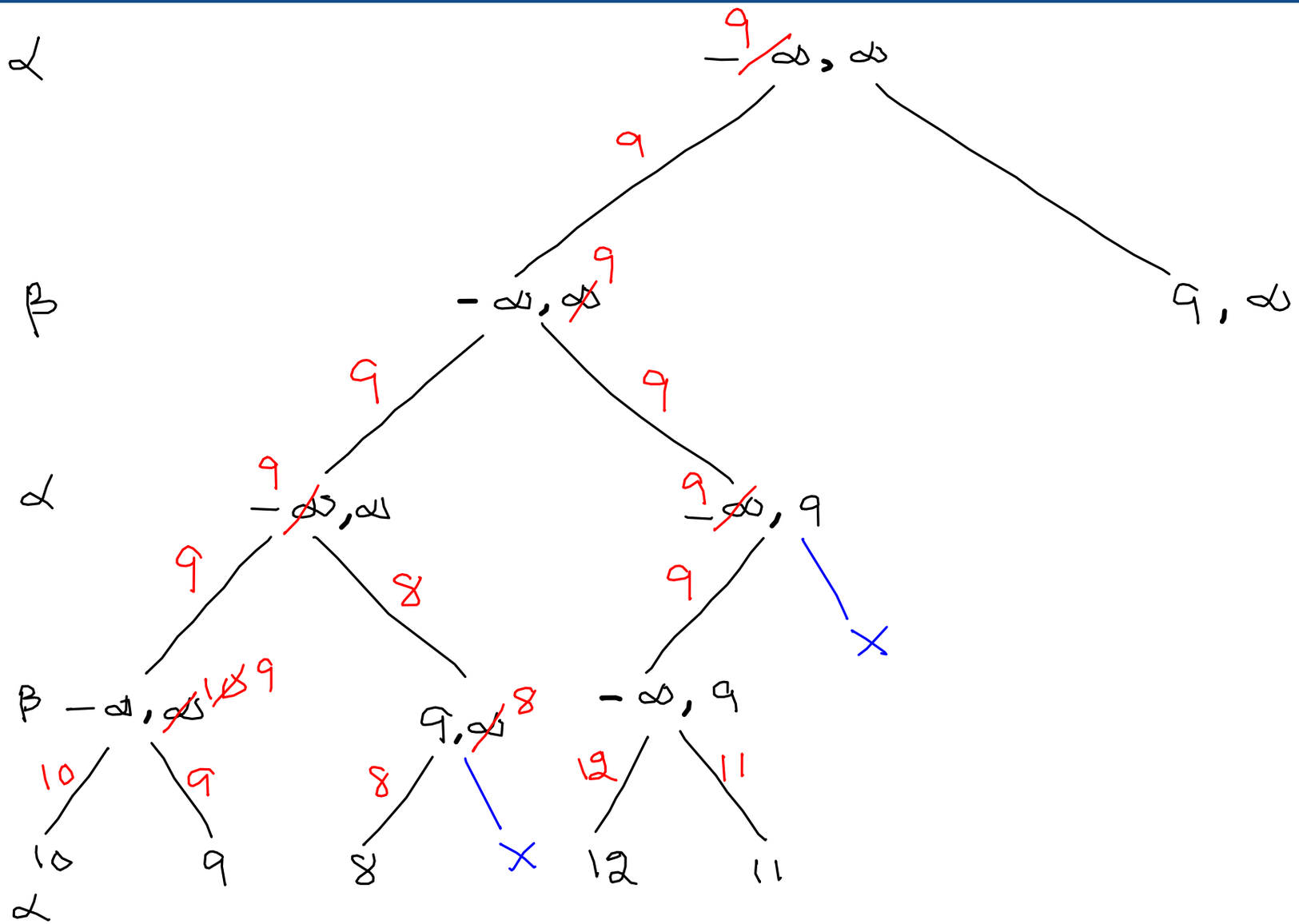

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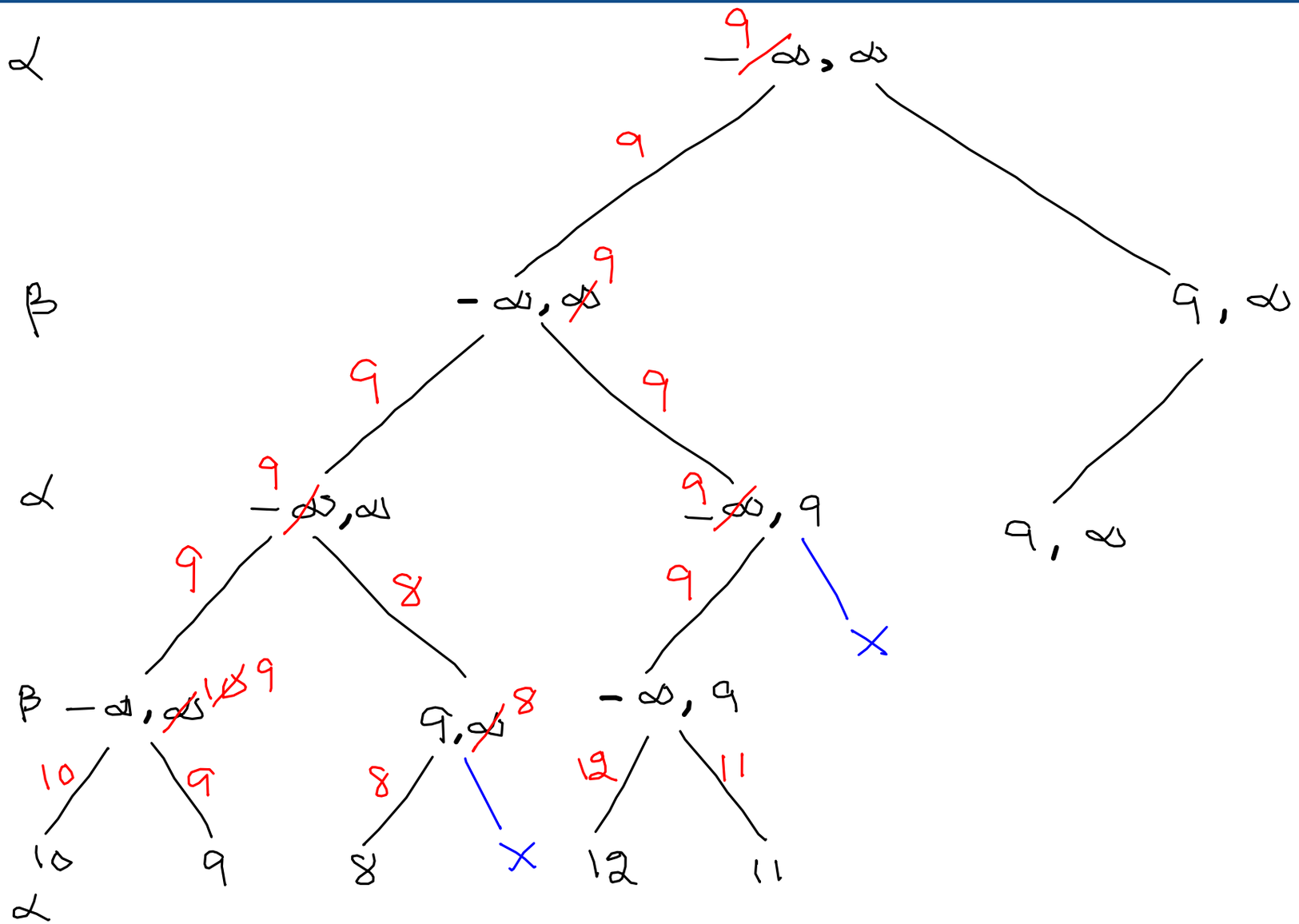

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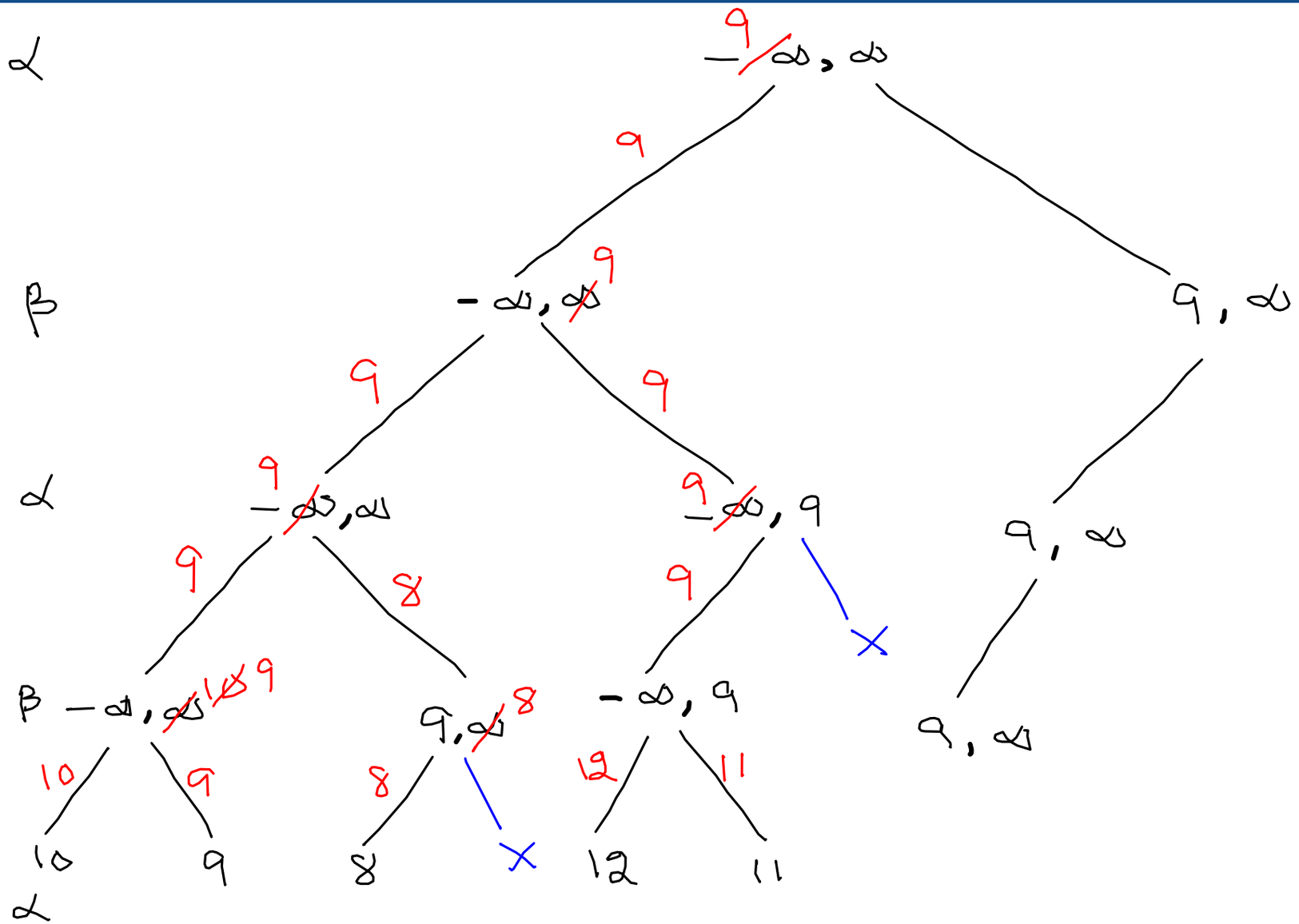

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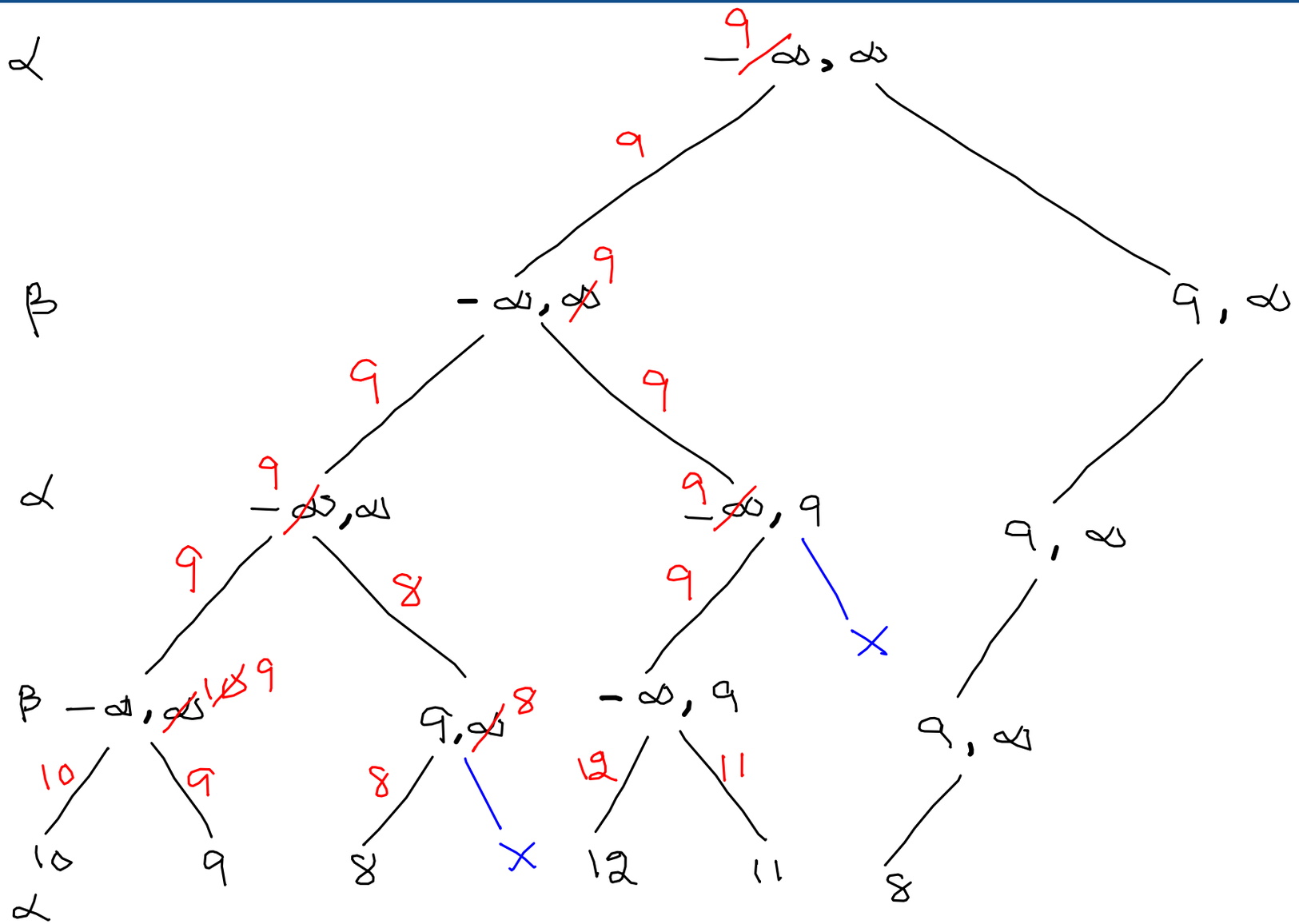

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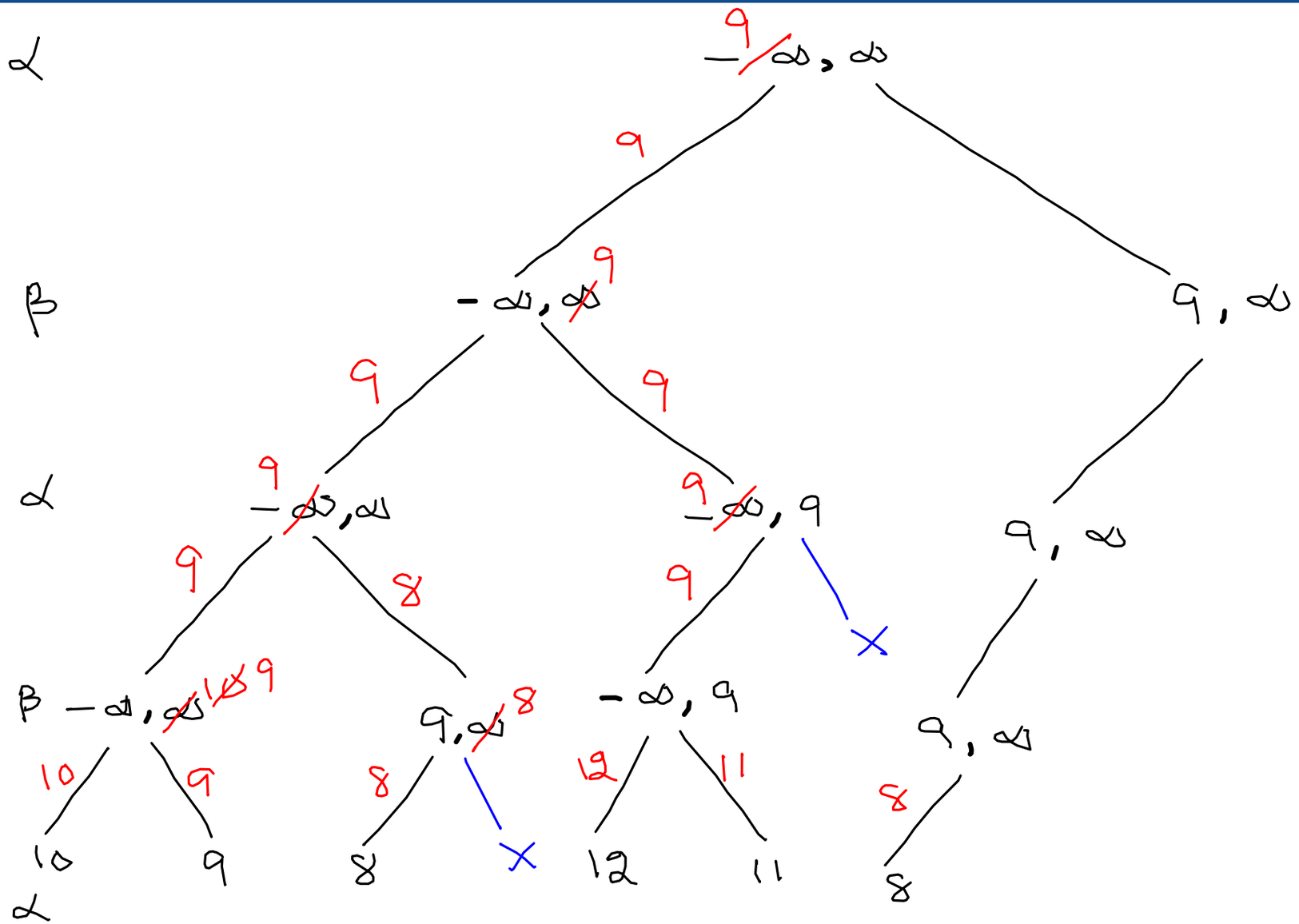

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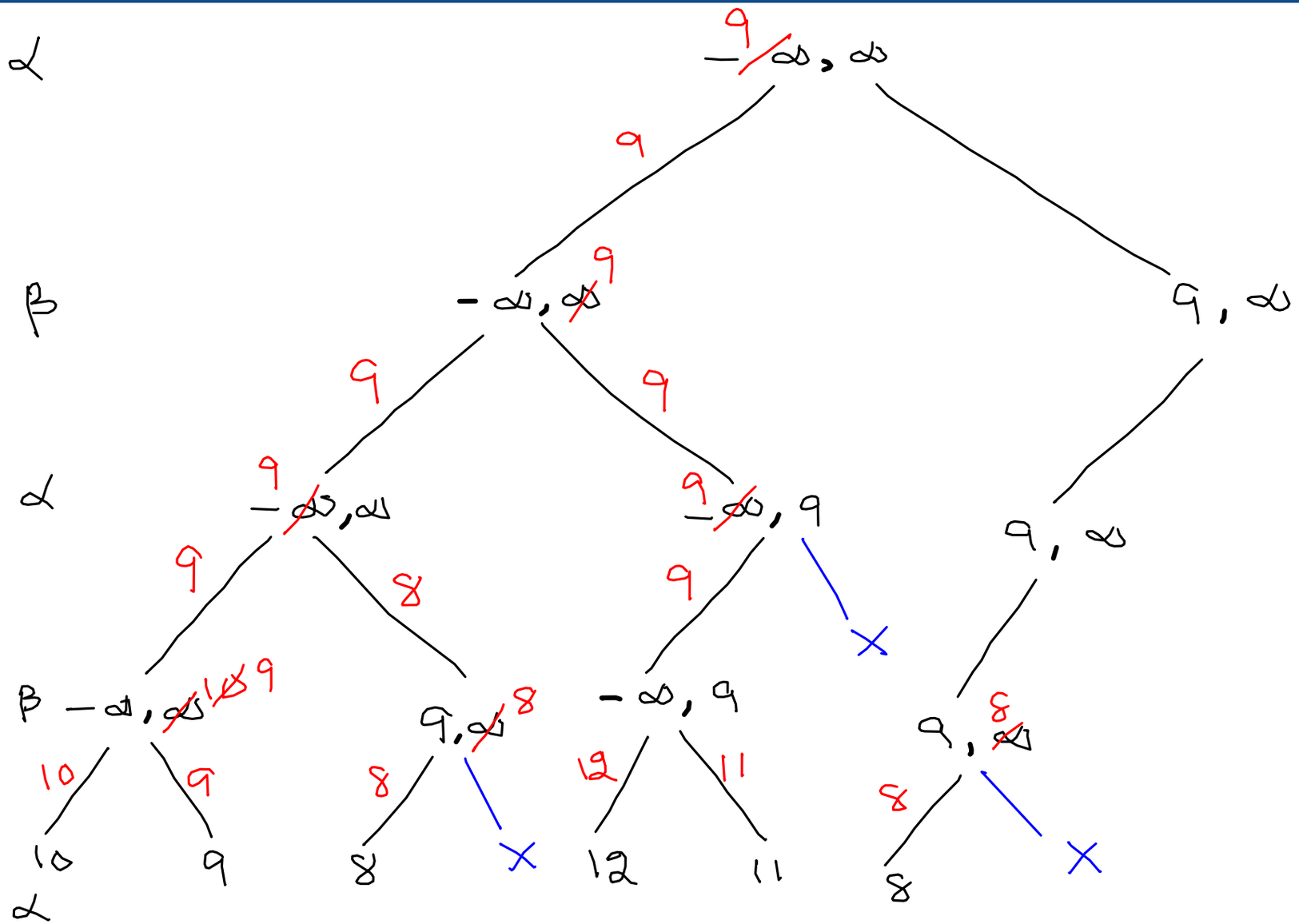

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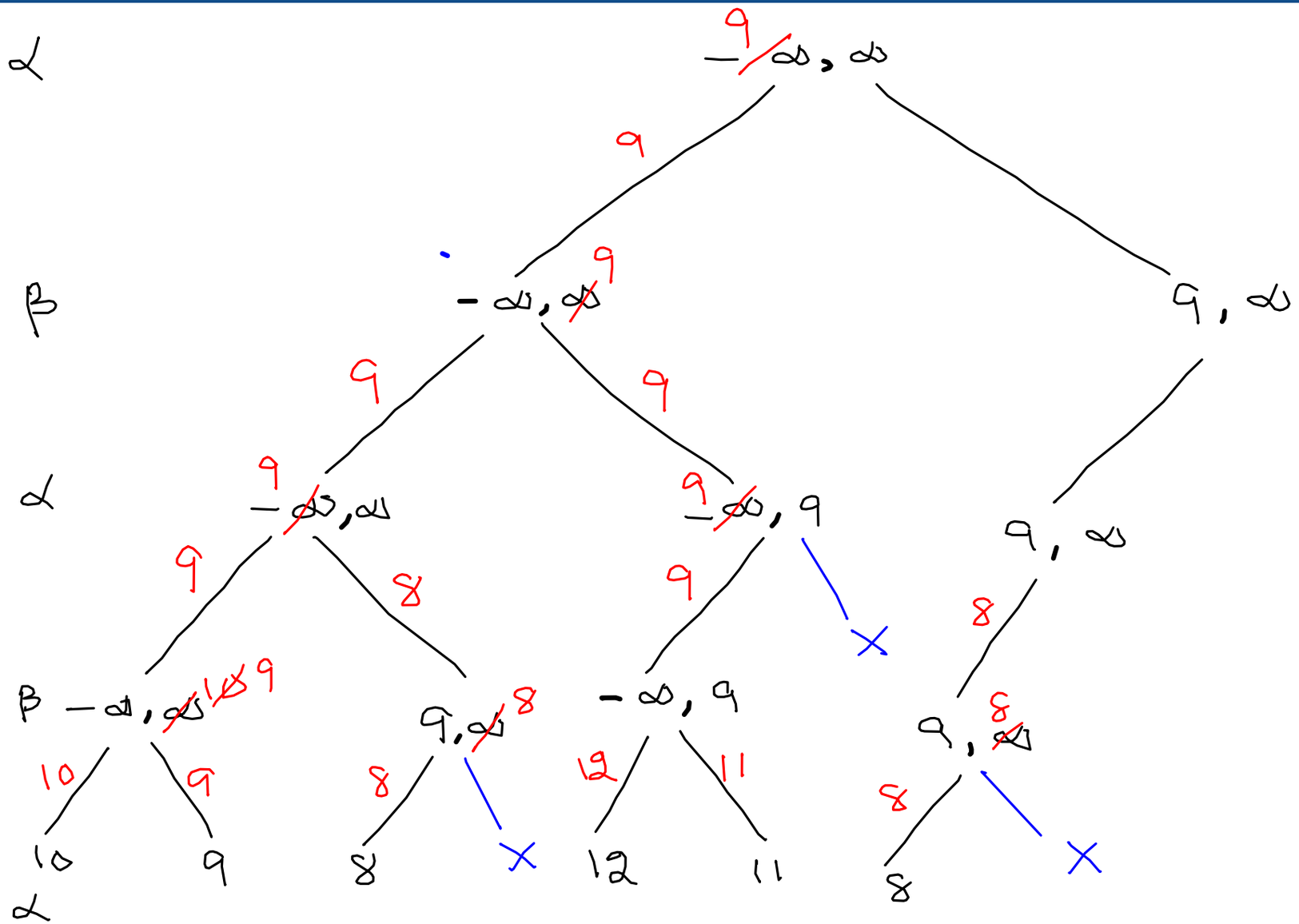

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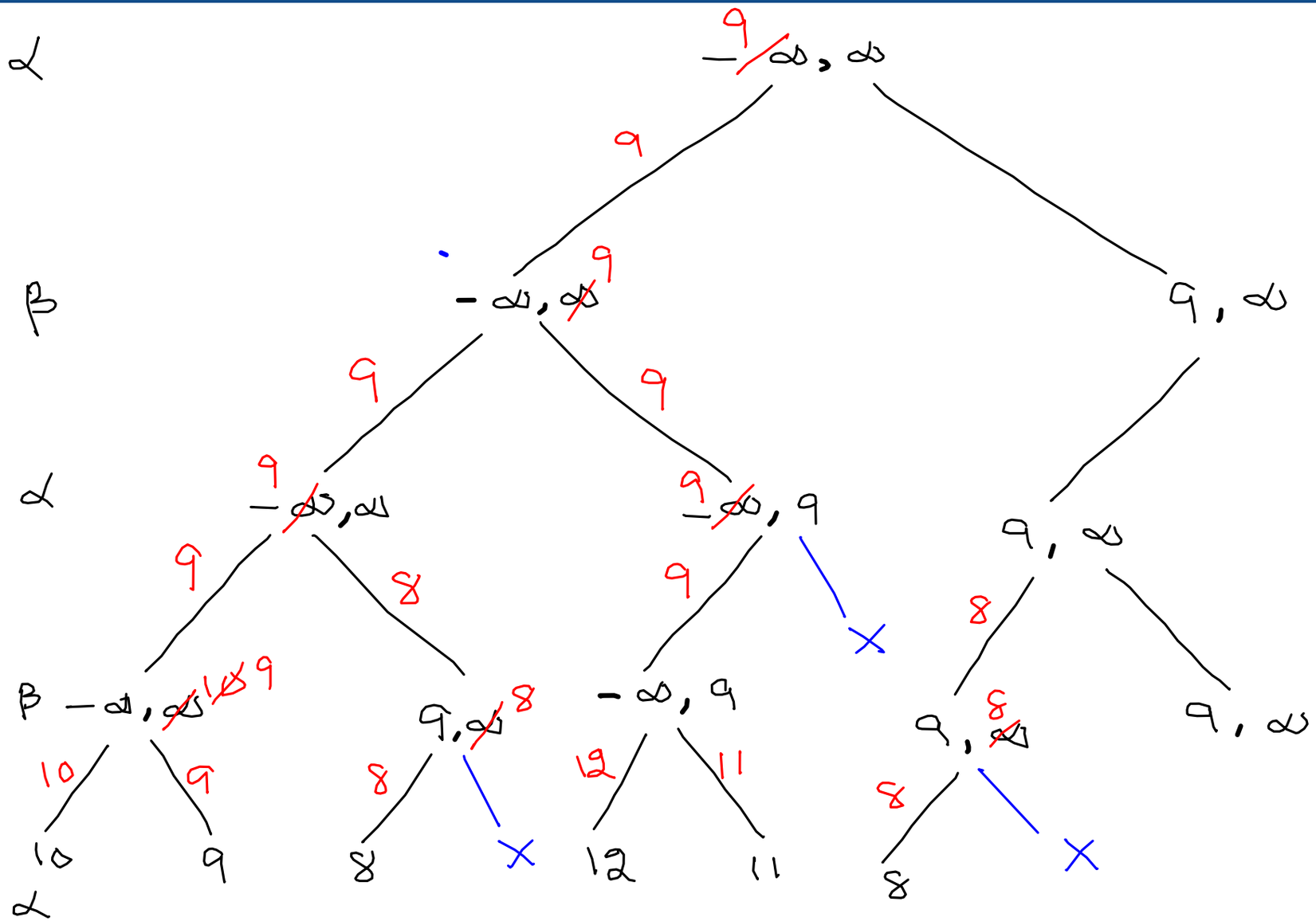

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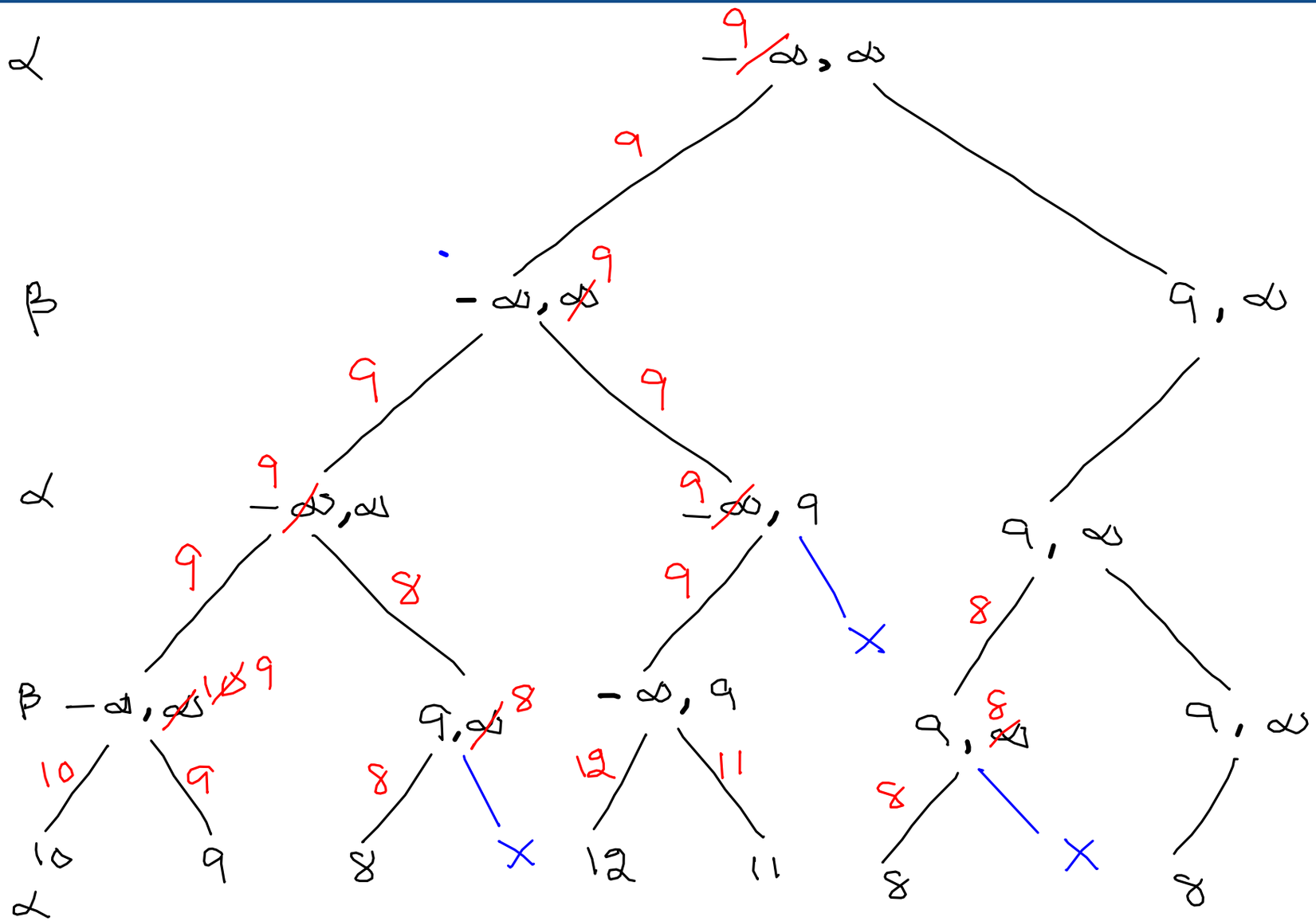

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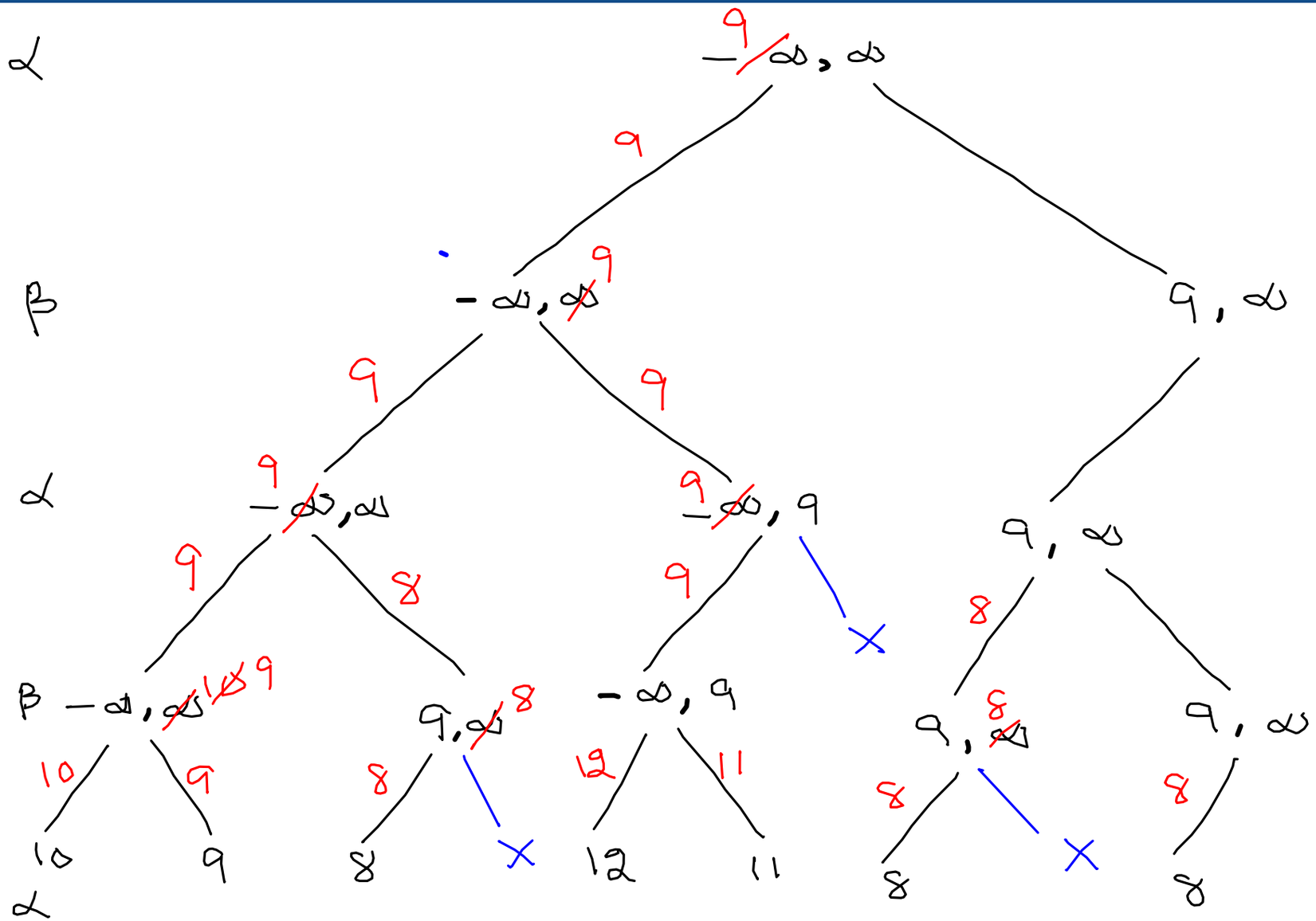

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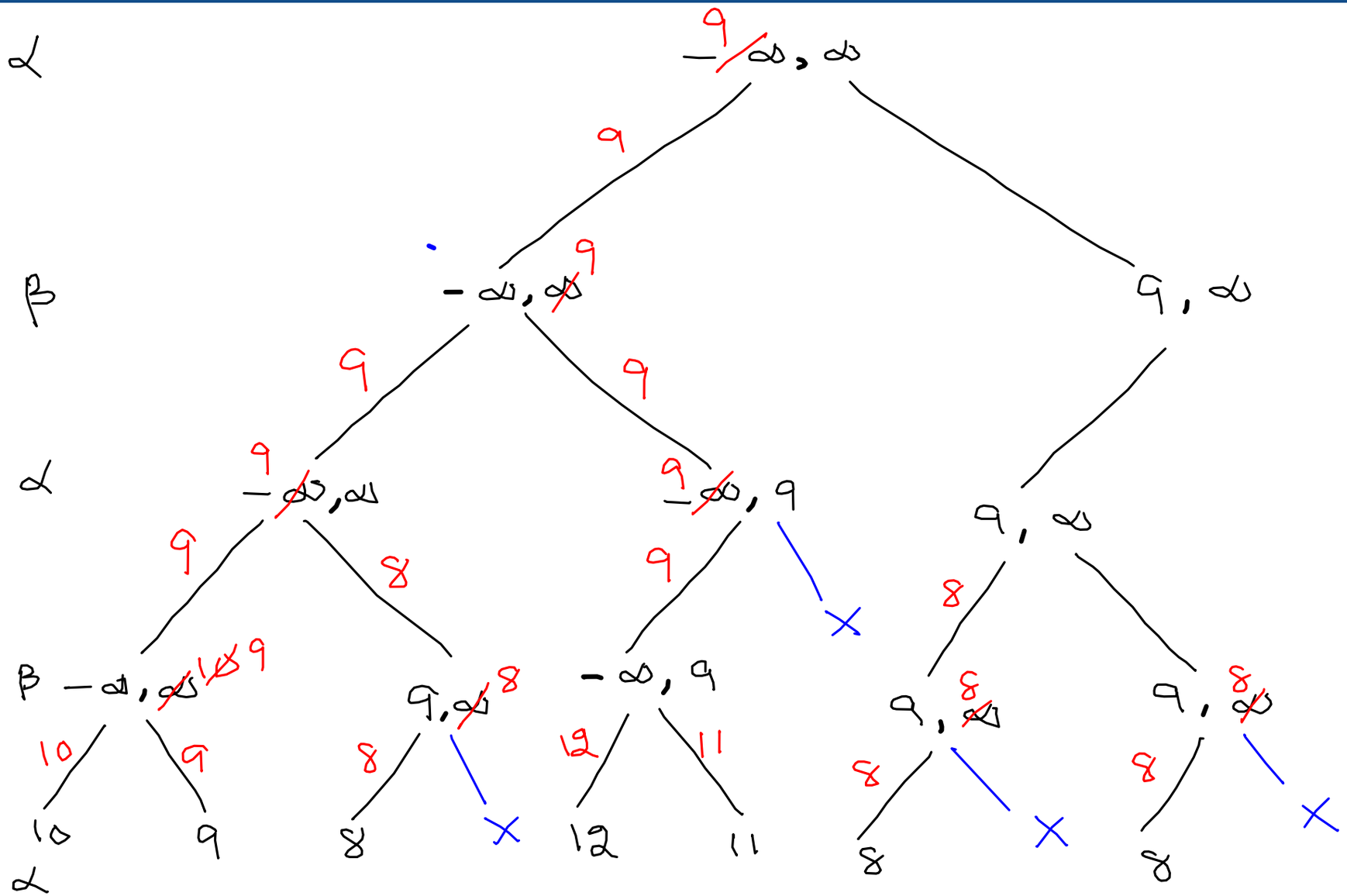

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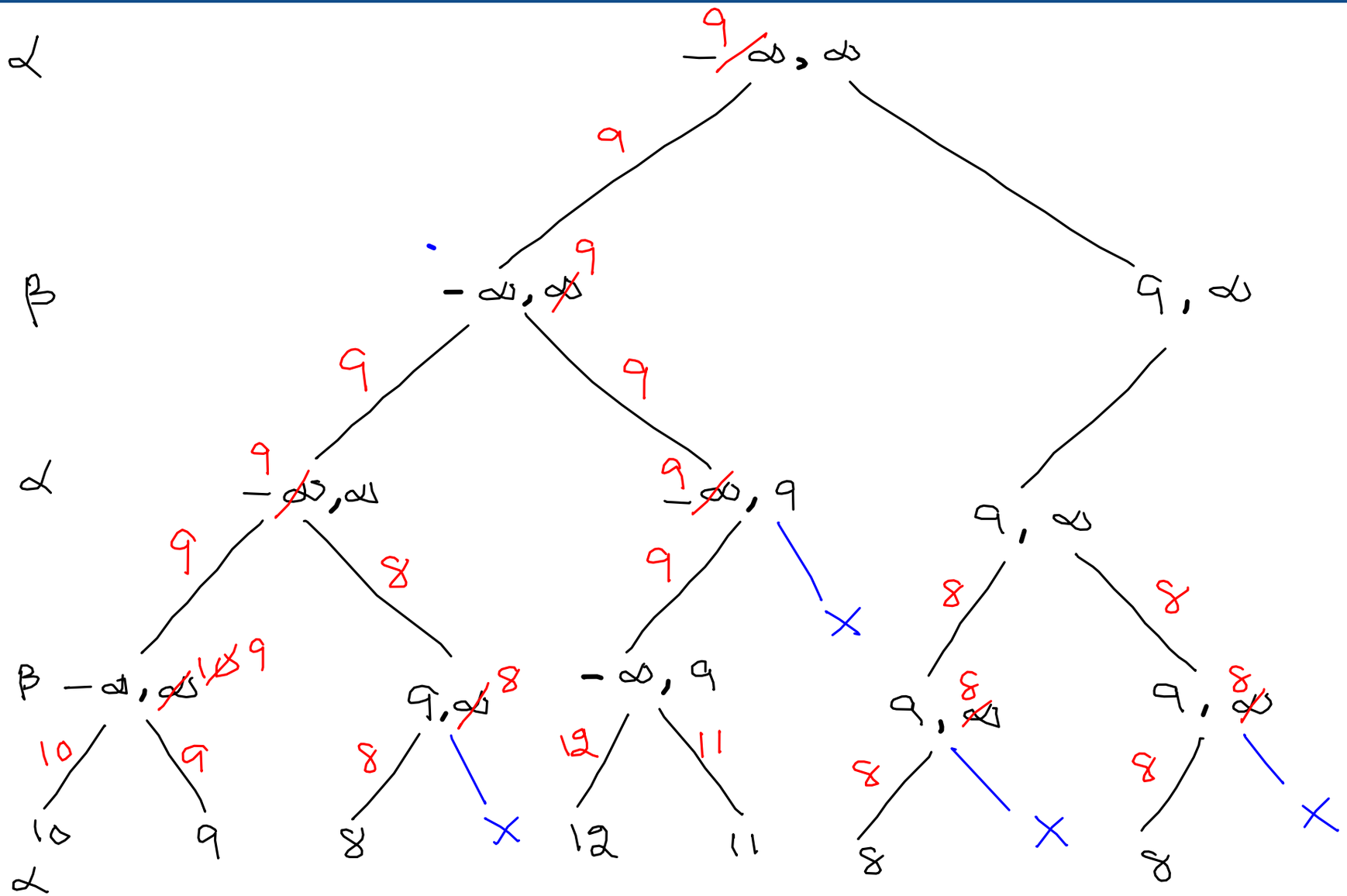

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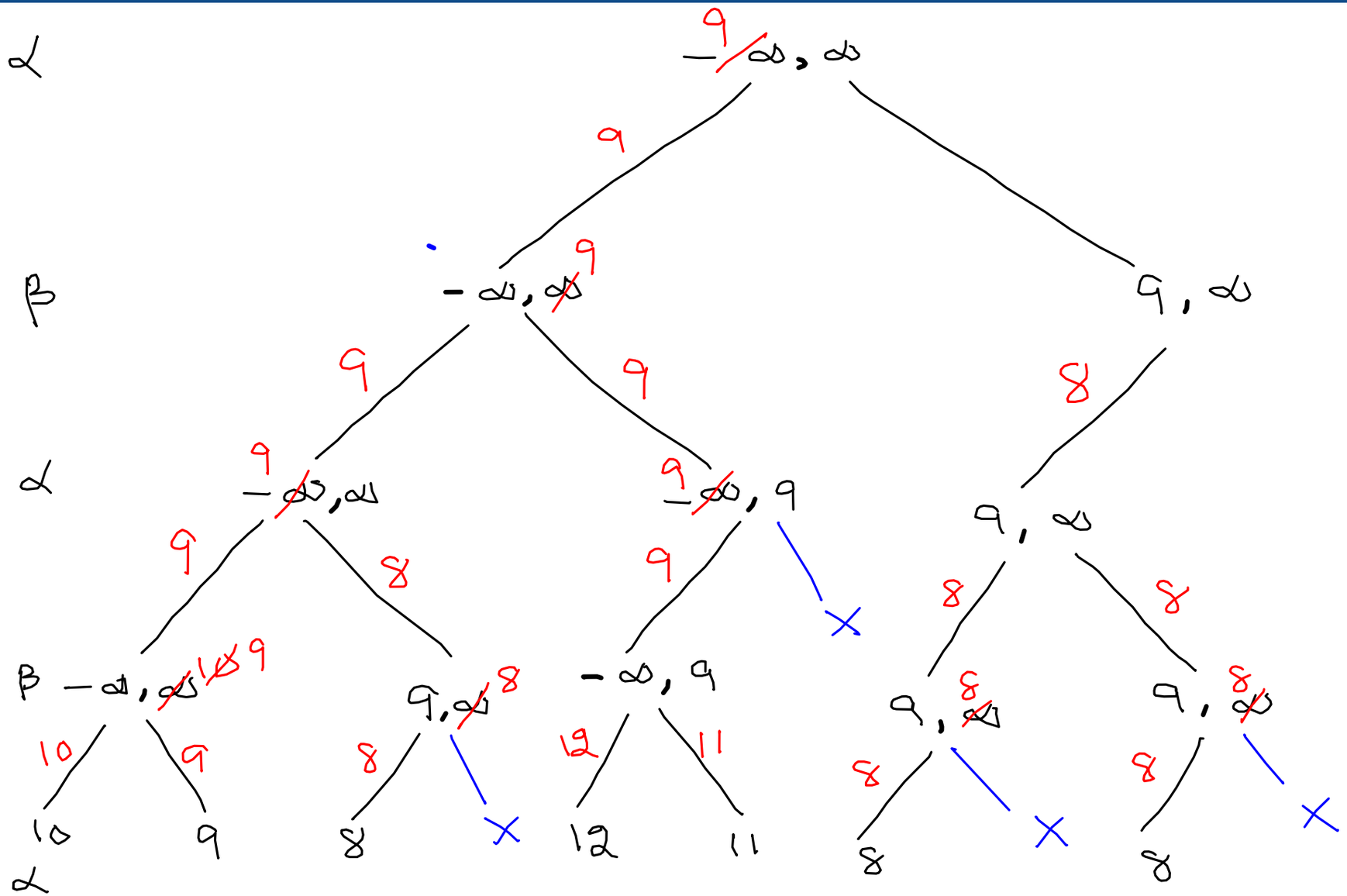

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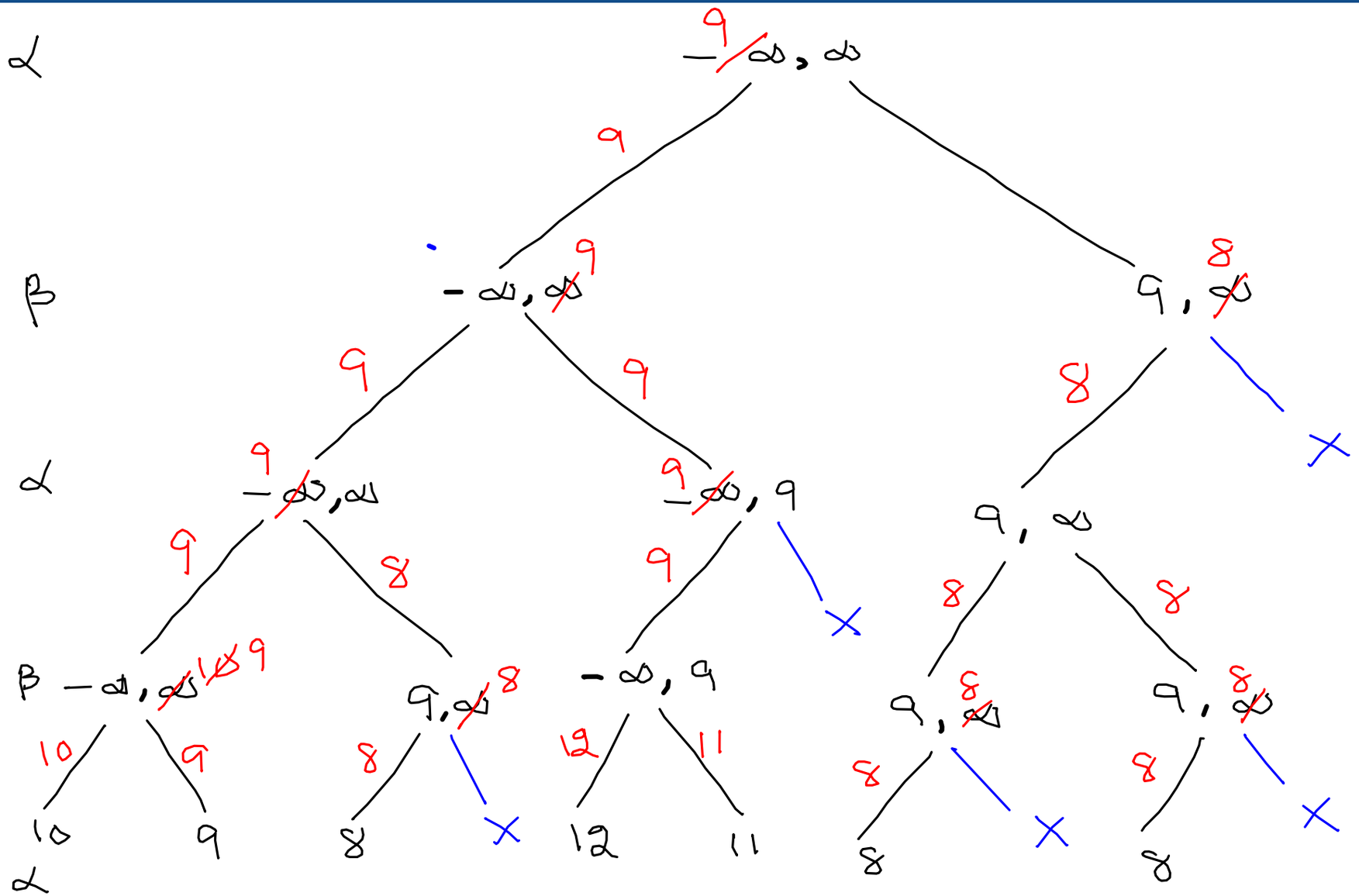

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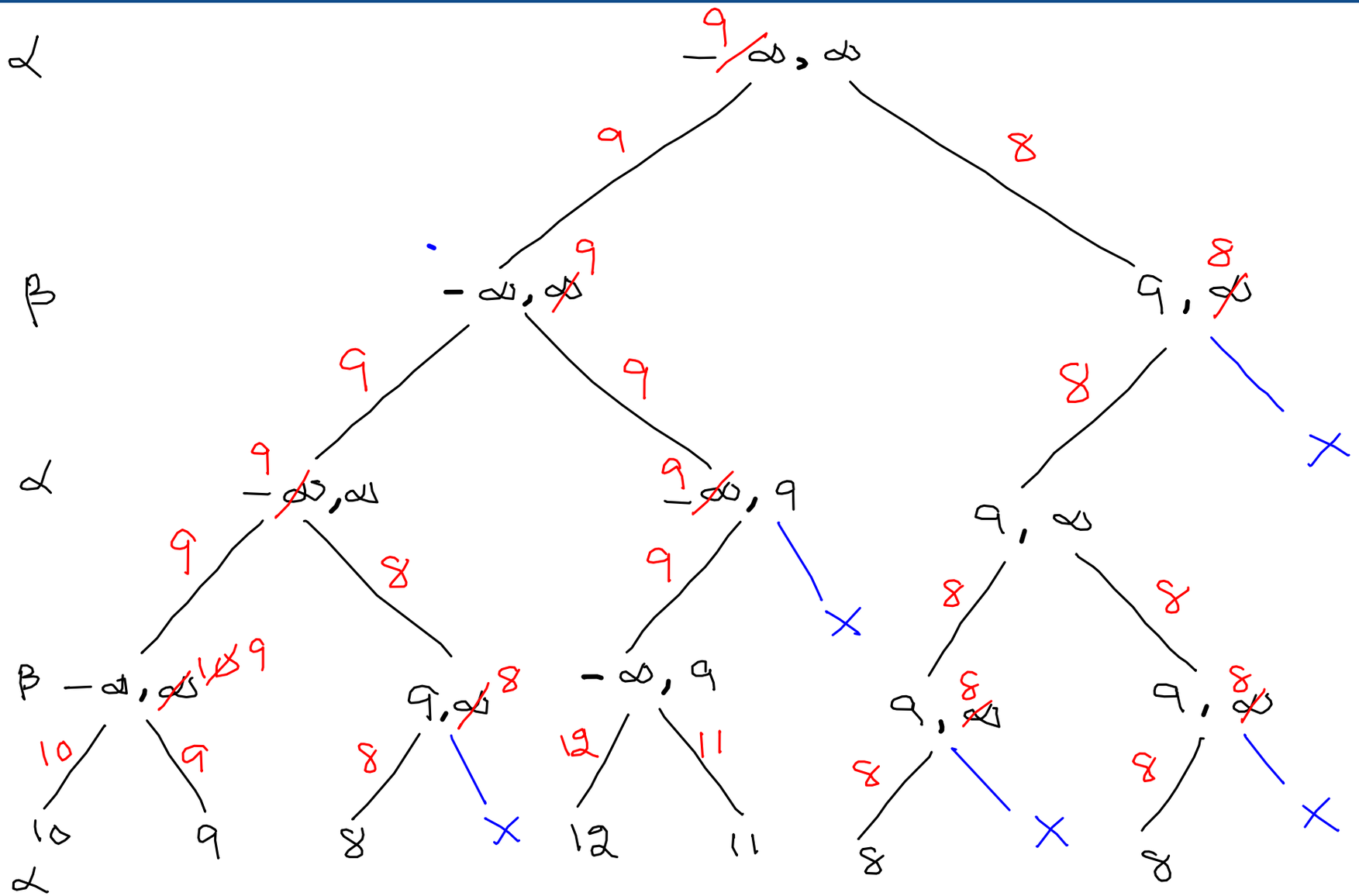
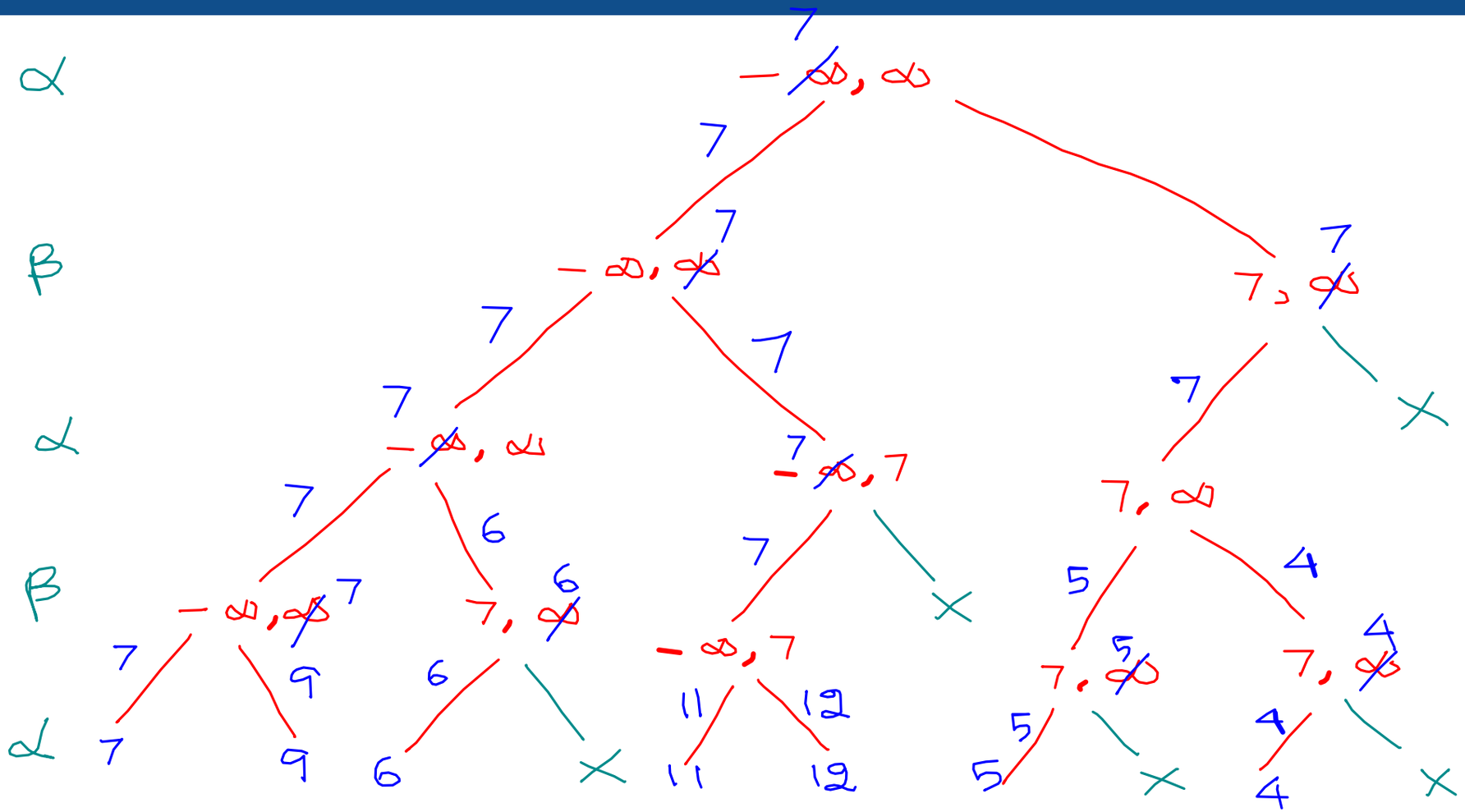


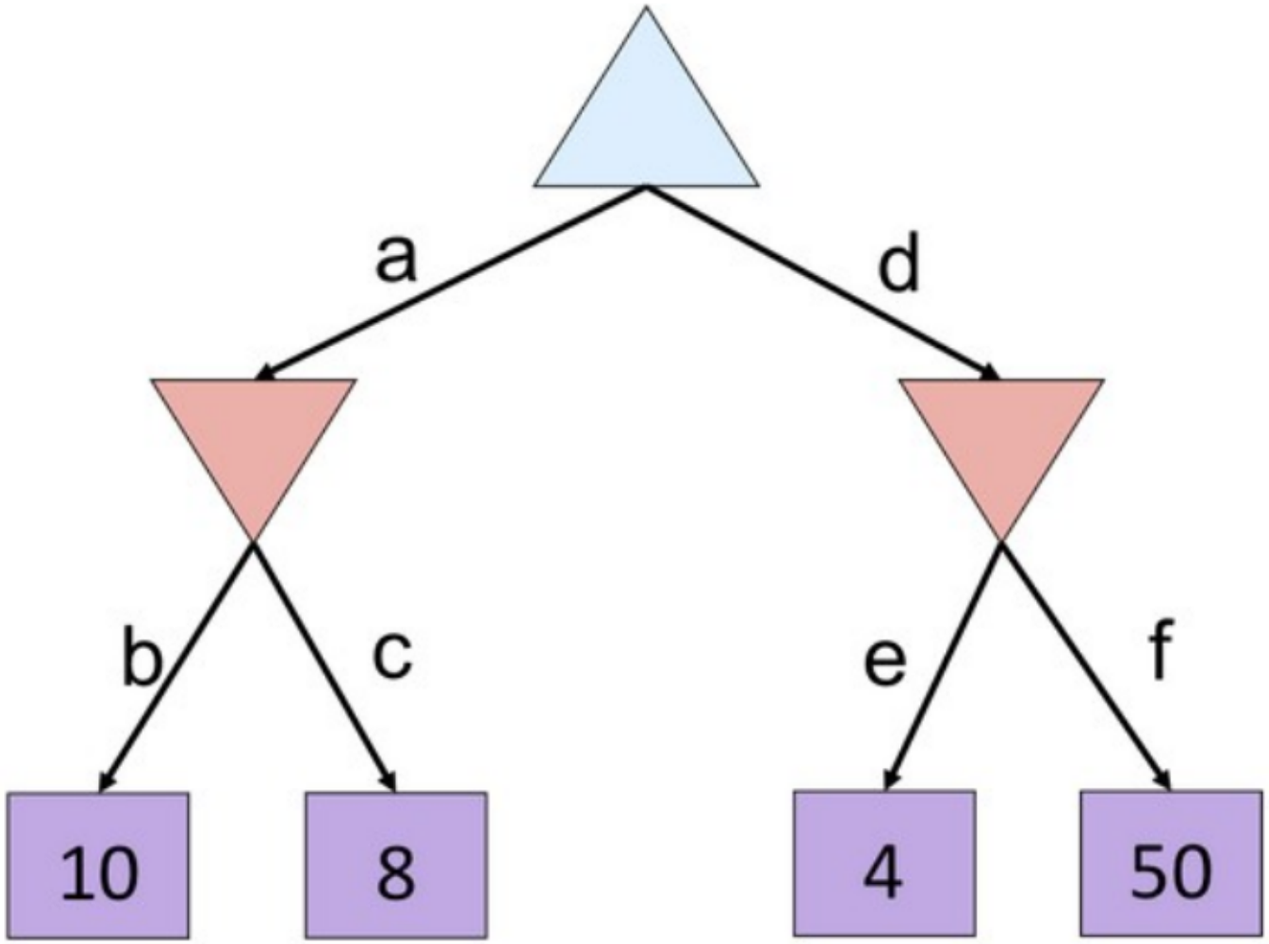
Move Ordering

- Effectiveness of alpha-beta pruning depends on the order in which the states (nodes) are examined.
- Examine the best successors first
  - $O(b^{m/2})$ instead of $O(b^m)$
  - Branching factor: from b to $\sqrt{b}$ (chess: 35 to 6)
  - Alpha-beta pruning can solve a tree twice as deep as minimax in the same amout of time
- Examine in random order
  - $O(b^{3m/4})$ for moderate b

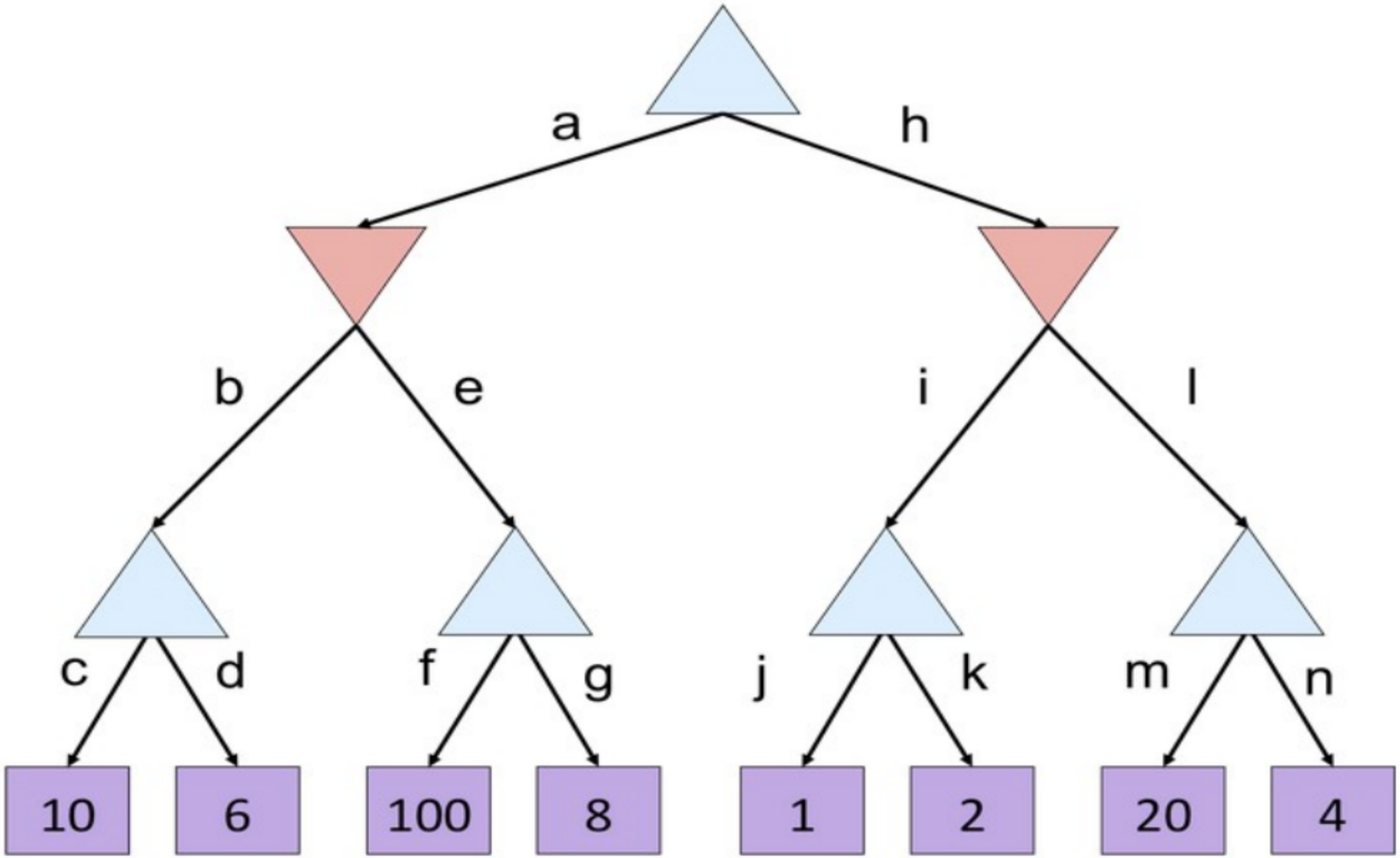
Alpha-Beta Pruning Trace



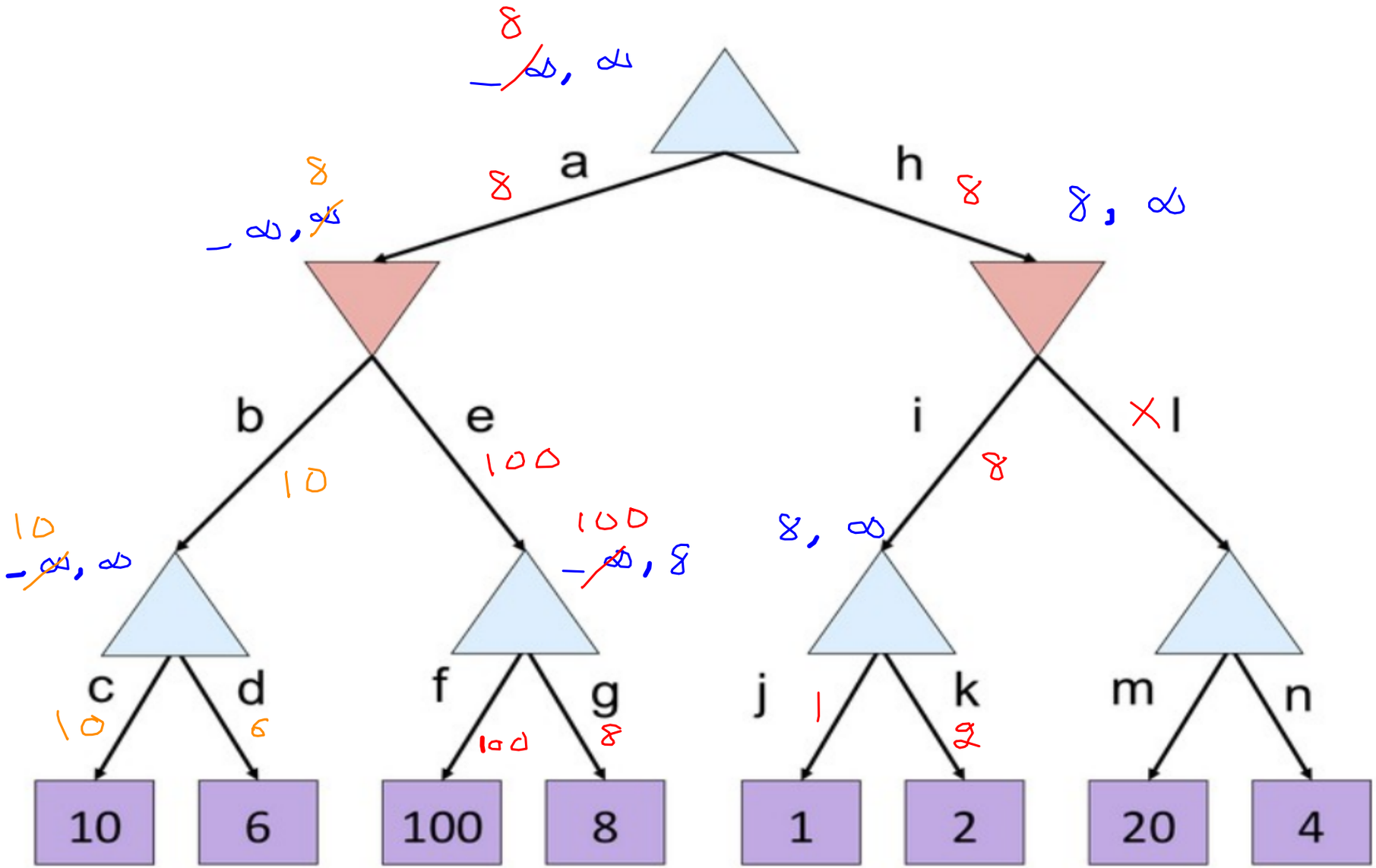
Alpha-Beta Question 1



Alpha-Beta Question 2



Alpha-Beta Solution



Killer Moves

- Dynamic move ordering
- Try first the best moves of the past
- Iterative deepening search: gain information from the current move
  - Search 1 ply first, record the best path of move
  - Search 1 ply deeper, use the recoreded best path of moves to inform the move ordering
  - Adds a constant fraction to search time
  - Made from better move ordering
- Best moves – killer moves
- Try best moves first – killer move heuristic

Transposition Table

- Repeated states in the search tree – exponential search time
- Transpositions – different permutations of the move sequence end up in the same position
- Transposition table
  - Store the evaluation of resulting positions in a hash table – look up
  - explored set in Graph-Search
  - May double the reachable search depth in chess

End